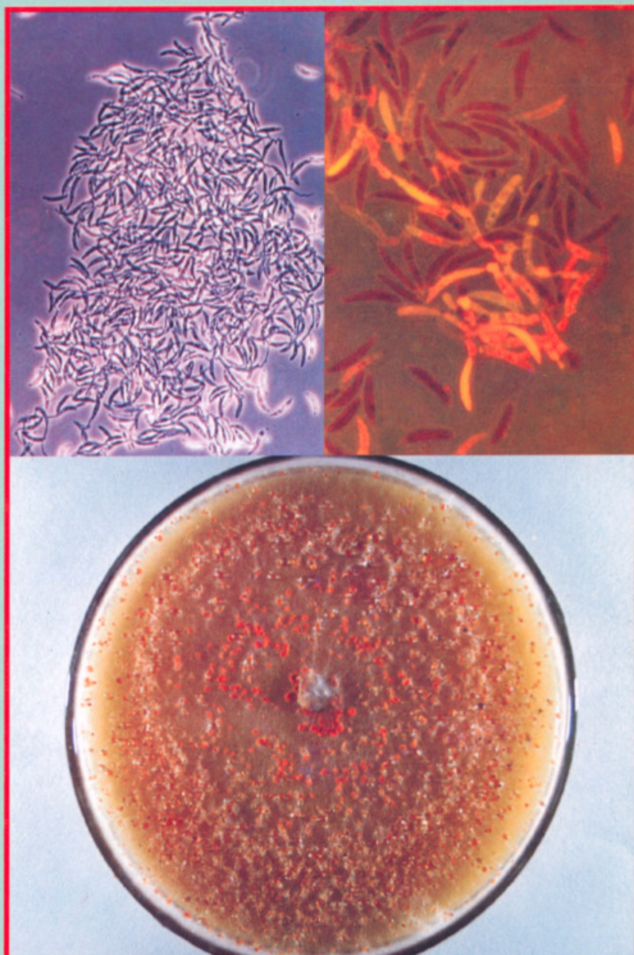
Red rot of Sugarcane



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LIST OF PLATES

Dedicated to

all

who bravely faced

the challenges of

Red rot

PREFACE

In the history of Plant Pathology, only a few diseases like rust of wheat, late blight of potato, ergot of cereals, rust of coffee, brown spot of rice and red rot of sugarcane have left longstanding impact on food availability which changed the face of human society. The potato murrain of 1842-45 in Ireland not only decimated over a million Irish people but also forced a mass exodus of another million or so to America.

In the Indian context, apart from the rusts of wheat and brown spot of rice, red rot of sugarcane is the most important disease that affected Indian agriculture badly with far reaching consequences. It is needless to say that red rot epidemic of 1938-39, along with brown spot epidemic of rice in 1942-43 augured the development of Plant Pathology as a subject in the curriculum of Indian Universities.

Red rot is the key menace of sugarcane in India, and the onus of its containment also squarely rests on the Indians. In spite of the best efforts, red rot is still posing challenges in stabilising sugarcane and sugar production. Trough the concerted efforts of breeders and pathologists, the disease has been contained to a manageable level and thus, the frequency and magnitude of red rot epidemic has been reduced to a great extent in recent years. In fact, entire sugarcane breeding in India is now geared around red rot and no sugarcane variety is released for general cultivation without resistance to the prevalent pathotypes of red rot.

In spite of its immense importance in the sugarcane agriculture, hardly any compilation has been produced so far on red rot except some odd reviews. This book is a humble effort in this regard to bridge the longstanding gap. Moreover, this book has been developed as a photo essay on red rot covering many facets of the host, disease and the pathogen. Errors and failings doubtless remain, and for these I owe the sole responsibility.

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Lucknow

July 2008

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I wish to acknowledge my gratefulness to Dr. Kishan Singh, the then Director of IISR, Lucknow, who initiated me in the realm of red rot in 1986 and reposed full faith to carry forward researches on this disease, in spite of my postgraduate specialization in bacterial plant pathology. My grateful thanks are also due to my teachers, who inspired and instilled in me the confidence to become a 'Plant pathologist' and not to get restricted in the boundaries of pathogens.

I am grateful to Dr. R. L. Yadav, the present Director of IISR, Lucknow for kindly agreeing to publish this effort in the form of a book. His inspiration and constant encouragement gave me the desired impetus to complete this book in time.

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It is my pleasant duty to acknowledge the support provided by my wife in writing the book. She not only read the book from a layman's perspective but also gave critical suggestions in improving the readability and clarity of the presentation.

Finally, I wish to acknowledge my gratitude to my parents who inculcated in me the confidence to follow the arduous path of truth '*it is truth and only the truth that prevails; one has to have patience and courage to attain the goal*'.

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July 2008

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INTRODUCTION

The sugarcane crop in India

The importance of sugar in human diet needs no introduction; it has become a part and parcel of daily life. Sugar is produced mainly from sugarcane and sugarbeet and more than 75 per cent of the world sugar comes from sugarcane. Throughout ages, sugarcane remained an important commercial crop of agriculture and trade in India contributing substantial revenue to the exchequer by way of tax and duties. Today, it is fast transforming into the most sought after renewable energy crop, as the demand for ethanol is increasing as an alternative green fuel for the automobile. This has become far more important in the backdrop of dwindling oil reserves. Currently Brazil is diverting 50 per cent of its sugarcane for the production of ethanol and blending ethanol with petrol to the tune of 25%. India is hoping to blend ethanol at least up to 10% in the near future. It has been estimated that India will need 495 million tonnes of sugarcane by 2025 AD to meet both sugar and energy demands (Yadav and Duttamajumder, 2007).

Sugarcane is a native of India and has been in use for '*gur*' making since prehistoric times. '*Gur*' was known as early as 3000 BC. The name of the raw sugar '*gur*' has originated from the word '*Gaura*' - a well-known dynasty that ruled Bengal. The raw sugar produced in Java (now Indonesia) is called '*Goela*' indicating that the process of *gur* making had travelled eastward from Bengal to Java. It is estimated that by 250 BC sugarcane had reached China from India and by 1 AD it reached the islands of Java (Indonesia). The westward journey of sugarcane started from India in 327-325 BC with the invasion of Alexander, the Great. The Greeks termed sugarcane as '*honey yielding reed*' as they were unaware of the existence of sugarcane and it was their prized catch. By the 8th century, sugarcane established itself in southern Europe (Italy, France and Spain).

In 1493, on his second voyage, Christopher Columbus took sugarcane and headed for the new continent–America. It reached Brazil in 1532 and now in the 21st century, Brazil has become the numero uno in the world, commanding maximum acreage and production of sugarcane. Today, sugarcane has established itself as an important cash crop in the world and around 115 countries are now actively engaged in growing sugarcane. Among the sugarcane producing nations, India occupies an important position with respect to cane acreage and cane production. Due to historical reasons world sugar trade is still governed by the United Kingdom.

Sugarcane occupies a prized position in the agrarian economy of India. About 45 million sugarcane farmers, their dependents and large mass of agricultural labourers are involved in cane cultivation. Besides, about half a million skilled and semi-skilled workers, mostly from rural areas, are engaged in the cane industry, the largest agro-processing industry of India (cotton industry is the foremost agro-industry of India). In recent times, increasing use of sugarcane and its by-products in the form of renewable energy source as fuel blend in petrol driven vehicles, and the production of electricity through co-generation using bagasse has opened up new opportunities for this crop. The area under sugarcane is hovering around 4 million hectares with a production of 240-295 million tonnes of cane with an average productivity of 65-68 tonnes/ha. Manufacturing of white sugar consumes nearly 60% of the cane produced in the country; 27% cane is utilized by '*gur*' and *khandsari* industries and the rest is utilized for other purposes such as chewing, juice vending and seed-cane. The performance of sugarcane, as judged by the acreage it commanded in the last

Table 1.1 Area, production and productivity of sugarcane in India

Period	Area (million ha)	Production (million tonnes)	Productivity (tonnes/ha)	No. of mills in operation
1930-31	1.176	36.35	30.9	29
1933-34	1.385	53.29	38.5	111
1940-41	1.617	51.97	32.1	148
1950-51	1.707	69.22	40.5	138
1960-61	2.456	110.54	45.0	173
1970-71	2.615	126.36	48.3	216
1980-81	2.667	154.24	57.8	314
1990-91	3.686	241.04	65.4	385
2000-01	4.316	295.95	68.6	436
2004-05	3.661	237.08	64.8	400

(Source : Indian Sugar, September 2006)

four decades, indicates that it has increased from a level of 1.70 million ha in 1950-51 to the level of 4.31 million ha in 2000-01. The production of sugarcane has increased from 69.22 million tonnes in 1950-51 to 295.95 million tonnes in 2000-01 (Table 1.1). There is a steady increase both in cane acreage and yield. Similarly, the numerical strength of the sugar mills has also gone up from 138 in 1950-51 to 436 in 2000-01.

The sugarcane growing area of India may be divided into two broad zones viz., tropical south and subtropical north, depending on the growing conditions. These zones are further subdivided, based on cane yield and recovery patterns. The subtropical region accounts for 60 per cent of cane acreage but only 45 per cent of white sugar production. Of late, this situation is gradually changing and contribution to both cane and white sugar is steadily increasing.

Sugarcane plant

Sugarcane, a marvel of the plant kingdom, is an efficient harvester of solar energy and uses C_4 pathway of photosynthesis like maize. It is a tropical perennial grass and is believed to have originated in Indo-Burma (now Myanmar)-southern China region extending to Java (now Indonesia). The primary economic product of sugarcane is sugar, a disaccharide, which on hydrolysis yields two monosaccharides i.e., glucose (dextrorotatory) and fructose (levorotatory). The sugar is stored in the parenchymatous tissues of the stalk, and by crushing the mature stalks, cane juice is extracted. In addition to sugar, the stalk bears bud and root primordia at the node, and thus is destined to act as the vehicle of vegetative propagation. The internode (the stalk portion between two nodes) contains the bulk of disaccharide saccharose (sucrose, sugar), which meets the immediate requirement for germination of bud and its establishment as a young cane plant with well developed root and shoot systems. A normal mature cane stalk, at harvest usually contains about 20-30 effective internodes for crushing. The vegetative/clonal propagation is used for all commercial purposes, and sugarcane is planted with stalk cuttings containing bud and root primordia called 'sett'. Sett may have one, two, three or more buds but as a general practice, a three-bud sett is used for planting. The cane stalk used for seed purpose is termed as 'seed-cane'. The other feature of sugarcane is the ratoon/stubble crop. This ratoon crop is raised by allowing the underground buds of the left over stalk to sprout after harvest and initiate another cycle of crop growth. Sugarcane, like other grasses, also produces true seeds (a product of sexual reproduction). The sexual means of reproduction is of very little direct consequence in the commercial cane cultivation for propagation of the crop. However, it is of prime importance to the cane breeders for generating new variability and genetic improvement of sugarcane. In fact, rediscovery of seed of sugarcane in late 1880s in Java transformed the face of sugarcane in the world, and today the sugarcane industry is solely supported by the hybrid canes. Genetically speaking, the present-day commercial sugarcane is highly heterozygous, complex polyploid, species hybrid (designated as *Saccharum* spp. hybrid) harnessing chromosomes/gene complements from diverse but related genera and species. A conservative estimate on the chromosome complements of modern sugarcane cultivars has indicated that 90% of chromosomes come from *S. officinarum* and rest are from *S. spontaneum* and others.

A look at the World Atlas of Sugarcane reveals that in spite of tropical origin it is successfully grown even 12^0 beyond the Tropic of Cancer and Capricorn ($23½^0$ N & S of the equator; 37^0N South Spain, 30^0 S, Natal, South Africa). However, in any geographical location, prevailing agro-climatic conditions and edaphic factors influence the duration and the productivity of the cane crop. Tropical climate permits longer period of crop growth and higher cane yield but less of sucrose in juice whereas, subtropical climate restricts both growth and yield but allows more of sucrose accumulation in juice. Usually, growing season of sugarcane varies from 10 to 12 months in subtropical region and from 18 to 24 months or more in tropical region.

Sugarcane belongs to the genus *Saccharum*. The word *Saccharum* owes its origin to the Sanskrit word '*sarkara*' or '*sakkara*' meaning sugar; this became '*sukkar*' in Arabic and '*Sakharon*' in Greek. In 1753, Linnaeus established the genus *Saccharum* along with the other crop plants in his master-piece '*Species Plantarum*'. (Incidentally in 1757, the battle of Plassey was fought, paving the way for the British colonial rule in India. In 1784, first colonial sugar factory was established by the British in Bengal). The genus *Saccharum* is placed in the grass family Gramineae, sub-family Panicoideae and tribe Andropogoneae. Traditionally, six species have been recognised in the genus *Saccharum* viz., *S. officinarum, S. barberi, S. sinense, S. robustum, S. edule* and *S. spontaneum.* Due to frequent intermixing (cross fertile/sexually compatible nature of sugarcane with the recognised species and allied genera) among the related species in nature, there was always some doubt about the distinctiveness of these recognised species (Parthasarathy, 1946). Irvine (1999) considered only two species in the genus *Saccharum* - the wild species *S. spontaneum* and the cultivated species *S. officinarum*. The rest of the four species viz., *S. barberi, S. sinense, S. robustum* and *S. edule* have been placed as the natural hybrids of *S. officinarum* and *S. spontaneum*. Probably such natural crossing took place around 1000 BC.

The growth pattern of sugarcane crop may be divided into five distinct phases viz., (i) germination (sprouting of axial bud of stalk cuttings; gives rise to a mother shoot), (ii) tillering (underground buds of young mother shoot grow to form daughter shoots), (iii) grand growth (the period of stalk elongation, a quantum jump in the height of the crop), (iv) maturity (accumulation of acceptable level of sugar in juice), and (v) flowering. The optimum temperature for germination of bud is about 25^0C. In subtropical India, in spring planted crop, germination usually takes place within one month of planting and may take another two weeks to complete (1½ months). Tillering starts after germination phase and continue for next two months. In this phase, from the mother shoot, new daughter plants (tillers) develop and increase the shoot population. Usually this phase switches over to the grand growth phase (GGP) with the arrival of monsoon in subtropical India. The monsoon months usually constitute the period of grand growth - with the availability of abundant rain and humidity along with favourable temperature and sunshine hours, the compressed cane stalk developed during summer months suddenly takes up the vertical growth and starts increasing in leaps and bounds (through elongation). Incidentally, this is the phase of the sugarcane crop (GGP) when the red rot pathogen flares up and takes a heavy toll. Usually, in subtropical India, a sugarcane produces about 50 buds/stalk in a crop cycle of 12 months. About 10-15 buds/stalk are usually left in the soil after harvesting of plant cane, and these stubble buds, on germination, give rise to a ratoon crop. Similarly, 10-15 buds always remain immature at the young nodes of crown leaves, and these are left out. Of late, these buds constituting the 'shoot tip' are being used in micro-propagation. Thus, the economic/millable portion of the stalk (millable cane) usually has 20-30 buds or internodes. Sugarcane is a typical tropical plant and requires an ambient temperature of 24-30^0C, enough sunshine and adequate water and humidity for growth. The affinity of sugarcane for water was aptly paraphrased "*Sugarcane loves water but doesn't tolerate wet feet*" by Sir Venkatraman. A temperature around 10-15^0C is suitable for maturity and peak accumulation of sugar. A temperature below 5^0C is harmful to the crop and it does not tolerate frost.

In any plant system, maturity of vegetative phase leads to the reproductive phase, and sugarcane is no exception. Depending on the availability of day length, moisture and temperature, sugarcane switches from vegetative to reproductive phase. Floral development is initiated by shortening of day length. The apical growing point of the stalk produces the flowers in an inflorescence. The inflorescence (tassel) of sugarcane is also called arrow **(Plate 1 a, b)** and so the flowering is also termed as arrowing. For all commercial purposes (except the cane breeding), the arrowing is taken as a bad character, as like most of the members of Gramineae, flowering also sounds the impending bell of death for the plant. In sugarcane, most of the stored sucrose is gradually utilized in the process of flowering and seed formation. The inflorescence consists of a main axis, and first, second and third order branches. Spikelets are arranged in pairs **(Plate 1c)**. The sugarcane flower consists of three stamens (pollen is very short lived having half life only 12 min after dehiscence) and a single carpel with a feathery stigma **(Plate 1d)**. Flowers take two to three months to mature after initiation.

Table 1.2 Important temperatures in the life of sugarcane

Temperature (^{0}C)	Behaviour of sugarcane
62	Heat adapted seed cane survives in water (20 min)
54	Hot air treatment of seed cane (8h), Moist hot air treatment (2½ h)
52	Hot water treatment (30 min)
50	Hot water treatment (2h)
40-45	Average summer temperature in subtropical India
24-30	Average growing season (Grand growth phase) This is the temperature when red rot flares up
25	Optimum for germination
<18.3	During photo-induction, flowering is prevented
<17	Pollen sterility, if occurs during microsporogenesis
10-15	Suitable for maturity and accumulation of sugar
<8.5	Buds and setts fail to germinate
<5	Frosting and reduction in sugar recovery
0	Cold chlorosis
-2.8	Young cane plants die
-3.5	Terminal buds and mature stalk die
-5.5	Entire above ground portion is killed
-11	Underground portion is killed

In India, flowering in sugarcane takes place during October to December (Coimbatore condition). It will not be out of place to mention that many of the *Saccharum officinarum* clones available in the world collection of sugarcane germplasm do not flower (probably due to the selection for non-flowering at the hand of aboriginals, who grew these at their kitchen garden). These noble canes therefore could not be further exploited commercially through conventional breeding. The mature seed of the sugarcane is termed as fluff/fuzz **(Plate 1f)** due to the tuft of hairs (callus hair) attached to the tiny seed for aerial dispersal. Mature fuzz consists of the mature dry fruit (caryopsis), glumes, callus hairs, anthers and stigma. The superfluous parts of the inflorescence are generally handled, stored and sown with the seed because it is not practical to separate them. The seeds do not have any resting period and germinate immediately to produce tiny grass like plants **(Plate 1 g, h)**. The seedlings differ from the common grasses in the tactile response and appear a bit stout on touch.

Table 1.3 Sucrose content in the genus *Saccharum*

Species of *Saccharum*	No. of clones evaluated	Sucrose % (fresh weight)	
		Range	Mean ± SD
S. spontaneum	40	1.3 - 6.5	3.96 ± 1.48
S. robustum	10	3.9 - 11.7	7.83 ± 2.67
S. sinense	2	13.1 - 13.9	13.45 ± 1.01
S. officinarum	25	15.0 - 20.2	16.05 ± 1.12

(After Bull and Glasziou, 1963)

Saccharum officinarum

This is the type species of sugarcane. It is identified and is best represented by thick-stemmed, colourful, soft and juicy sweet canes. It has never been found in the wild. Available clones of this species were part of the kitchen gardens of primitive tribes in the Melanesian islands who used these canes for chewing. Owing to the majestic appearance and quality, the Dutch settlers in Java named the cane of their choice as **'noble cane'**. Noble canes got different names in different countries like Creole, Otaheite, Lahaina, Badila, Black Cheribon, Fiji, Ashy Mauritius, Red Mauritius, etc. It is an octoploid (2n=80), with a basic chromosome number of x =10. Before the advent of hybrid varieties in the 20th century, pure cultivars of *S. officinarum* served the sugar industries around the globe. However, these genotypes off and on suffered badly from the attack of various pests and pathogens and thus, necessitated frequent replacement of planting material from disease/pest free area/region/country. *S. officinarum* was present in India from the prehistoric times; however reintroduction of this type of cane took place in a regular fashion. Like, Major Sleeman (based in Jabalpur) introduced this type of cane from Mauritius in 1827.

Saccharum barberi and *Saccharum sinense*

The famous thin-stalked north Indian canes, mostly used for jaggery (*gur*) making, were studied in detail by Barber (1916, 1918). He separated these into five groups, based on morphology of stem, leaf and root. These groups are Sunnabile, Mungo, Nargori, Saretha and Pansahi. The Chunnee, Kansar *(S. barberi)* and Uba (*S. sinense)* are well known due to their utilization in the early cane breeding programmes of both Java (Indonesia) and Coimbatore. Both Barber and Jesweit were of the opinion that the *barberi* type has arisen from *S. spontaneum* through the process of mutation and selection. The *sinense* type was brought to India from China by the East India Company around 1796 and was received by Roxburgh, who treated it as a new species and named as *S. sinense*. The clones of these groups, in general, possess deep root system with good tillering habit and are moderate in girth and weight, medium in sugar, high in fibre, and well adapted to the biotic and abiotic stresses of the subtropical conditions.

With the publications of Irvine (1999), D' Hont and Glaszmann (2001) and D' Hont *et al.* (2002) providing further supportive molecular evidences confirmed the hypothesis that the so called north Indian and Chinese canes, in real terms, are the natural hybrids of *S. officinarum* and *S. spontaneum*. These natural hybrids have established themselves under the guidance of evolutionary pressure exerted by the prevailing conditions of subtropical India and China. The *S. barberi* type was evolved to withstand the harsher biotic and abiotic stresses prevalent in subtropical India. Obviously, the nature has favoured the hardy thin canes with good root system in the evolutionary process. These natural hybrids were selected and reselected so much in the past that traditional taxonomists placed them as new species. The *sinense* type, that flourished in China, is cold tolerant, whereas, the *barberi* type that thrived well in the harsher climate of subtropical India is tolerant to drought. In addition, these canes possess ample resistance to ward off various pests and diseases. Probably, due to this hybridity, no immediate success was obtained by Barber and Venkatraman in their initial effort at the very beginning of Sugarcane Breeding Institute (SBI), Coimbatore.

Saccharum spontaneum

S. spontaneum, the *Kans*, is well known for its wide variability and adaptiveness to diverse climatic and edaphic conditions. It exists throughout the length and breadth of Indian subcontinent, Southeast Asia (China, Taiwan, Japan, Indonesia, Philippines, and New Guinea) and in parts of Africa. It is a polymorphic species and diversity ranges from annual to perennial type with both sexual and asexual reproduction taking place freely. Many forms are highly rhizomatous; rhizomes growing to several metres even. In general, the group as a whole possesses pithy stalk and hardly any extractable juice. Tillering is profuse, and in general, they exhibit marked resistance to moisture stress, low temperature and many of the diseases and pests. Chromosome numbers vary from 2n=40 to 128 (x =8, most frequent count, 2n=64). Meiosis in *S. spontaneum* is generally normal with bivalent formation, regardless of the chromosome number. It will be interesting to note that chromosome number in *S. spontaneum* varied widely depending on its geographical distribution. A gradual increase in chromosome number is observed from India to Java (Himalayan form 2n =56; Bihar 2n=64, 72; Assam 2n=80; Burma 2n= 96; Java = 112, 120, 128). It is often assumed that the forms bearing higher chromosome number have originated from the forms bearing lower chromosome number.

The story of POJ 2878

Early sugarcane breeders realized that the resultant F_1 progeny of *S. officinarum* and *S. spontaneum* was distinctively different from either of the parents. When *S. officinarum* clones were used as the female parent, progeny tended to be taller stalked, higher in sucrose level, and generally more vigorous than when *S. spontaneum* clones were used as the female parent. Reciprocal difference in vigour was attributed to '2n+n' transmission in *S. officinarum* (female) x *S. spontaneum* (male) cross combination. This phenomenon of irregular transmission was termed as **'female restitution'** by Bremer. The present day successes of sugarcane dates back to the early part of the 20th century when active use of this species with noble cane (*S. officinarum*) yielded the much-desired tolerant hybrids at the two main centres of cane breeding i.e., Java (Proefstation Oost Java) and India (Coimbatore). In fact, these spectacular hybrids provided desired fillip to the sugarcane and sugar industry worldwide (POJ 2878 in Java, Co 205, Co 213 in India, POJ 213 in USA), which were plagued with '*Sereh*', red rot and mosaic. No doubt, in the history of agriculture these events remained path-breaking achievements (use of wild species in the improvement of crop plant, harnessing of hybrid vigour, a quantum jump in acreage of the crop and yield) and ushered the *'first green revolution'*, both in India and elsewhere in the early part of 20th century. Due to the development of POJ 2878 alone, Java became a sugar surplus country in no time. In the history of sugarcane breeding, the Javan variety POJ 2878 stands out as a major achievement. Both as a commercial variety in its own merit and as a parent of many commercial varieties, it has established a remarkable record (Mangelsdorf, 1959). The genotype POJ 2878 was first raised in 1921 and by 1929, it established itself as the ruling variety of Java, surpassing all previous records. The breeding history of POJ 2878 is highlighted here as an example of utilisation of wild species in the development of disease resistant cane genotypes. The '*Sereh*' which was plaguing sugar industry of Java was a boon in disguise; it forced a comprehensive study of sugarcane–its breeding, diseases and the various management options. To name a few, like rediscovery of fertile seed of sugarcane, heat treatment of cane, discovery of diseases like red rot and leaf scald, development of hybrids and use of other related species in breeding are the most significant. The development started with the collection of Kassoer (a natural hybrid of unknown noble cane and unknown Glagh, *S. spontaneum*) from near the base of Tjerimai volcano in Java in about 1890. Similarly, POJ 100 was developed from an open pollinated tassel by Wakker in 1893 (Mangelsdorf, 1959). Miss Wilbrink, the pathologist who first reported leaf scald disease, tried her hand in producing clones resistant to '*Sereh*'. By crossing POJ 100 as female and Kassoer (was resistant to *Sereh*) as male parent she obtained POJ 2364 in 1911. POJ 2364 inherited immunity to '*Sereh*' from Kassoer but its sucrose content was too low to qualify as a commercial genotype. In 1912, the young Jeswiet joined the staff of Proefstation Oost Java as sugarcane breeder. In 1917, he crossed POJ 2364 with EK 28 (popular noble cane variety of Java). In 1921, he from 32 tassels grew 2266 seedlings, one of which received the number POJ 2878, the wonder cane of the world. It yielded 35% more than its predecessor and put a new lease of life to the ailing Javan sugar industry. Later, POJ 2878 became the most sought after parent, giving birth to several successful sugarcane varieties of the world.

In spite of the spectacular success of early hybrids, one finds the involvement of only a handful of *S. spontaneum* in the parentage of present day sugarcane cultivars. In the past 100 years of cane breeding, due to inherent difficulty in crossing, obtaining viable seeds and selection of desired type in the progeny always remained a major bottleneck in the effective utilisation of the veritable gene goldmine, the *S. spontaneum,* available in India. Newer findings on the origin and taxonomy of different species of *Saccharum* acknowledge the pivotal part played by this species in stabilizing *S. officinarum* progenies in the harsher sub-tropical environment of both India and China. Obviously, the progenies that could tolerate the onslaught of red rot and wilt pathogens survived, and naturally, these were selected for '*gur*' making. However, spread of these genotypes was restricted, especially in U.P., due to prevalent attitude towards the flowering of cane. In common folklore, flowering of sugarcane was taken as a curse and it foretold the death of the head of the family. In the area where cane was grown, the sight of flowering was obstructed by the servants. Early in the mornings, servants made a round of the cane crop and if any sign of flowering was noticed, it was immediately destroyed. Watt (1893) had also mentioned this in his book on Economic products of India and he wrote, "…*if cane flowers and seen by the head of the family, divine curse will surely befall on him and death is foretold*." Probably due to this reason, sugarcane farmers over the generations preferred non-flowering types.

2

SYMPTOMS

The pathogen, *Colletotrichum falcatum* Went, can attack any part of the sugarcane plant; be it stalk, leaf, buds or roots. *C. falcatum* completes its life cycle on the sugarcane leaf and usually the damage to leaf does not pose a serious threat to cane or cause much harm to the plant. The most damaging phase of this disease occurs when the pathogen attacks the stalk. Depending on the age of the stalk, time of infection and susceptibility of the cane genotype, it produces different types of symptoms. The typical stalk symptoms i.e., presence of white spots in otherwise rotten (dull red) internodal tissues and nodal rotting appear when the crop is at the fag end of the grand growth phase during August-September in subtropical India. In the early stages of infection, it is difficult to recognise the presence of the disease in the field, as the plant does not display any external symptom or distress. At a later stage, some discolouration of rind often becomes apparent when internal tissues have been badly damaged and are fully rotten. This is more pronounced in the stalk of light coloured genotypes **(Plates 9a, 38 b, c)**. At the end, affected plant dies. At the field level, this may be observed as the death of a few plants or clumps to the failure of entire crop **(Plate 2)**.

The infected cane setts carry the primary infection to the field. Depending on the nature of infection and availability of favourable environment, pathogen starts taking toll by killing the bud. This affects the germination and initial establishment of the crop. Poor germination leads to a gappy crop stand and reduction in yield. If, at all, the buds of the infected setts are able to sprout and grow, then above ground symptoms appear. The type of symptoms varies depending on the prevailing weather conditions. At first, symptoms appear as the death of young and emerging shoots **(Plate 3a, b)** without any conspicuous identifiable symptom (in March-April-May in north Indian condition, spring planting). Occasionally spindle infection also apears with heavy sporulation. However, in ratoon crop spindle infection appears much earlier **(Plate 3d)**. The dying symptoms are often confused with the damage of young shoots by rats / termites, especially in the ratoon crop. However, a close microscopic examination of these dying shoots, if affected due to red rot reveals the presence of fruiting structures of *Colletotrichum falcatum* in the form of acervuli and chlamydospores in the leaf folds **(Plate 6a, b, c)**. Occasional spindle infection showing heavy sporulation also appears in the emerging leaf (sett inoculation, adequate moisture). With the passage of time, if the infected shoots survive, one encounters tiller mortality **(Plate 3c)**. This is just the extension of death of mother shoots due to primary infection. Thereafter, the spindle infection **(Plate 4a, b, c, d)** appears and its frequency is more during the onset of monsoon and gradually the frequency tapers. Spindle infection was also observed during the 1938-39 epidemic of red rot (Chona, 1942) and in the epidemic of Godavari Delta (Barber, 1901), as evident from the furnished drawings and photographs of different symptoms. However, at that period this type (spindle infection) of infection was taken as the mid-rib infection in young crown leaves. Spindle infection contributes significantly in the spread of the disease. Experiments conducted at IISR, simulating spindle infection by inoculating young mid-ribs with the hypodermic syringe (damaging the epidermal layer and placing the inoculum simultaneously; heavy sporulation appeared on the mid-rib in 7 days) showed that July inoculations are more potent in spreading the disease (through secondary spread) to the adjacent clumps than the inoculations carried out in late August/September (Duttamajumder and Misra, unpublished observation). The spindle infection is characterized by the infection reaching the crown from the stalk portion, affecting the unfolded leaves quickly. At this time *C. falcatum* hardly proliferates in the stalk tissues and quickly reaches the crown. The spindle infection essentially starts from the primary inoculum present in the sett. However, such symptom frequently develops in nodal inoculations (Parafilm method). Generally, infection traverses from the sett to the stalk without producing any conspicuous damage or reddening of the stalk tissues and finally reaches the crown where it expresses itself. On the way it may occasionally express and damage the leaf sheath **(Plate 5c)**. Only one or two odd discoloured vascular bundles indicate the path which *C. falcatum* followed during its passage through the immature stalk. On the crown leaf, it starts from the base as water-soaked elongated lesion (oval, eye shaped), usually at the abaxial side (back side) of the mid-rib **(Plate 4a, b)**. On the leaf lamina, it also produces elongated lesion surrounded by a yellow halo **(Plate 5a)**. In three to four days, these water soaked lesions turn light brown and eventually become dark coloured due to the bearing of abundant setae and spore masses **(Plate 5b, d)**. Sporulation on this spindle infection is very fast, and in a week's time, millions of spores (conidia) are formed, whereas in case of mid-rib infection it takes more than a month to sporulate. In the damaged unfolded leaves of the crown, abundant chlamydospores are also formed **(Plate 6a, b, c)**. The spindle infection is also one of the very characteristics of this disease, which one can identify very easily in the field due to its typical appearance. Further, presence of abundant setae **(Plate 5d)** gives a prickly feeling on touching, and thus helps its confirmation. Spindle infection is a confirmatory symptom of red rot like internal rotting with white spot. Occasional spindle infection also develops in artificial inoculations (plug method), especially when inoculation is done in the early phase of crop growth— especially during July.

The characteristic symptoms, as described in textbooks, appear as the grand growth phase progresses accompanied with sufficient stalk formation. This starts with the yellowing of one or two leaves of the crown **(Plate 7a)**. The third or fourth open leaf, being the most prominent leaf of the crown attracts one's attention (which is incorrectly written as the yellowing of 3^{rd} or 4^{th} leaf as the first appearance of the disease in many writings). There is no hard and fast rule of yellowing of a particular leaf. The yellowing as well as withering of leaves starts from the margin. Later on, the entire crown becomes light orange yellow and dries **(Plates 7b, 8a, b)**. External symptoms of stalk rotting appear as discolouration of rind. This is more pronounced in light coloured type **(Plates 9a, 38b, c)**. At this time, if the stalk is split open longitudinally, the typical red rot symptoms as described in the literature become visible in the internal tissues **(Plate 10a,b, c, d)**. Reddening of the internal tissues interspersed with white patches (white spots), which are usually at right angle to the stalk axis, along with the presence of sourly alcoholic smell confirm the disease. Probably, these white spots in the internodal tissues of cane stalk act as the inn, where the pathogen safeguards itself from the host defences **(Plate 9a, b, c)**. The pathogen detoxifies the phenolic compounds from a portion of the cane tissue for a comfortable stay in that area. In highly susceptible genotypes, the border of white spot is indistinguishable with the background tissue, whereas in moderately susceptible genotypes the white spot when present is definitely surrounded by a dark red zone. Isolation from white spot area invariably yields virulent and sporulating form of *C. falcatum*. In fact, in the white spot area, tissue damage is most prominent and the tissues are usually devoid of cell contents and filled with air and web of fungal hyphae **(Plate 17a, b, c, d)**. When the stalk tissues with prominent white spots are allowed to dry, the white spot area becomes sunken, indicating greater moisture loss and tissue degradation of this particular area (Duttamajumder, unpublished observation). The redness of the internal tissues of the cane is due to the phenolic compounds, which are invariably produced due to any type of injury/invasion. However, the intensity of accumulation of the phenolics varies depending on the host-pathogen combination. *Colletotrichum falcatum* has taken this host defence in its own stride to avoid competition from other microorganisms that follow suit.

Reddened or discoloured vascular bundles run through the internodes in a characteristic manner in early symptoms (spindle infection). During the early phase of crop growth one or two odd discoloured vascular bundles provide the quick transit route to the pathogen for reaching the crown. The fungal hypahe run along the vascular bundle, rarely venturing out in the parenchymatous tissues. At this time, no white spot or internal reddening, as seen in the typical symptom (August/September), develops and the pathogen hardly spreads sideways in the internode. In the late stage, when ambient temperature goes down, pathogen again restricts itself to the vascular bundles. However, the vascular bundles are definitely red and occasionally the pathogen ventures out in the surrounding internodal tissues, sometimes giving the appearance of serial spotting. Serial spotting also develops due to the passage of conidia through the vascular bundles, proliferating at different sites. This situation specifically arises in plug method of inoculation, wherein a large number of conidia placed at the damaged end (due to boring of a hole in stalk) of the vascular bundles are taken up due to upward transpiration pull. Another peculiar situation is often encountered if inoculation is carried out using plug method during the colder months (October & November, low ambient temperature). It has been noticed that due to proximity of inoculation point to the ground, maturity of cane and unfavourable ambient temperature, there is a certain tendency of *C. falcatum* to traverse more towards the bottom of the cane from the inoculation point in comparison to that of the normal period (July, August inoculation). As the inoculation is just at the third internode from the base of the cane, *C. falcatum* tries to reach the stool base, as if looking for relatively warmer condition for survival. However, the response is not uniform among the genotypes and the relative frequency also varies from year to year.

The sourly alcoholic smell in the badly affected tissues is due to the preponderance of acetic acid bacteria that enjoy the hospitality of *Colletotrichum falcatum*. With further progress of the disease, death of the cane takes place. It is important to note that sporulation usually appears after the death of the plant (formation of acervuli, the asexual fruiting structures). These appear externally on the root primordia **(Plate 13f, g, h)**, and on the stomata present on the stalk and at any conceivable area where the pathogen finds or is able to create an exit point **(Plate 13a, b, c, d, e)** and internally in the cavities formed in the stalk

(Plates 10d, 11b, c). It is to be pointed out here that sporulation is blazingly fast on cut or damaged portion of the diseased cane. Once the diseased cane is cut, a good amount of sporulation appears on the cut surface even in an overnight incubation, provided the cut surface does not dry up.

The other important point that is quite characteristic of this disease is the rotting of the node. This nodal rotting also kills the bud and root primordia and renders the node unfit for germination and establishment. This typical nodal rotting helps to differentiate red rot from wilt **(Plate 11b, d)**, wherein the buds remain unaffected and retain viability. Moreover, it is the nodal tissue, which the *C. falcatum* has to overcome first to establish itself in the standing cane (natural infection). Until and unless the pathogen possesses the capability to damage the nodal region, it gets little chance to infect internodal tissues of the stalk in nature. In the nodal rotting, however, the vascular bundles running through the node remain largely unaffected, only the other connecting tissues are preferentially attacked **(Plate 12a, b)**. Sometimes in some cane genotypes, the cane breaks from the damaged node (growth ring) and topples the crown down. Through artificial inoculation, Duttamajumder and Misra (2004) have shown that *C. falcatum* can enter through the immature internode via stomata present on the stalk. After the entry, it quickly reaches the node (up or below), growing along with the vascular bundles, just below the epidermis, without displaying any conspicuous symptom. In the nodal tissues, pathogen consolidates itself and then spreads to the new areas of the stalk. It has also been shown that through these portals *C. falcatum* invariably comes out and produces acervuli bearing conidia and setae on the surface of the stomata **(Plate 13a, b, c, d, e)**. Like many other members of *Colletotrichum* affecting different crops, *C. falcatum* hardly pushes away the epidermis of the stalk for the production of acervulus underneath; rather it takes the route of any available avenue, be it natural or artificial.

The other most important phase of the disease is the infection on leaf mid-rib and leaf lamina. *C. falcatum* produces reddish lesions on the adaxial (dorsal) surface of the mid-rib **(Plate 14a, b, c, d)**. The lesions gradually increase in size and may elongate up to the entire leaf mid-rib **(Plate 15a, b)**. Aging/old spot develops ashy white centre where the pathogen produces acervuli laden with conidia and setae **(Plate 14c, d)**. The mid-rib lesions, for a long time have been taken as the chief source of inoculum for causing infection in the standing cane stalk, as indicated earlier. The spindle infection was also taken as mid-rib infection and thus, resulted in conflicting claims regarding the capability of mid-rib isolates in infecting stalk. Thus, it added some confusion about the role of mid-rib infection in the spread of the disease. The inoculum that causes spindle infection is essentially a stalk infecting isolate, which passes through the cane stalk and expresses on the crown leaves in the early phase of the crop growth and helps in the spread of the pathogen. Incidentally, infection in the crown was also common in the red rot epidemic that hit Godavari Delta (Barber, 1901) and in the epidemic of 1938-39 (Chona, 1943). However, later workers have found that actual mid-rib isolates hardly possess the capability to infect the stalk and have concluded that genes governing leaf susceptibility and stalk susceptibility are different. Duttamajumder (2002) arrived at the inference that the leaf population of *C. falcatum* is the major population, which thrives well in nature, and the stalk population is the aberrant population like the dacoits/rogues/ mercenary of human society who have acquired the nuisance value. The mid-rib population is the reservoir, silently contributing to the development of virulent pathotypes, most of which fizzle out due to their lack of fitness. Only, a few are able to muster the capability of stalk infection and thus, able to proliferate in cane stalk. This initial stalk infection becomes sett-borne and unleashes a fresh onslaught on the cane genotype. The mid-rib infection usually starts appearing in the pre-monsoon phase (in June in subtropical belt). However, it is not clear which type of inoculum is responsible for the initiation of mid-rib infection early in the crop season. Nevertheless, the mid-rib infection never puts any discernible effect on the cane plant in terms of juice quality and yield. Several trials conducted at IISR (Anon. 1953-2005) and elsewhere with different fungicides as well as removal of affected leaves by various workers did not yield any dividend, and therefore, it has been concluded that the mid-rib population of *C. falcatum* does not pose any serious threat to cane cultivation directly.

The third type of infection that affects the leaf is termed laminar infection. On the lamina, numerous tiny reddish lesions appear on the adaxial surface of the leaves and sometimes the lesions are devoid of any red or dark colouration (Singh and Alexander, 1970). Later on, at the infection site, acervuli are formed, especially on the dying/dead cane leaves. Laminar infection is often encountered in fields where sugarcane breeding experiments are carried out and a large number of cane genotypes with varying resistance/susceptibility to *C. falcatum* is usually grown. The most inconspicuous symptom and which is mostly overlooked is that of perithecia formation (symptom of perfect state). The symptoms appear just after the rains (Chona and Bajaj, 1953), and it may easily be seen in the month of August in north India when ambient temperature is around 25-26^{0}C and intermittent rains provide the requisite relative humidity for the development of sexual structures (Duttamajumder, 2002). Asci are initially seen as tiny black dots present between the two veins on the abaxial side of the mid-rib of a dying leaf

(Plate 29a, b, c). Later on such black dots, (perithecia) produce asci and ascospores **(Plates 30a, b, c, 31a, b, c, d)**, which come out through the ostiole **(Plate 29d)**. The conspicuous absence of imperfect state (along with the perfect state) in most of the cases casts some doubt of its accurate identity. Only the observation of the ascospores and their germination **(Plate 31e, f, g, h)** behaviour confirms the identity. Sometimes development of sexual stage has also been observed in nature during March with the availability of required humidity and ambient temperature (Duttamajumder unpublished).

Another phase of the disease, i.e., cane/sett-rotting phase at planting, which was common in USA during first quarters of 20^{th} century, is not frequently encountered in India. In Louisiana, planting takes place in September-October and during the harsh winter, the above ground parts die due to frost and ice. The underground cane becomes more vulnerable and easily gets damaged by root rotting fungi including *Colletotrichum*. Edgerton and Moreland (1920) described it in detail. They observed that in such conditions *C. falcatum* spreads very rapidly in the cut cane. Affected tissues in the setts of the susceptible variety become slightly reddish and later take various shades of red, brown, greyish white. White spots sometimes develop but are not as prominent as seen in the standing stalk in the field. Usually the tissues of infected cut stalks deteriorate quickly. Infected setts, while more or less inactive in the soil, are frequently overrun by secondary rotting organisms and this often makes identification of the disease more difficult in the field.

The condition of the node due to red rot infection needs some explanation. As indicated earlier, the node is the focal point for secondary infection of red rot in the field and *C. falcatum* has to overcome this barrier of nodal tissues to set foot in the cane successfully. In susceptible varieties, all of the internal tissues of the nodes appear affected but with the Parafilm method of inoculation, it has been found that vascular bundles of the nodes retain their identity whereas the sclerenchymatous tissues are destroyed exposing the vascular bundles **(Plate 12a, b)**. As the disease progresses, the internal tissues change in colour from the normal white to shades of red, brown or greyish white. At the other end, nodes of the resistant varieties remain free, except occasional passing of red vascular bundles and that too in cases of artificial inoculation with plug method. Varieties with limited nodal rotting are usually considered somewhat resistant. Duttamajumder and Singh (1999) have introduced the evaluation of nodal rotting in the grading system of Srinivasan and Bhat (1961). It has also been observed by various workers that *C. falcatum* stays alive longer in the nodal tissues, in comparison to that of the internodal tissues.

3

LOSS AND ECONOMIC REPERCUSSIONS

From the time of Went, red rot has been recognised as a disease that can cause serious losses. The losses vary depending on the susceptibility of the variety and also from place to place. It is taking some toll every year. Serious epidemics have occurred, at times in different countries like USA, West Indies, Formosa (now Taiwan), Australia, and South Africa besides the Indian subcontinent.

No doubt, the disease occasionally assumed annihilating proportion in the past, yet very limited information is available with regard to the extent and magnitude of losses caused by this pathogen. Losses can be of two types viz., (i) Direct loss and (ii) Indirect loss. Direct loss starts from the very beginning. The presence of red rot in seed-piece often leads to germination failure and thereby affects the total number of millable canes (NMC) and yield. The gappy stands provide opportunity for the weeds to grow and compete with the cane for nutrition and space and this accounts for most of the indirect losses. The setts, carrying the primary infection to the field, depending on the intensity and location of infection, exercise different effects on the cane plant. The appearance of the disease in the form of infection reaching the spindle (spindle infection) provides necessary inoculum for the quick spread of the pathogen. In favourable environment, this plays havoc; damages the crop, reduces the number of millable canes (NMC) and yield.

As the *C. falcatum* extensively grows in the internodal parenchymatous tissues, chemical composition of the juice is bound to change. Changes in the sucrose and glucose content in the juice from the red rot affected canes were first observed by Went. He reported that the increase of the glucose and reduction of sucrose content were due to the inversion of sucrose. Butler later confirmed the observation of Went and further pointed out that the damage is due to the inversion of sucrose and not the actual consumption of sugar by the fungus. Later on, it has been confirmed by various workers that invertases play a major role in the breakdown of sucrose. At the beginning of the disease expression, even a full-blown red rot affected cane possesses ample sugar. There is little drop in sugar content in the red rot affected cane, until and unless the cane starts drying. Depending on the extent of damage and duration of affliction, juice extraction percentage, purity percentage, total soluble solid in the juice as well as sucrose content go down. However, there is a substantial increase in gum content and in the content of titrable acidity in juice. It is also known that juice from red rot affected canes, when used for sugar making does not crystallize well and thus affects recovery and thereby the sugar production.

The main indirect losses are due to the imposition of severe limitations of growing high sugar, high yielding sugarcane genotypes that easily fall prey to this disease. Currently, as a standard practice, no sugarcane genotype is released in India until and unless it possesses resistance against the prevailing red rot flora of the target region. Greater emphasis for breeding disease resistant cane genotypes forfeits the chance of many excellent ones with low level of resistance,

Table 3.1 Changes in juice caused by red rot pathogen in sugarcane genotype CoL 29

Parameter	Days after inoculation	Healthy	Diseased
Sucrose (%)	15	16.47	16.17
	45	18.34	17.29
	90	18.15	14.34
	Mean	**17.59**	**15.93**
Reducing sugars (%)	15	0.44	0.74
	45	0.26	0.54
	90	0.24	0.99
	Mean	**0.31**	**0.67**
Purity (%)	15	87.90	85.40
	45	88.70	83.10
	90	84.80	53.70
	Mean	**87.10**	**74.10**
Titrable acidity (cm^3 0.1 N NaOH)	15	7.54	9.92
	45	7.33	11.24
	90	8.14	13.52
	Mean	**7.67**	**11.56**
Gums (%)	15	2.23	2.82
	45	2.52	3.23
	90	2.64	4.00
	Mean	**2.46**	**3.35**

(After Singh and Waraitch, 1981)

and thereby substantially increases the developmental cost of any sugarcane variety. Further, frequent failure of cane variety against the onslaught of red rot pathogen adds insult to the injury. Such failures of a variety in field especially when it has acquired reasonable acreage, hit hard both the farmer and the sugar industry, as both have to bear the loss and the burden to replace the affected genotype to check the disease as well as to sustain sugar production and remain competitive.

Sugarcane is also used in many areas as green fodder for the cattle, especially when a scarcity of green fodder arises. The damaged cane hardly has any green top left for its use as fodder and cattle avers to take dry cane leaves. However, feeding of diseased sugarcane plants to the cattle did not make any perceptible change in the cattle behaviour.

The available data on the quantum of damage is sometimes quite misleading. The damage depends on the period (month) of artificial inoculation, prevailing ambient temperature and moisture availability and finally the aggressiveness of the pathotype used for such study. These discrepancies have become more apparent in different publications. One such is highlighted in Table 3.1. As it is known that in a highly susceptible-virulent pathotype combination, the cane starts drying within one month of inoculation (plug method) with concomitant drop in sucrose content in the juice. The drop in one unit of sucrose at the end of 45 days of inoculation in a highly susceptible genotype like CoL 29 is not at all acceptable; however, the data furnished for 90 days are usually encountered in 30-45 days in most of the susceptible-virulent pathotype combinations as estimated by various researchers. At 90 days, the cane simply dries out and hardly any juice is left to carry out such estimations.

4

HISTORICAL BACKGROUND

> *"A historical knowledge of one's own branch of science, however profound, is no guarantee against the making of new mistakes in a new situation. But history also shows that it is only too easy to make the old mistakes in a recurrent situation that a thorough acquaintance with past work could have taught us to recognize."* - Garrett, S.D., 1959.

The prophetic remark of Garrett aptly outlines the chequered history of red rot. Over and over again, pathologists of different genre have made similar mistakes in similar situations due to a lack of proper understanding of the disease. This situation was partly due to the non-availability of seminal literature at one's disposal (at one place), and partly due to the haste in having a quick-fix solution. Red rot, though prevalent in India since time immemorial, drew the attention of the scientific community only when FAFC Went chance encountered this malady in Java. He was deputed to Java (now Indonesia), to investigate the notorious '*Sereh*' disease, which was threatening the sugar industry of Java in 1880's. In search of the cause and cure of '*Sereh*', Went stumbled upon a situation of cane dying in 1892 at the Tjomal estate in Java, and in the following year (1893) he published an account of this problem of stalk rotting of sugarcane plants and thus, this sugarcane disease came to light in the scientific world.

He studied the malady, and described the causal fungus as *Colletotrichum falcatum* Went and named the disease as "*het rood snot*", meaning **red smut**. Obviously, the name '**red smut**' is not a happy one as 'smut' is caused by an entirely different group of fungi and produce remarkably different type of symptoms. The species is named '*falcatum*' due to its typical falcate/sickle shaped conidia **(Plates 19, 21)**. The accepted name, **'Red rot'** was given to this cane disease in 1906 by Sir E. J. Butler, the celebrated Imperial Mycologist of India, who was then working at Pusa, Bihar. He wrote the first major account of this disease in 1906. It was by sheer chance that just after three years of Went's report, in 1896 Spegazzini, from the other side of the globe (Argentina), reported the ascigerous stage (perfect stage) of this fungus and named it *Physalospora tucumanensis* Speg. He did not mention the nature of its imperfect stage (conidial state). The specific epithet "*tucumanensis*" was assigned after the place Tucuman, where the Sugarcane Research Station was situated. Almost half a century later Carvajal, one of the students of Edgerton from Louisiana State University established a link between these two stages. As they described :

> "Early in March 1941 an ascigerous stage was produced upon sterilized cane leaves in test tubes 10-20 days after inoculation with conidia from pure culture of *C. falcatum* and during May and June 1941, the perithecial stage was produced again by inoculating leaves of standing cane in the green house. The perithecial stage was first found in the field in July 1941 on dead and drying leaves previously inoculated."

In 1944, they published a detailed account of their findings in the journal Phytopathology (Carvajal and Edgerton, 1944).

After the first report to the scientific world from Java (now Indonesia), the disease was reported from many sugarcane growing countries of the world. The prominent countries were West Indies, India, USA and Australia. In the same year, Massee (1893) also isolated the same fungus (*Colletotrichum falcatum*) from specimens of cane received at Kew Gardens from West Indies. The report of Massee was quite confusing and the identity of the disease was jumbled up with the rind disease (*Melanconium sacchari* = *Phaeocytostroma sacchari*). It was left to the systematic work of Howard (1903) to prove that the disease in question in West Indies was actually caused by *Colletotrichum falcatum* Went and the infection by the 'rind disease' fungus followed the injury by *C. falcatum*. In the mainland U.S.A., Edgerton observed the presence of the disease in 1908, and reported it in 1910 (Edgerton, 1959). Lewton-Brain (1908) also observed this disease in Hawaii. In India, the discase was first reported by Barber (1901).

Initially, the major work on this disease and pathogen was carried out in two main centres i.e., India and USA. The Indian work resulted in the detailed description of the disease and its epidemiology by Butler in his publications of 1906, 1913 and 1918. The American work led to the publication of a comprehensive technical bulletin in 1938 by Abbott. Unfortunately, in the same year i.e., in 1938, red rot rocked the Indian subcontinent in a big way and engineered an epidemic on an unprecedented scale in the popular interspecific hybrid cane Co 213, pushing sugar factories of Eastern Uttar Pradesh and Northern Bihar on a ruinous course. Though the march of red rot in the hybrid canes was started with Co 210 in 1932, the 1938-39 epidemic year was the watershed year in the history of red rot in India. A reduction of 70,000 tonnes of sugar production in the country was attributed to the red rot epidemic. This epidemic was a major jolt to the popularity of the hybrid sugarcane as well as to the fast expanding sugar industry that got a boost after the withdrawal of differential tariff. The red rot epidemic of 1938-39 and the epidemic due to brown spot of rice in 1942-43 (The Great Bengal Famine) in Bengal and wheat shortages due to rust in 1946-48 infused the requisite impetus to the progress of Plant Pathology in India. For the first time, Plant Pathology, as a full-fledged discipline, stood on its own in India shedding the restrictive folds of Botany.

It is to be borne in mind that during the early phase of sugarcane development in India, sugarcane was mainly grown in the subtropical region. The first sugar factory was established in 1784 in Bengal. The second factory was established in 1791 in Bihar (then a part of Bengal Province). Sugarcane industry as a whole faced several ups and downs, and in the 150 years of its existence in India, it could hardly muster 29 mills in operation by 1931. Out of 1.176 million hectare of sugarcane, 80% was the share of subtropical north zone. In 1931, Govt. of India introduced tariff protection barrier against the imported sugar from Java. It will not be out of place to mention that the spectacular success of sugarcane industry of Java (Indonesia) was due to the development of the hybrid POJ 2878, which later on ruled the world directly as a popular cane variety and as a much sought after parental stock in breeding. The number of sugar factories rose from 29 in the pre-protection regime to 111 within two years (Table 1.1) and to 134 by 1936-37 producing 11 lakh tonnes of white sugar as compared to mere one lakh tonne prior to 1932. It was soon realised that with the increased number of sugar factories, the requirement of sugarcane also increased several folds, which in turn could not be met with the thin indigenous types, even though it commanded maximum acreage. The thick *officinarum* types did not survive well under subtropical environment of northern India. The spectacular success of the new hybrid sugarcane drew the attention of all and sundry and everybody who had any chance to lay hand on these new hybrids, put them in cultivation. Thus, the seed consciousness took a back seat, and every single stick of hybrid cane available at that time was planted to encash the new opportunity. This indiscriminate spread of cane varieties provided the undesired impetus to the red rot pathogen to make inroads in the cane crop. Thus, both the farmers and the industry had to pay a heavy price for this single folly at the hand of *C. falcatum*.

Before 1960s, the entire subtropical belt of India was mostly dependent on the cane varieties bred at Coimbatore (with the prefix Co) but with the advent of new sugarcane research stations in this region with emphasis on *in situ* selection of cane genotypes in the target area itself, the share of exclusive Co canes has gradually declined in these areas. The locally selected varieties are now commanding maximum acreage. However, Coimbatore still remains the focal point of breeding cane varieties as the National Hybridisation Garden (NHG) has been established there, and every cane breeder visits this 'Mecca' for crossing. Varieties selected at the U. P. Council of Sugarcane Research, Shahjahanpur (also established in 1912) are now having the largest acreage in this region.

A closer look at the varietal scenario of sugarcane in subtropical India in different decades after the development and release of first hybrid Co 205 is given in Table 4.1.

Table 4.1. Varietal spectrum of sugarcane in the service of sugar industry in subtropical India

Period	Prevalent cane genotype
Before 1918	Hemja, Katha, Khakai, Pathri, Saretha and noble canes
1920-1930	Co 205, Co 210, Co 213, Co 214, Co 223, Co 281, Co 290
1930-1940	Co 205, Co 213, Co 223, Co 244, Co 281, Co 285, Co 290, Co 312, Co 313
1940-1950	Co 213, Co 312, Co 313, Co 331, Co 356, Co 453
1950-1960	Co 312, Co 313, Co 453, Co 951, CoS 245, CoS 510
1960-1970	Co 312, Co 975, Co 1107, Co 1148, BO 17, CoS 510
1970-1980	Co 312, Co 1148, Co 1158, BO 17, CoS 510
1980-1990	Co 1148, Co 1158, Co 7717, BO 91, BO 99, CoJ 64, CoS 687, CoS 767
1990-2000	Co 1148, Co 1158, Co 7717, BO 91, BO 120, CoJ 64, CoJ 85, CoS 8436, CoS 767, CoS 91269, CoLk 8102, CoSe 92423

It is on record that Co 213, which was badly affected by red rot in 1938-39, commanded 1.4% of the acreage even after a decade of its failure (Table 4.2). Farmers in unaffected areas continued to grow this variety in spite of high risk of crop failure due to red rot. They were not willing to discontinue this variety, as a suitable replacement was not readidy available. This was due to the low seed multiplication rate in sugarcane (1:8-10), which hinded quick replacement. Only a few varieties commanded a significant acreage for a longer duration. In the subtropical India, Co 1148 is one such variety that dominated the scene commanding more than 45% of the cane acreage. However, in tropical India, the wonder cane, Co 419 at certain point commanded more than 90% area. Varietal diversity in the subtropical belt is much wider than that of the tropical belt. Obviously, red rot is playing a decisive role in this regard.

Table 4.2. Area(%) occupied by major sugarcane varieties in subtropical India

Variety	Year of varietal census						
	1941-42	1948-49	1952-53	1959-60	1965-66	1970-71	1975-76
Co 213	16.5	1.4	0.1				
Co 312	21.2	33.0	28.7	24.7	22.6	3.0	1.2
Co 313	10.3	10.6	14.5	8.6	1.9	1.4	0.4
Co 453		6.0	14.0	3.4	2.0		
CoS 245			3.1	11.3	1.3	1.5	
CoS 510				9.9	5.6	3.5	
BO 17				2.7	9.8	13.3	8.6
Co 1148					1.3	23.4	39.5
Co 1158					0.3	9.0	17.7
Others	52.0	49.0	39.6	39.4	55.2	46.3	32.6

The hybrid cane variety, though started replacing the thin canes belonging to *Saccharum barberi* and *S. sinense* from 1920s, even at the time of independence, these old varieties continued to have a major share in cane acreage in subtropical India. Their acreage drastically reduced in 1950s (Srivastava, 1993). These varieties in general were tolerant to red rot. However, on artificial inoculations many of these varieties showed high degree of susceptibility (Abbott, 1938; Chona, 1954).

Red rot prior to 1893

From the description of Watt (1893) in the *"A Dictionary of the Economic Products of India"*, it was quite apparent that red rot was prevalent in India and was affecting the canes throughout the length and breadth of the country, wherever noble canes (*Saccharum officinarum*) were planted.

Mention of red rot in Buddhist literature

Red rot, as was thought by many workers in the past, is not really an introduced disease in India. On the contrary, one also finds its mention in the Buddhist literature (Deerr, 1949; Daniels and Daniels, 1976). The disease was so prevalent in the sugarcane growing belt of eastern Uttar Pradesh and Bihar (where Gautam Buddha started preaching after enlightment) that Lord Buddha used the description of the devastating effect of this disease on sugarcane to prevail upon his followers. Deerr (1949) concluded that 'this mention' of the disease of sugarcane appeared to be the earliest record of a sugarcane disease and his passage is quoted below. (The contention of the earlier workers that red rot is an introduced disease in India is not appropriate; however, the pathogen has got repeated chance of introduction in India along with the noble canes from countries like Mauritius, Java, Burma (Myanmar), Barbados, etc. The red rot epidemic of 1895 in Godavari Delta may have received the inoculum from the introduced cane).

> "Mahaprajapati, the foster-mother of Rahula, Gautama's son, requested the Buddha to allow women to enter an order of nuns. After a first refusal, she followed him from Kapilavastu to Vesaili and, arriving footsore and weary, again urged her request. In this she was supported by Ananda, the favourite disciple of Sakya Muni, who at last relented, but with the statement."
>
> *"Just as when the disease known as manjitthika*%% *falls on a field of ripened sugar-cane, that field does not last long, even so, Ananda, in whatever discipline of Dhamma women are allowed to go forth from the home to the homeless life, that godly life will not last long."*
>
> %% (Deerr's footnote) "The word *Manjitthika* means, 'colour of madder.' Madder or manjit is native to, and still used as a dye in India. It affords the deep red dye known as Turkey red. The most widespread cane disease is that known as 'red rot,' caused by the fungus *Colletotrichum falcatum*. At the time of writing, it is very prevalent in the area where Buddha came to this decision. A feature of the disease is that often it only makes itself manifest when the cane is ripe for harvest. It does not seem unlikely that this is the very disease referred to by Buddha, and that the passage forms the first mention of a cane disease."

Daniels and Daniels (1976) have given more supportive evidence for the presence of red rot in India from the Chinese literature. They suggested a working hypothesis that

> "... the centre of origin of noble cane, *S. officinarum* could have been in the Terai region of North-east India (Assam), but that the presence of red rot disease, to which it is susceptible, prevented this area from becoming the centre of diversity."

It is quite natural that red rot was prevalent in India from the very beginning of the sugarcane cultivation. Probably for this reason, the so-called north Indian canes (*S. barberi*) are in general tolerant to red rot. These canes are essentially thin canes with narrow leaves with good tillering habit. The pure *S. officinarum* canes were also grown in pockets with much care and used for chewing purposes, and in some religious ceremonies. With the advent of sugar industries, cultivation of sugarcane in the mill command area increased several folds and this paved the way for large scale cultivation of *S. officinarum* canes due to good sugar, tonnage and extraction (softness). Increase in acreage of *officinarum* canes, along with cultivation practices that prevailed under the sugar mill command area, favoured the proliferation of red rot pathogen. *C. falcatum* attacked the crop off and on, leading to the abandonment of the affected genotype. Thus, it necessitated frequent introduction of *officinarum* type cane from disease free area (district, state, country) to sustain the sugar industry.

The "*Sereh*" in Java and search for new resistance source

It was only through calamities that threatened the very existence of the sugar industry, the progress in the knowledge of sugarcane and its cultivation usually took place. The outbreak of "*Sereh*" in Java in 1880s, being one of the greatest calamities that ever befell the sugar industry has proven to be one of the greatest blessings, since it led to such an extensive study of sugarcane, and especially of methods of cane breeding, thus wrote Earle (1928) in his famous book '*Sugar Cane and its Culture*'. The disease '*Sereh*' was of prime importance in Java. In '*Sereh*', vascular bundles and the tissues of the nodes turn red, the internodes fail to elongate normally and all of the lateral buds sprouts, resulting in a worthless bushy stool resembling some big grass. For this reason the disease is called '*Sereh*' - the local name for one of the big Andropogons.

A historical event of great importance in sugarcane is the rediscovery of the fact that sugarcane sometimes produces fertile seed, and that new varieties may be produced by growing these seeds. Modern interest in the subject dates from the rediscovery of fertile seeds in Java by Soltwedel in 1888. To overcome the menace of '*Sereh*' in Java, Mr. R. D. Kobus (he also introduced heat treatment for disease control in sugarcane) from Java Sugar Industry visited India in search of resistance and took a number of sugarcane genotypes including "*Chunnee*" (*S. barberi*) along with him on his return voyage. *S. officinarum* x *S. barberi* crosses were initiated for the first time in Java by Kobus in 1897 to transfer the disease resistance character in the thick noble canes. The "*Chunnee*" proved to be immune to "*Sereh*" but was of no immediate use commercially in Java due to both low sugar and low tonnage. However, this "*Chunnee*" ushered in a new era of hybrid canes and the use of different species of the genus *Saccharum* in the improvement of sugarcane.

Watt (1893) wrote:

> "In the report of the Botanic Gardens of Saharanpur (Uttar Pradesh) for the year ending March 1891, mention is made of the visit of Mr. R. D. Kobus, a Dutch gentleman, who had been sent to India by the Government of Java in order to secure stock of Indian varieties of sugar-cane. Mr. Kobus explained that the sugar cane had been attacked in Java by a disease supposed to be of fungoid nature, which threatened to extinguish the sugar cane industry there. Mr. Kobus recognised all the *paunda* forms (the class specially grown in India to be chewed) as the same as the cane grown by the Dutch in Java for sugar manufacture. The *ek* or *ganna* canes (the class specially grown in India for sugar making) were, he said entirely new to him. These are very much more hardy than the *paunda* canes and accordingly, Mr. Kobus took back with him a large supply of these and was hopeful that he had thus secured a stock that might prove able to resist the disease."

Red rot epidemic in Godavari Delta and its impact

Large-scale mortality of sugarcane was noticed in the Godavari Delta (Ramachandrapuram *taluk* of Andhra Pradesh), then a part of Madras Presidency, due to an unknown disease in the last decade of 19th century. Two varieties namely "***Namalu***" and "***Keli***", then largely grown in the Godavari Delta, were affected by the disease (In some publications the affected variety was wrongly mentioned as Red Mauritius, whereas 'Red Mauritius' was one of the resistant varieties that replaced the affected ones). The situation aggravated further in the Godavari Delta and by 1895, it shook the backbone of the sugarcane farmers. Because of this malady, regular cane area was reduced from about 2500 ha in 1895 to about 500 ha by 1899. The plight of the farmers forced the person, in charge of the *taluk*, to seek help from state machinery to find out the cause and provide necessary solution to this problem, so that the native '*gur*' and sugar industry could be saved. In March 1900, Government of Madras introduced sugarcane cuttings from Hospet (Bellary district, Karnataka), where the disease was rare, but this did little good. Mr. C. A. Barber, the then Economic Botanist of Madras Presidency was accordingly deputed to make a thorough investigation of the crop and the disease.

In his report, dated 24th April 1901, he ascribed the reason of sugarcane crop damage to a fungus named *Colletotrichum falcatum* Went (Barber, 1901). In the report, he further pointed out that an intensive study was necessary into both local practices and characteristics of different varieties against this fungus before any such introduction or acclimatisation was attempted. Based on his recommendations, about five hectares of land was taken on lease in 1902 at Samalkota (Andra Pradesh), and several varieties, obtained both from different parts of India and abroad were screened (Barber, 1906). Later on, the area became popular as Samalkota farm. At that time varieties like Red Mauritius, Gillman sport and 247B were found tolerant to red rot. In fact, Red Mauritius was one of the tolerant varieties that effectively replaced the affected local varieties in the red rot ravaged area of Godavari Delta. The major supply of seed-cane of variety 'Red Mauritius' was obtained from the erstwhile state of Bhopal.

Establishment of Sugarcane Breeding Institute, the 'Mecca' of sugarcane breeding in India

> *"With the outbreak of the Great Fire in 1666, the city of London was practically gutted but the dreadful disease plague ravaging the city disappeared and the proud metropolis of the world developed."*

The outbreak of "*Sereh*" in Java in 1880s, being one of the greatest calamities that ever befell the sugar industry has proven to be one of the greatest blessings, since it led to such an extensive study of sugarcane, and especially of methods of cane breeding, identification of various cane diseases like red rot, leaf scald, management of sugarcane crop, use of heat treatment, etc. (Earle, 1928). Similarly, the occurrence of red rot in an epiphytotic form in the Godavari Delta caused serious irrecoverable loss to the farmers but resulted in the initiation of an extensive search for resistant/tolerant varieties (at Samalkota). It ultimately led to the establishment of the Sugarcane Breeding Institute (SBI), Coimbatore in 1912, and to the development of indigenous interspecific hybrids of sugarcane which assured the prosperity of sugarcane farmers in India and abroad (Prasadarao, 1987).

Thus started the Sugarcane Breeding Institute (then a research station) at Coimbatore in 1912 with only two persons on roll viz., Mr. C. A. Barber as Sugarcane Expert and Mr. T. S. Venkatraman as his assistant who accompanied him from Madras. The Institute was the first in the world to attempt at evolving suitable sugarcane varieties for the subtropical sugarcane growing belt of the country, the other two then existing institutions in Java and Barbados being solely devoted to developing varieties for tropical condition. Faced with this onerous task, the duo attempted to cross the tropical sugarcane *S. officinarum* with the indigenous varieties in cultivation in subtropical north India, classified later as *S. barberi*. This work, carried on for a few years, was not successful. While in a dejected mood, sometime in 1914-15, Venkatraman noticed the wild species of sugarcane, *S. spontaneum* (*Kans*) on the channel bunds adjoining the Institute, in profuse flowering and the idea flashed in his mind 'why not to give it a try' in breeding to evolve the much needed hardy varieties for the subtropical belt. Again, for the first time in the world, the idea of utilising a wild species for improvement of cultivated crops was thought out, which has now become a common practice in many crops (Prasadarao, 1987). Incidentally, the first batch of hybrid-sugarcane varieties came out in 1918; the year Mr. T. S. Venkatraman took charge of Coimbatore station after the retirement of Dr. Barber. Due to the spectacular success of the hybrid sugarcane varieties, Venkatraman was knighted in 1942 by the then Imperial Government. His Excellency, Viceroy of India, Lord Irwin visited Coimbatore in 1927 and he was so impressed with the work carried out there that he remarked "*I have not so far seen anything so wonderful as this*" and asked everybody working on sugarcane to visit this 'Mecca' and to emulate the path of Mr. Venkatraman.

This solitary event ultimately laid the foundation of the first Green revolution in the world. With the acumen and zeal of Sir T. S. Venkatraman, the first 'Co' hybrid Co 205, a progeny of *S. officinarum* (Vellai) x *S. spontaneum* (Coimbatore) came into being in 1918. It became an instant success in the red rot ravaged area of subtropical India particularly in the Punjab, where it recorded 50 per cent more yield than the indigenous varieties (like Katha) in cultivation. The other interspecific hybrid genotypes like Co 213, Co 244, Co 281, Co 285, Co 290, Co 299, Co 312, Co 313, etc., followed suit and provided the requisite fillip to the ailing Indian Sugar Industry, which was suffering badly due to differential tariff, making Indian sugar dearer in London market. Within the decade (in 1927), Coimbatore bred variety (Co 285) reached American shore for cultivation. Later, NCo 310 became the true international variety (crossed at Coimbatore, selected at Natal, South Africa and became the main cane variety of many countries, including USA). As a testimony to this success of hybrid sugarcane, during the 9th ISSCT Congress in 1956, it was decided to pay tribute to the first commercial hybrid sugarcane variety by erecting a monument in the place where the genotype Co 205 was spotted.

Before the advent of the interspecific hybrids, red rot was ravaging most of the introduced noble canes in the sugarcane mill zones and thus, was putting enormous pressure on the profitable existence of the mill. These newly bred hybrid cane varieties brought the necessary respite from this unstable situation. However, the boom of hybrid sugarcane crop did not last long, and in 1938-39, an unprecedented red rot epidemic engulfed eastern Uttar Pradesh and northern Bihar shattering the backbone of sugar industry in this region.

BUTLER AND RED ROT

The entry of Sir Edwin John Butler at Pusa, Bihar and the first comprehensive work on red rot in India

Dr. Butler, a trained medical man, came to India and joined service in 1901 as the Cryptogamic Botanist to the Govt. of India in Calcutta and he was attached to The Botanical Survey of India. In 1902, the section in which Butler was posted was transferred to Dehradun as a part of the Dept. of Agriculture, Govt. of India. With the establishment of India's premiere research institute on agriculture at Pusa, Bihar, his department was again shifted there. Butler, working as Imperial Mycologist at Pusa, published the landmark research paper on sugarcane diseases of Bengal in 1906, which contained detailed description of red rot. In this paper he wrote :

> "The history of the usual outbreak is somewhat as follows. A concrete example studied in 1902 is selected being typical of the rest. In this case, about the beginning of August several fields separated from one another by distances of about half a mile showed obvious signs of the disease. The diseased field had all been planted for the first time with a yellow cane of the Bourbon type, the seed cane for which had all been obtained from one locality. Other varieties of cane growing in close proximity to the diseased ones were not attacked. The following season (1903) this localisation of the disease to only one of the several varieties of cane growing on the estate was still more marked. Wherever the yellow Bourbon of the diseased stock was planted, disease broke out, while none of the other varieties was affected. Even in neighbouring villages to which the seed had been distributed, and where cane had not previously been grown, disease appeared.
>
> The external appearance and progress of the disease as seen on this estate and elsewhere are very characteristic. The upper leaves of a shoot, usually one approaching maturity (and therefore rich in sugar), begin to lose colour and droop slightly. Then they wither at the tip, and the withering progresses down the margins leaving the centre green. The top leaves were not usually first affected but the third or fourth from the unopened tip, these being the most prominent leaves of the cane. Later the whole crown withers and droops. At this stage, there is no visible alteration in the cane itself, and the withering does not differ from that due to drought, except that withered and green canes are commonly seen arising from the same stool. When one cane of the stool shows the disease, the whole stool is invariably, though often slowly, destroyed.
>
> On splitting open a cane, in the early stages of the disease nothing abnormal can be found in the upper portion. As the base is reached, however, the tissues are found to be reddened in one or a few of the lowest internodes. This discoloration appears first in the vascular bundles, which show as red dots or streaks, and then extends to form blotches; finally, much of the cane pith is reddened. Whitish areas are often to be seen, due to the occurrence of groups of dead cells containing air.

In many instances, however, such as that figured, definite red blotches with a white centre transversely elongated are found in the internode, which, when present, are characteristic of the disease; this appears to be more or less general in certain outbreaks such as that described by Mr. Barber in Madras.

As the disease progresses, large patches can be seen with every stool destroyed. By the end of the season, little or nothing of the original cane is left. In this respect the outbreak in bad cases causes the most complete destruction of cane."

He also traced the life history of the fungus *Colletotrichum falcatum* Went, as it occurs in nature and in artificial culture and I quote:

"In the living cane, the microscope reveals it as an irregular network of fine colourless filaments ramifying in and through the cells of the pith. These filaments are so like those of many other fungi that their presence alone tells nothing of the nature of the parasite. One peculiarity, however, they possess, that within them can be seen numerous little bright globules which are probably droplets of oil. Spores are not ordinarily produced until the cane dries up, and their formation and further details of the life history can best be studied in artificial culture.

The spores of *Colletotrichum* are produced within two days in cultures obtained as described above, by cutting out a part of the diseased tissue under aseptic conditions, or generally in any culture on living cane material. It is a simple matter to obtain sterile slabs from the interior of living healthy canes by the same method. These should be incubated for a couple of days to test the absence of other organisms. On dead, boiled cane slabs, boiled cane extracts, gelatine, dextrose-gelatine, agar-agar and other nutrient substances, ... spores are often not produced within this period.

A culture examined on the second day shows the following appearance. The filaments of the fungus extend in a whitish fluffy mass from the surface of the slab. Spores are first seen produced slightly on short lateral branches. They arise at the tip of the branch and as soon as completely formed fall off. On the second to fourth day definite spore-producing areas known as "stromata" are formed. Each consists of densely well-woven felt of filaments with short branches, matted together until the filamentous character of the fungus is almost lost. The stromata are visible to the naked eye as tiny black dots by the third or fourth day. Their component filaments are soon differentiated into two kinds at the surface of the stroma. The majority are short, crowded and turned more or less at right angles to the surface with a dense layer. Others, fewer in number, growing out into dark hairs sometimes scattered but more usually forming a sort of fringe round the edge of the stroma.

The spores are more or less sickle shaped, undivided and measure from 20 to 30 thousands of a millimetre in length by 5 to 7 in breadth. They contain a large refractive dot in the centre. The hairs are dark greenish-brown with several septa, and are about four to six times as long as the spores.

At an early period, a second sort of spore is formed in the cultures. This is what is known as a "chlamydospore" ("gemma" of Went). It is formed out of an ordinary filament of the fungus, either at the tip or in its length. A portion of the filament is cut off by septum, turns dark in colour, and becomes thick-walled. The chlamydospores are better adapted to stand unfavourable conditions of preservation than the ordinary spores; though precise part they play in carrying on the disease is not known, yet their presence in large numbers has probably of much significance. The quantities that are found in some cases in the interior of the diseased canes may be the means by which disease is disseminated when the latter are left to rot on the ground.

On germination, the spores (conidia) sometimes divide into two cells by a transverse wall. They germinate by giving out one or a couple of filaments from near the ends. These filaments may develop directly into a new plant, or may give rise at the tips to secondary chlamydospores. After a period of rest, these secondary spores themselves germinate."

The work of Kulkarni (1911) from Bombay needs special mention as most of the authors have left the contribution of this gentleman to oblivion. He was the third man in India to study the red rot. His findings supported those of Butler (1906) that the disease is basically a sett-borne one and the pathogen gets entry inside the cane through the root eyes and leaf scar region of the bud.

The other important point may also be highlighted from the publication of Butler and Khan (1913) regarding the epidemiology of the red rot.

"At the Samalkota Sugar Station in Madras, sett selection has been regularly carried out for the past ten years, at the instance of Dr. C. A. Barber, now Government Sugarcane Expert, Madras. Dr. Barber was, we believe, the first to advocate this method of fighting red rot. In the beginning, it was combined with pickling the setts in a strong mixture of limewater and carbolic, with a view to checking moth borer and the Queensland Acarus "rust", but this was subsequently not considered necessary. In 1906, Dr. Barber stated that it was not possible by mere rejection of red-marked setts to root out the disease. In spite of all care, all but two of the local kinds were found gradually to become worse, until they had to be replaced by new seed from outside. Once these had been discarded, however, better results were obtained. Thus, the 1907 report states that selection had a satisfactory effect, disease becoming less every year among the best varieties.

In 1909-10, red rot was still present on the farm but to a far less extent than here before. In 1910-11, very little disease was noted either on the farm or in the nursery, though it was still prevalent outside the farm. It is fair to assume that the local varieties, which were at first grown, were very largely from diseased stock. So long as these varieties were retained, sett selection did not give satisfactory results. When they were replaced by other, comparatively healthy varieties, sett selection was effective in keeping the disease under control.

All that can safely be stated, so far, is that planting setts from obviously diseased canes leads to a considerable development of disease in the resulting crop, and that provided the variety is fairly free from disease to start with, sett selection keeps red rot within reasonable limits, unless some untoward circumstance intervenes.

As was pointed out formerly this fungus is not, in Northern India, provided with as suitable mechanism for spore distribution as in the case with most parasitic fungi. As long as the cane is growing, there is comparatively little risk of air-borne contamination from the stems of diseased plant in the crop. Dead and rotting canes are, however, frequently well provided with spores. Enormous numbers are often found in the pith cavities of old canes. These may contaminate the soil or get into the irrigation water. They may thus reach the newly planted setts. Several experiments have been carried out to ascertain whether infection of sound setts may take place in this manner.

In all these cases, the presence of *Colletotrichum* was demonstrated in several of the withering shoots, and they established fully that true red rot cane arises from infection of the planted canes. Not only is the disease perpetuated by planting previously diseased setts, but also healthy setts can be infected at the time of planting, if reached by the fungus, and, no doubt, subsequent infection from below ground can also occur. It is well known from previous work that the fungus can enter at wounds exposing the pith, such as the cut ends of the setts, and as will be shown, infection through the root readily occurs. The course of the infection up into the stem can be traced in many cases and direct connection between the mycelium in the sett and that in the new shoot was established.

At the cane node, there are two other points of discontinuity in the rind, where the shoot bud (the "eye") comes through and where the eyes of the adventitious roots occur. These were found to admit the fungus readily.

Systematic and thorough selection of the setts used for planting must then be done each year or the new varieties will not maintain their freedom from disease for long. It is considered that there is no single operation in the cultivation of thick canes in most parts of India of greater importance than this. It is not to be anticipated that the disease can be got rid of by a single selection nor, indeed, usually got rid of entirely by annual selection; the most that is claimed is that it can be kept within reasonable limits in normal years and with good cultivation.

Of lesser importance, but still worth going in most cases, is the regular removal of all withering clumps during the growing and ripening season. Such clumps, if left, dry up and produce spores, sometimes in considerable quantity. Infection of even perfectly sound cane through the aerial root eyes and through injured buds has been shown to occur, and though experience in Northern India has been that such infection is not common, it is perhaps more frequent in Madras, where the disease appears to be more virulent and rapid in its onset than with us. There is, also, the danger of infection through the soil, especially in irrigated cane, by means of the shed spores ..."

"... When setts cut from diseased cane (and, therefore, usually allowing reddened pith at the cut ends) are planted, they germinate in most cases as well as if sound "seed" were used, unless the infection is severe and of long standing. However, many of the young shoots begin to wither when they are a few weeks old, and the presence of actively growing *Colletotrichum* can be demonstrated in them. Many

> shoots, however, for a time grow out from the fungus, which remains dormant in the base of the cane. When the crop is about six months old and is beginning to contain appreciable quantity of sugar, the parasite resumes activity, and from October on (in northern India) withered shoots are found in increasing numbers. By the end of the season many clumps, and even large fields, may be entirely destroyed. Periodic examination enables the fungus to be traced in the young shoot and, later on, rising from the under ground parts into large canes. Fields in which the "seed" is known to have come from a diseased crop will show much red rot, while adjacent fields planted with healthy "seed" (even of varieties known to suffer) may have little or none. This occurs to a marked extent in land not recently under sugarcane, but is also often seen where the crop has been repeated after three or four years. There is, therefore, no room to suppose that infection from the soil, and not from the diseased setts, can be responsible for the disease in these epidemic outbreaks. Not only has it been found that naturally infected setts can carry over the disease, but perfectly healthy, growing canes have been artificially inoculated with pure cultures of the fungus obtained both from the leaves and the stem, and when subsequently cut up into setts and planted, have given rise to shoots infected with red rot. …
>
> It early becomes obvious to any observer of sugarcane diseases that there are immense differences in the susceptibility of canes to attack. The thin canes, so commonly grown in India, are on the whole, much less liable to red rot than the thick. Their resistance seems to be inherent, whereas the temporary resistance of certain classes of thick canes, which has been observed from time to time, appears to break down readily under unfavourable conditions of the environment."

Butler's critical observations found support much later by the subsequent workers in India, though his findings did not receive the needed appreciation at that time from outside India, especially from USA. As it is known today, the difference of cane growing climate drastically altered the host physiology, which was reflected in the pests as well as the pathogen problems in both these countries. Red rot is essentially a disease of standing cane in India, whereas in USA *C. falcatum* accentuates the sett rotting caused by *Pythium* and other soil fungi. The fungus *C. falcatum* does not like cold weather and the harsh cold climate of USA restricts its growth severely.

5

THE MARCH OF RED ROT IN INDIA

Red rot is not an introduced disease in India. It was prevalent since pre-historic times, as one finds its mention in Buddhist literature. The disease systemically ravaged the *S. officinarum* canes and thus, hindered the proliferation of 'noble canes' in their primary centre of origin (Daniels and Daniels, 1976). Before the advent of present day hybrids in 1918, sugar mills were mostly growing 'noble canes' and were suffering heavily from the frequent crop damage due to red rot. The indigenous thin canes *(S. barberi)*, though less in sucrose and yield tolerated the red rot menace well and were the predominant genotypes used by the native *'gur'* making cottage industry. The 1985 epidemic of red rot in Godavari Delta, after its first report by Went (1893) from Java, remains the first major catastrophe of the disease in India. Since then red rot assumed serious proportion and damaged the sugarcane crop with impunity.

Red rot ran through many excellent varieties during its course. In fact, it has practically left no sugarcane cultivars untouched. Eventually, all the varieties fell prey to red rot and had to be withdrawn from the general cultivation, or had to be replaced by the new more tolerant genotypes. The popular cane genotypes that served the Indian sugar industry well, such as, Co 213, Co 281, Co 290, Co 312, Co 313, Co 419, Co 421, Co 453, Co 527, Co 658, Co 975, Co 997, Co 1148, Co 7717, CoC 671, CoC 85061, CoC 86062, CoC 92061, CoJ 64, CoJ 82, CoJ 84, CoLk 8001, CoLk 8102, CoS 245, CoS 510, CoS 770, CoS 767, CoS 802, CoS 8436, CoSe 93232, CoSe 92423, BO 3, BO 10, BO 11, BO 14, BO 17, BO 29, BO 34, BO 54, BO 120, BO 128, etc. succumbed to the onslaught of red rot and the list of such casualties is an ever increasing one (Table 5.1). No sugarcane genotype could stand erect against the threat of red rot. Probably this is the reason that entire sugarcane breeding in India is geared around red rot resistance. Moreover, this perpetual threat has taken a heavy toll on sugar production. Greater susceptibility of high sugar genotypes to red rot and wilt has deterred the cultivation of such genotypes on a large scale, and has forced the cultivation of more tolerant cane genotypes with moderate sugar, especially in subtropical belt of India.

Table 5.1. The march of red rot in India, since it hit the Godavari Delta in 1895

Year	Area	Major genotype affected
1895-1901	Godavari Delta (Andhra Pradesh)	*Namalu, Keli*, Bourbon, Ashy Mauritius, Striped Mauritius
1902-1913	Champaran, Muzaffarpur (Bihar)	Bourbon, Striped Mauritius
1922	Jammu (Kashmir)	
1925	Punjab	
1932	Pusa (Bihar)	Co 210
1936	Inthiathoke (Gonda, U.P.)	
1937	Gola, Nagina (U.P.)	
1938-40	Uttar Pradesh, Bihar	Co 213
1946-47, 1951-68	Uttar Pradesh, Bihar	Co 312, Co 313, Co 419, Co 421, Co 443, Co 453, Co 513, Co 997, CoS 510, CoS 514, BO 3, BO 11, BO 17, Mamchuah (local variety)
1960-70	Andhra Pradesh	Co 421, Co 419,Co 997, Co 62175
1971-75	Tamil Nadu	Co 419, Co 658, Co 6304
1978-83	Andhra Pradesh	CoC 671, CoA 7701
1981-85	Kerala	Co 997, Co 419, Co 785
1970-85	Uttar Pradesh, Haryana, Punjab, Bihar	Co 1148, Co 7717, CoJ 64, CoS 527, CoS 659, CoS 770, BO 29, BO 32, BO 54, BO 70
1992-97	Gujarat, Tamil Nadu	CoC 671, CoC 85061, CoC 86062, CoC 90063, CoC 92061
1991-95	Uttar Pradesh	CoS 767, CoS 802
1990-99	Andhra Pradesh	Co 7508, CoC 86062
2000-2005	Uttar Pradesh	CoS 8436, CoSe 92232, CoSe 92423, BO 120, BO 128

RED ROT IN HYBRID CANE- EPIDEMICS AND AFTERMATH

Red rot research in India on hybrid cane varieties

It is amply clear that a major share of our knowledge of red rot in India stemmed from the researches of Butler. However, his research was confined to the available thick canes (*S. officinarum*) and the popular local canes of *S. barberi* type. After the departure of Butler from the Indian scene, the situation changed altogether with respect to the genetic make-up of the host. Then came the hybrid sugarcane (the fruits of breeding efforts at Coimbatore) – the Co 205 became the first successful hybrid, yielding more than double of the popular canes like 'Katha' (*S. barberi*) in the Punjab and revolutionised the sugarcane and sugar industry of India. However, the euphoria of resistance to red rot did not last long and within a span of a few years, these first generation hybrid canes were also attacked by red rot. It is on record that Co 210 got severely affected at Pusa in 1932, and thus started the misadventure of red rot again (in hybrid cane) in India.

Indigo vs Sugarcane

A historical event of first order was the discovery of synthetic indigo dye in Germany in 1880s. By 1897, cheap commercial production started in a bigway. In the back drop of World War I, synthetic indigo dye flooded the European market leaving no room for natural indigo. The easy availability of cheap dye forced a steep fall in price and as a result, *neel* (indigo) dye, an important commodity of commerce in British India, could not stand up to the challenge. Thus, *neel* farming became uneconomical and was abandoned. In 1931, Pandit Madan Mohan Malviya successfully pleaded in the Assembly the plight of farmers after stoppage of indigo cultivation in U.P. and Bihar; as an alternative to *neel* (indigo), hybrid sugarcane was introduced in this *neel* growing area to save the farmers. At that period, newly evolved hybrid canes showed great promise and most of the new plantings were of hybrid canes. Excellent performance of the new hybrid Co 213 in northern India prompted its spread like a wild fire, and within a very short span covered thousands of hectares in eastern Uttar Pradesh and northern Bihar (predominantly *neel* growing area). In fact, first 'Green revolution' was ushered in sugarcane in 1920's when Co 205 (first hybrid cane released from Coimbatore) yielded more than the double in red rot ravaged area of the Punjab. This, along with tariff protection by Govt. of India in 1932, boosted sugarcane cultivation. In 150 years of sugar manufacturing in India, the industry could muster only 29 mills in operation up to 1930-31. The industry grew in leaps and bounds, and within a span of two years, the number rose to 111 in 1933-34! The honeymoon period of sugar industry did not last long and red rot spoiled the euphoria.

The epidemic of 1938-39 and aftermath

This quick spread (to get high dividend from sugarcane cultivation by the newly established sugar factories) subverted the normal precautions of good and healthy seed. This simple folly took a very heavy toll. The red rot hit hard the Indian Sugar Industry in 1938-39 flooring the excellent hybrid genotype Co 213 in thousands of hectares in eastern Uttar Pradesh and northern Bihar (incidentally its female parent, POJ 213 also suffered badly in USA during late 1920's and 1930's). This epidemic changed the entire face of red rot in India. It was observed that contrary to the prevailing ideas, the epidemic was not confined to low lying or ill drained fields only. Co 213 was affected everywhere from upland to lowlands and from acidic to alkaline soils. From the prevalent dark, less sporulating *Colletotrichum falcatum* (pre epidemic isolates mostly prevalent on noble canes) arose a highly sporulating virulent light culture (isolate No. 78, IARI) attacking the new hybrid canes. This culture was quite different from that of the earlier reported ones. However, it may be pointed out that though Louisiana, USA was also growing hybrid canes at that period and faced red rot problem in 1920s and 1930s, it did not meet with such virulent isolates. In fact, Dr. Abbott commented "*when I was in India in 1956, I was shown some cultures of Colletotrichum that seemed very different in cultural characters from the fungus we have in Louisiana.*" During the visit to Louisiana in 1992, the author also noticed the difference – the cultures obtained on isolations from the spindle infections (in June) were quite similar to the cultures readily obtained on isolations from mid-rib in India — dark, velvety with sparse sporulation.

The epidemic of 1938-39 is the turning point of Indian sugarcane and sugar industry. This forced the administrators to think about entrusting the sugarcane research to a specialized group, and in 1941, The Central Sugarcane Committee was formed to look after this specific need. To further the advance of sugarcane research, on the recommendations of Sir T.S. Ventakatraman, a Central Institute exclusively for sugarcane was established in Lucknow in 1952 (now known as the Indian Institute of Sugarcane Research) to look after all aspects of sugarcane culture other than cane breeding. Dr. B. K. Mukherjee, a well-known agronomist from Sugarcane Research Station, Shahjahanpur took the charge as Head of the Institute and Mr. S. A. Rafay moved from Pusa to Lucknow to start research on Sugarcane Pathology at the Institute.

Epidemics prior to 1938-39 were mostly restricted to thick canes - those belonged to *Saccharum officinarum*. The thin canes typified by *S. barberi* usually escaped red rot infection or the damage was not to such an extent that could draw immediate attention (established an endemicity). Though some amount of infection was always there in the thin north Indian canes, it did not pose any serious problem to the cultivation of sugarcane. Duttamajumder and Misra (unpublished observation) have indicated that the thin *barberi* canes possess high degree of nodal resistance/tolerance as indicated by the non-rotting of the nodes against many *C. falcatum* isolates through nodal inoculation. However, when inoculated using plug method, the internodal tissues were severely damaged showing S/HS reactions. Moreover, most of these isolates failed to infect when inoculated using nodal methods.

Table 5.2. Parentage of a few early sugarcane genotypes indicating the involvement of limited source of disease resistance

Genotype	Parentage	Year of release/ identification	Station
Co 205	Vellai x *S. spontaneum*	1918	SBI, Coimbatore
POJ 213	Black Cheribon x Chunnee		POJ, Java
POJ 2878	POJ 2364 x EK 28	1922	POJ, Java
Co 213	POJ 213 x Co 291	1918	SBI, Coimbatore
Co 244	POJ 213 x Co 205	1922	SBI, Coimbatore
Co 285	Green Sport x *S. spontaneum*	1924	SBI, Coimbatore
Co 312	Co 213 x Co 244	1928	SBI, Coimbatore
Co 313	Co 213 x Co 244	1928	SBI, Coimbatore
Co 419	POJ 2878 x Co 290	1933	SBI, Coimbatore
Co 421	POJ 2878 x Co 285	1934	SBI, Coimbatore
Co 449	POJ 2878 x Co 331	1938	SBI, Coimbatore
Co 453	Black Cheribon x Co 285	1938	SBI, Coimbatore
Co 740	P 3247 x P 4775	1949	SBI, Coimbatore
Co 975	Co 527 x Co 617	1953	SBI, Coimbatore
Co 997	Co 683 x P_1 63/62 (POJ 2725)	1953	SBI, Coimbatore
Co 1148	P 4383 x Co 301	1963	SBI, Coimbatore
Co 1158	Co 421 x Co 419	1963	SBI, Coimbatore
Co 7717	Co 419 x Co 775	1978	SBI, Coimbatore
CoC 671	Q 63 x Co 775	1975	Cuddalore
CoJ 64	Co 976 x Co 617	1971	Jalandhar
CoL 9	Co 312 x Co 285	1945	Lyallpur
CoL 29	Co 312 x Co 285	1954	Jalandhar
CoS 510	Co 453 x Co 557	1954	Shahjahanpur
CoS 527	Co 349 x Co 312	1946	Shahjahanpur
CoS 687	Co 976 x Co 312	1980	Shahjahanpur
CoS 767	Co 419 x Co 313	1979	Shahjahanpur
BO 3	Co 312 x Co 326	1948	Pusa
BO 11	Co 331 x POJ 2878	1948	Pusa
BO 17	Co 331 x Co336	1955	Pusa
BO 54	BO 17 x BO 3	1969	Pusa

These new cane genotypes being the first generation interspecific hybrids, had a good share of *S. spontaneum* component (due to 2n +n transmission). The pathogen gradually spread its tentacles to other genotypes like Co 290 (it replaced Co 205), Co 370, Co 393, Co 421, Co 357, CoS 5, etc. Obviously, these genotypes succumbed to the existing race flora of north India that was thriving on *S. barberi* and *S. officinarum* types. The traditional '*gur*' making cottage industry used mostly the cultivars of *S. barberi*. The noble cane best served the religious and chewing purpose. The luxuriant growth, excellent juice quality and high yield potential of noble canes attracted the attention of both sugar industry and the farmers. To increase the cane supply quickly and to reap a good sugar harvest, a rampant introduction of these canes took place and in most cases, the basic philosophy of seed health was relegated to the side line. This neglect helped the spread of *C. falcatum* along with the planting material "seed cane" from one place to another. As a consequence both millers and farmers had to pay a heavy price to bear with and in many cases it put the sugar mills to total bankruptcy. The same *'boom and bust cycle'*, which was indicated earlier by Watt (1893) is still continuing, and a good number of excellent genotypes are regularly being phased out of cultivation due to the breach of resistance by red rot. With the release of a new variety of sugarcane, its acreage gradually increases and it becomes the variety of choice; then red rot breaks out to mock at the claims of resistance made at the time of its release, forcing a denotification and withdrawal of the variety from cultivation.

In fact, most of the cane genotypes, so far withdrawn/denotified from commercial cultivation in India were due to the arrival of new races/pathotypes of red rot pathogen in the scene, making the survival of the variety difficult in the field. To counter the 1938-39 red rot epidemic, the work of seed replacement was started on a war footing. Strong emphasis was given on reaching out to the farmers and probably this was the first epidemic of any plant disease in India where street plays were organised to educate the farmers for managing the malady. Songs were composed and direct interactions with farmers through street plays were organised. Pamphlets and other literature were distributed freely. The red rot epidemic of 1938-39 affected the genotype Co 213 and it was replaced with Co 313 in north Bihar, Co 513 in South Bihar and Co 312 in Uttar Pradesh. Co 313, the variety that replaced Co 213, fell prey to this pathogen in 1946-47. The other cane genotypes that also felt the heat of red rot were Co 301, Co 312 and Co 385. Incidentally, the female parent of Co 312 and Co 313 was Co 213 (which was affected in 1938-39) and there was no fresh involvement of any *spontaneum* clone directly for the red rot resistance (Table 5.2). The *spontaneum* component for red rot resistance came from Co 244, which in turn received it from Co 205. In Bihar, Co 453 largely replaced the affected Co 313. The fall of

Co 453 gave rise to the appearance of "D" strain (isolate No. 244). The fall of BO 11 yielded the "I" strain. In Bihar, the Coimbatore-bred varieties were replaced with the Pusa-bred genotypes like BO 3, BO 10, BO 11, BO 14 and BO 17 (Cane breeding at Pusa is possible as sugarcane flowering and seed setting also takes place over there, though in much lesser number, in comparison to what is obtained at Coimbatore).

The 1938-39 epidemic prompted many researchers to look into the red rot problem. At Pusa, the place where Butler did the monumental work, Mr. K.L. Khanna and later on Mr. S. A. Rafay took up work on red rot fungus again. However, most of the researches of Mr. Khanna are only in the pages of Annual Report of Central Sugarcane Research Station, Pusa (from 1940 onwards). His findings are mostly the extension of Butler's findings (Butler, 1906; Butler and Khan, 1913). It was apparent that the new hybrid canes with different resistance genes (from *S. spontaneum*) also behaved in the same fashion as the noble canes (*S. officinarum*). The only difference that was evident was the intensity of the disease, which was more in the hybrid canes. The explanation of this situation came much later with other host-pathogen combinations. With the gain of matching gene(s) for virulence in the quiver to counter the genes governing host resistance, pathogen attacks in vengeance and produces susceptibility to the highest degree, which is not even encountered in the universal susceptible type. Later on, Vaderplank (1968) tried to explain this situation as the erosion of horizontal resistance due to breeding for vertical resistance and termed it as the *'vertifolia'* effect. However, the interplay of genes in sugarcane is still shrouded in mystery. Due to the complexity of sugarcane genome, the extension of various theories in this pathosystem is more of conjecture in nature than the actual experimental provings.

The epidemic of 1946-47 and breeding for disease resistance

After the red rot epidemic of 1946-47 in Uttar Pradesh that affected popular varieties Co 312 and Co 313, research efforts were geared up to test the sugarcane genotypes against red rot pathogen before their actual release for cultivation. A strong group of workers under the able leadership of Dr. B. L. Chona started working on this disease at the Indian Agricultural Research Institute, New Delhi. Chona (1954) gave a detailed account of his work on screening of the sugarcane genotypes. The Table 5.3 (adapted from Chona, 1954) is highlighted here to drive home the point of variability of *C. falcatum*, which changed from season to season, and relatively limited field life of a variety. Due to the narrow genetic base (genes governing red rot resistance apparently came from the same parents used a few generations back for the development of interspecific hybrids), sooner or later, care genotypes succumbed to the onslaught of the pathogen (Table 5.1 & 5.2). Chona (1954) recorded that:

> "The proper selection of the *C. falcatum* isolate to be used in these varietal resistance tests is obviously of utmost importance. The possibility of existence of *C. falcatum* strains of varying virulence was kept in mind from the very beginning of the red rot resistance tests. *C. falcatum* isolates from different cane varieties and localities were always studied for their virulence and only the most virulent one, available in our collection, was always used for the varietal resistance tests. Prior to 1938-39 red rot epidemic in eastern U.P. and northern Bihar tract, only dark types of *C. falcatum* isolates were met with in the various isolations that were not highly virulent. During the epidemic, two morphologic races, one light coloured with profuse sporulation and the other dark coloured and sparsely sporing, similar to the old pre-epidemic type, were met with (Chona and Padwick, 1942). The light type was found to be highly virulent and was selected and used in the red rot varietal resistance tests for about six seasons (1941-47); when its virulence was found to be falling off, and it was replaced in further tests of the last two seasons (1947-49) with a fresh virulent isolate obtained from 1946-47 red rot epidemic areas of U.P. The existence of physiologic forms of the red rot fungus *C. falcatum* Went, certainly complicates the work of varietal resistance tests. A large number of *C. falcatum* isolates obtained from different varieties and localities have been tested. The differences in the various isolates are mainly of their virulence. Generally speaking, the light type isolates are more virulent than the dark type. …
>
> One more point, however, has to be kept in mind, i.e., the maintenance of the selected isolate in a pure form, as some of the isolates have been found to mutate in culture. The fungus after a few subculturings may be quite different from the parent culture, both morphologically as well as in pathogenicity."

The loss of virulence (stability) in culture has always posed a serious problem in the identification of isolates in physiologic groups based on pathogenicity. This riddle has to be solved to prolong the life of a cane genotype in commercial run. If pathogen is so quick to shed off the genes governing pathogenicity in artificial culture, it is difficult to contemplate the mechanism(s) pathogen employs to maintain these genes in natural population. Moreover, how these pathogenicity genes are gained and how quickly they are transferred in the population, remain the moot questions. *C. falcatum* is a transient visitor of the soil and hardly has any existence outside the sugarcane debris. It is not comfortable to venture out of the boundaries of debris in the presence of strong antagonistic flora in the soil environment. It appears that this may be one of the reasons that moderated *C. falcatum* to evolve to a near obligate type of relationship with the host sugarcane. The semi-perennial nature of the crop has helped in the evolution of the pathogen in this direction. In this situation, obviously to increase its survival value, pathogen has to have enormous array of variability or method of accumulation of variability from natural population to have an effective parasitic life (i.e., to reproduce in sufficient number). The enormous variation in the virulence of *C. falcatum* is the perpetual problem in the development of resistant sugarcane clones and in their cultivation for a reasonable period of time (field life).

Table 5.3 Summary of the statement showing resistance and susceptibility to red rot of the various cane varieties tested at I.A.R.I., during 1936-49

Resistant			Moderately resistant			Susceptible		
Cane variety	Reaction (different season)	No. of years tested	Cane variety	Reaction (different season)	No. of years tested	Cane variety	Reaction (different season)	No. of years tested
Co 349*	3R	3	Co 205	4M-1R	5	Co 213	2R-3M-8S	13
Co 356	9R-4M	13	Co 214	5M-6R-1S	12	Co 221	2S	2
Co 441	6R-2M	8	Co 229	3M	3	Co 223	1R-2M-10S	13
Co 532	5R	5	Co 243	2M	2	Co 291	3S	3
Co 547	6R	6	Co 244*	6M-6R	12	Co 299	11S	11
Co 548	5R	5	Co 281*	8M-3R	11	Co 312*	3R-8M-2S	13
Co 549	5R	5	Co 285*	5M-8R	13	Co 313*	6R-5M-2S	13
BO 4	5R	5	Co 290*	11M-2R	13	Co 331	7M-6S	13
CoK 34	3R	3	Co 300	3M	3	Co 361	1S	1
Chunnee	4R-2M	6	Co 301*	1M-2R	3	Co 362	1M-8S	9
S.spontaneum (Coimbatore)	2R	2	Co 352	4M-6R	10	Co 373	2R-4S	6
S.spontaneum (Burma)	2R	2	Co 360	9M-1R	10	Co 417	2M-8S	10
S.spontaneum (Glagah, Java)	3R	3	Co 370*	6M-3R	9	Co 426	3M-7S	10
			Co 371	4M-3R	7	Co 445	1M-6S	7
			Co 385*	8M-2R	10	Co 457	1M-6S	7
			Co 386	4M-2R	6	Co 458	4S	4
			Co 393*	2M-5R	7	Co 467*	1M-2S	3
			Co 395*	4M-2R	6	Co 469	2S	2
			Co 396*	4M-3R	7	Co 471	1S	1
			Co 402	6M-3R	9	Co 472	2S	2
			Co 408	1M-2R	3	Co 473	2S	2
			Co 411	5M-6R	11	Co 474	1M-1S	2
			Co 412	10M-2R	12	Co 526	7S	7
			Co 413*	6M-5R	11	Co 528	3M-4S	7
			Co 419*	6M-6R	12	Co 540	1S	1
			Co 421*	8M-5R	13	Co 552	2M-1S	3
			Co 432	7M-3R	10	Co 556	3S	3
			Co 433	4M-2R	6	Co 557	1M-2S	3
			Co 434	3M-4R	7	Co 642	2S	2
			Co 439	4M-2R	6	B.H.V	2S	2
			Co 442	3M	3	CAC 87	1M-2S	3
			Co 443	8M-2R	10	CoK 25	3R-1M-2S	6
			Co 444	3M-3R	6	CoK 26	1R-3M-2S	6
			Co 449*	4M	4	CoK 29	4S	4
			Co 453*	2M-2R	4	CoK 31	2M-2S	4
			Co 459	2M-3R	5	CoK 35	3S	3
			Co 460	4M	4	CoK 38	1M-2S	3
			Co 461	3M-1R	4	HM 661	1M-1S	2
			Co 462	2M-2R	4	K 1216	1S	1
			Co 463	3M-1R	4	P 1587	1M-2S	3
			Co 464	6M-2R	8	SG 63/32	2S	2
			Co 466	3M	3	POJ 213	2S	2
			Co 468	3M	3	POJ 1499	3S	3

Resistant			Moderately resistant			Susceptible		
Cane variety	Reaction (different season)	No. of years tested	Cane variety	Reaction (different season)	No. of years tested	Cane variety	Reaction (different season)	No. of years tested
			Co 475*	3M-1R	4	POJ 2725	2S	2
			Co 508*	5M-5R	10	Striped Mauritius	1S	1
			Co 513*	4M-5R	9	Uba Marrot	3S	3
			Co 523	3M	3	Surkha saharanpuri	1S	1
			Co 524	6M-3R	9	Dehra Dun Paunda	3S	3
			Co 527*	4M	4	Peshwar Paunda	1S	1
			Co 529	3M-4R	7			
			Co 531	4M-2R	6			
			Co 538	4M	4			
			Co 534	2M-1R	3			
			Co 545	3M-1R	4			
			Co 546	5M-4R	9			
			Co 550	1M-2R	3			
			Co 551	3M-1R	4			
			Co 553	2M-2R	4			
			Co 554	3M-1R	4			
			Co 555	2M-1R	3			
			Co 558	1M-2R	3			
			Co 605	3M	3			
			Co 617	3M	3			
			Co 627	1M-2R	3			
			CoK 10	3M-2R	5			
			CoK 22	4M-2R	6			
			CoK 27	3M	3			
			CoK 28	1M-2R	3			
			CoK 30*	2M-2R	4			
			CoK 36	3M	3			
			CoK 37	3M	3			
			CoK 39	2M-1R	3			
			CoS 5	3M-2R	5			
			CoS 76*	2M-4R	6			
			G 1051	2M-1R	3			
			SC 851	2M-1R	3			
			POJ 1410	3M	3			
			POJ 2961	3M	3			
			Hemja	3M-2R	5			
			Pansahi	2M-3R	5			
			Saretha	5M-1R	6			
			Uba	3M-1R	4			

*R= Resistant, M= Moderately resistant, S= Susceptible, *= Important commercial cane varieties (Adapted from Table IV of Chona, 1954)*

6

THE FUNGUS

Description of *Colletotrichum falcatum* Went

(= *Glomerella tucumanensis* von Arx and Muller)

The fungus causing red rot of sugarcane is commonly known by its imperfect state, i.e., *Colletotrichum falcatum* Went. The disease was first described to the scientific world in 1893 by Went (Went, 1893) from Java (now Indonesia). Incidentally, the perfect state (*Physalospora tucumanensis* Speg.) was first recorded by Spegazzini from Argentina in 1896 (Spegazzini, 1896) but it took almost half a century to establish the association between *Colletotrichum falcatum* Went and *Physalospora tucumanensis* Speg. (Carvajal and Edgerton, 1944). Ten years later von Arx and Muller from Germany transferred this fungus *Physalospora tucumanensis* to the genus *Glomerella* and renamed it as *Glomerella tucumanensis* (Speg.) von Arx and Muller (von Arx and Muller, 1954). The identifying characteristics of the imperfect state, the disease-causing phase, as described by different authors (Abbott, 1938; Sivanesan and Waller, 1986; Abbott and Hughes, 1961; Sutton 1980) are given below:

Colony greyish white with sparse aerial mycelium, occasional small dense felty patches, reverse white to grey, conidial masses **(Plate 18a, b)** salmon pink (light race); some cultures have abundant greyish white aerial mycelium with poor sporulation and no distinct acervuli (dark race). Sclerotia not produced by both the races, setae sparse, conidia falcate (but not markedly so), fusoid, apices obtuse, 15.5 (25-26.5) 48 µm x 4 (5-6) 8 µm **(Plates 19, 20, 21, 22)** and contents are granular and sometime contain oil globules. The central white area, when stained with appropriate stains indicates the presence of nuclear material. A conidium usually contains single central nucleus **(Plate 20)**. The occurrence of more than one nucleus in the conidium is often observed **(Plate 21)**. With the addition of nitrogenous substances like peptone in the medium the conidial shape and size changes quite significantly. Conidia appear more falcate, **(Plate 19e)** the average length goes up and ranges in 35-40 μm and the breadth comes down to 4-5 μm. Occasionally the presence of aberrant conidia measuring more than 60 μm **(Plate 25g)** was also noticed in peptone containing media (Duttamajumder, unpublished observation). Conidia do not have any resting period and can germinate immediately after their production. Most of the conidia follow a normal pattern of germination. Usually it registers sub-polar germination **(Plates 22, 23)** from one or both ends of the conidium. Sometimes another germ tube comes out from the central portion of the conidium. Usually the conidium develops a median septum before germination but germination without septa is also common. Development of septa without germination was also noticed occasionally **(Plate 19)**. In rare cases, germination of conidia directly leads to development of conidiophore yielding daughter conidium **(Plate 25)** (Duttamajumder, unpublished). After germination, the hypha may immediately produce appressoria (1, 2, 3, the number may vary) or may continue to grow depending on the available environment **(Plate 23)**. From the appressorium infection peg develops which enters the host tissue and initiates infection process **(Plates 23, 24)**. During the germination of conidium the central nucleus divides in two and each migrates to the respective distal end and thereafter a central septation develops dividing the conidia into two halves. Thereafter, the nucleus undergoes one more division and then one daughter nucleus travels to the hypha.

Under stress, like exposure to toxic chemicals like fungicides (sub-lethal dose) or under waterlogged situation or even in old culture (spore mass in acervulus), conidia germinate and produce a fusion aggregate **(Plates 33, 34)** to consolidate the available variability to overcome the stress situation (Duttamajumder *et al.,* 1990). From the fusion aggregates, a new crop of conidia develops **(Plates 33, 35)**. In the stress situation like waterlogging, walls of the thin hyaline conidia get thickened and become dark coloured **(Plate 33b)**. Contrary to the common description, occasional dark coloured conidia are often observed in old cultures.

Appressoria medium brown, clavate or circular, edge entire, 12.5-14.5 μm x 9.5-12 μm (6-21 x 6-17 μm). The appressorium bears a pore/light area from where the infection peg comes out **(Plates 23, 24)**. These structures (appressoria) have little or no resting period and immediately germinate to develop infection peg. The other type, produced mostly in/on the cane tissue/apical leaf folds **(Plates 6, 24)** (especially during spindle infection) and also observed in the backside of agar media in petridish culture **(Plate 24a)**, often stays for a longer time without germination and acts as chlamydospore. These spores usually show definite content within it when stained with appropriate dye **(Plate 24c)**. The appressoria are quite typical in the genus *Colletotrichum*. In fact, Sutton (1966) took the smoothness, area and type of development of appressoria as an important character in the differentiation of species in the genus *Colletotrichum*. Chlamydospores may remain viable for sometime and may serve as perpetuating structures. They appear to be more important as appressoria and germinate to form an infection peg capable of penetrating the cane plant (Butler, 1906; Edgerton and Carvajal, 1944). However, the reason for their preponderance in spindle infection is not yet clear.

The asexual fruiting structure of *Colletotrichum* is acervulus **(Plates 13, 16, 26)**, which develops due to the aggregation of short and compressed conidiophores producing large mass of conidia embedded in a gelatinous matrix. In the acervulus, dark brown setae measuring 100-200 µm also develop. The setae are septate stout structures, sometimes with bulbous base, tapering toward the tip **(Plate 26)**. Development of synnematous setae **(Plate 28)** has also been observed with many isolates. Occasional development of colourless setae has also been noticed in some isolated cases. The setae are usually sterile structure and do not contribute to the multiplication of the pathogen. However, it has been noticed that under favourable condition, when tip remains hyaline, it can contribute in the production of conidia **(Plate 27)** (Edgerton, 1959, Duttamajumder, 2002). This type of situation is encountered in the leaf folds of spindle infection and in case, when sporulation takes place within the cavities formed in the internodal tissues of the cane. Sometimes the hyaline tip grows to produce a hyphal mass leading to the production of an acervulus. The exact role, these often-sterile structures play in nature is not clearly understood. In the view of the author, the setae serve twin purposes. On one hand these hide and hinder mites from devouring up all the conidia and conidiophores from an acervulus and on the other hand these act as vibrating structures (like tuning fork) which vibrates in air current and aids in the dissemination of conidia (through spattering effect) embedded in a gelatinous matrix.

Glomerella tucumanensis is an ascomycetous fungus and produces ascospores in a typical 8-spored ascus in perithecium. Perithecia are inconspicuous and are almost entirely embedded in the cane tissues **(Plate 29)**. They are found in abundance on dead or dying leaf lamina, mid-rib (under side), and leaf sheaths. In the leaf tissue, they are crowded between the vascular strands, and measure 100-260 µm in width by 80-250 µm in height. Asci are clavate, measuring 70-90 µm x 7.5 (13)-18 µm. Hyaline, straight to slightly fusoid, one-celled ascospores measuring 18-22 µm x 7-8 µm are arranged in a biseriate pattern **(Plates 29, 30)** in the ascus (Carvajal and Edgerton, 1944; Chona and Bajaj, 1953;

Duttamajumder, 2002). Ascospores follow the same germination pattern **(Plate 31)** as the conidia (Duttamajumder, unpublished).

von Arx (1957) included most of the described species of *C. falcatum* on Gramineae as synonyms of *Colletotrichum graminicola* (Ces.) Wils. and accepted *Glomerella tucumanensis* (Speg.) von Arx and Muller (syn. *Physalospora tucumanensis* Speg.) as the name of the perfect state of this species. However, due to the strong and specific pathogenic reaction of *C. falcatum* to sugarcane, it is considered different from *C. graminicola* (Sutton, 1980; Abbott and Hughes, 1961, Singh and Singh, 1989) and *C. falcatum* has been retained as a separate species.

Physiology and growth

Abbott (1938) summarized the work carried out on different culture media for growing *C. falcatum*. The fungus grows well on a variety of media including PDA, OMA, Cornmeal, Carrot, Czapex-Dox, Bean-dextose. Oatmeal agar was found most suitable for the growth and sporulation of *C. falcatum*. However, OMA is opaque and thus hinders many observations, which normally are taken from a clear growth medium. Abbott found that a medium consisting of decoction of 100g sugarcane leaves, 5g $NaNO_3$, 1g KH_2PO_4, 1g $MgSO_4$ and 10g sucrose is the most suitable alternative to OMA. Duttamajumder (unpublished) has found that a medium containing 10g soluble starch, 5g peptone, 1g K_2HPO_4 and 0.5g $MgSO_4$ is suitable for induction of sporulation in sporulation-shy isolates. Regarding the pH of the media, the fungus grows well in acidic media and a pH of 5.0 to 6.0 was found most suitable. It is also important to note that the pH of the cane juice is around 5.5 and *C. falcatum* has adapted to this natural environment of the host tissue.

Abbott (1938) has also shown that a temperature between 27.5 to 32.5 ^{0}C is optimum for the growth of *C. falcatum* isolates (Table 6.1). In a temperature below 10^{0}C, hardly any growth takes place even after 30 days of incubation. However, it is possible to gradually train *C. falcatum* to grow at a lower a temperature (Duttamajumder unpublished). Similarly, at temperature above 40^{0}C, no growth took place. However, subtropical isolates of *C. falcatum* that withstand much higher temperature in the field during summer continue to grow at 40^{0}C. There is an apprehension that exposure to high temperature (>40^{0}C) increases virulence in *C.falcatum*.

The average mean radial growth of eight light type isolates was 6.1 cm, whereas it was 6.0 cm for the nine dark type isolates at the end of 6 days of incubation at 30^{0}C (Abbott, 1938). The average growth rate of *C. falcatum* is about 10mm/day on OMA, whereas in the susceptible genotype a growth rate of

35mm/day of *C. falcatum* (vertical spread in the cane) is fairly common in inoculation with plug method. Why the pathogen attains 350% higher growth rate in the cane in comparison to OMA or sterilized cane is not clear. As far as sugar depletion in the cane due to the invasion of *C. falcatum* is concerned, it hardly drops beyond 4-6 units (%brix value), majority of sucrose remains unutilised at the beginning of the appearance of full blown symptoms.

Table 6.1 Growth of two isolates of *Colletotrichum falcatum* at various temperatures on oatmeal agar

Temperature (^{0}C)	Diameter of colonies (cm) reached in days							
	Isolate L-7				Isolate F-1			
	3rd	4th	6th	8th	3rd	4th	6th	8th
0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
5.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
10.0	0.0	0.0	G	G	0.0	0.0	0.0	0.3
15.0	G	G	1.9	3.3	G	G	2.3	3.6
17.5	0.7	1.9	3.8	5.1	0.9	2.2	4.3	5.5
20.0	1.5	2.6	3.9	5.1	1.9	3.0	4.9	6.4
22.5	2.2	4.0	4.7	8.2	2.4	3.9	5.8	8.3
25.0	3.4	5.2	6.8	F	3.2	5.3	8.2	-
27.5	2.6	6.0	8.8	-	3.4	5.9	8.9	-
30.0	3.0	6.0	-	-	4.1	6.2	-	-
32.5	3.3	4.7	7.3	-	4.0	5.8	8.5	-
35.0	1.8	2.5	3.6	4.9	1.7	2.8	3.4	4.6
37.5	0.5	1.5	2.5	3.0	G	0.6	0.9	1.5

G= germination only, F= filled entire plate
(Adapted from Abbott, 1938)

In India, Ramakrishnan (1941) studied the physiological aspects of growth and behaviour of *C. falcatum*. He found that *C. falcatum* grows well in a number of standard media, including French bean and Richard's agar and found oatmeal agar to be the best. A pH of 4.5- 5.0 was found optimal by him for the growth. Spores formed both at 15^{0}C and 30^{0}C were of normal size (19.9 -27.6 µm). Thermal death point of the conidia was five minutes at 51^{0}C. As regards the source of carbon and nitrogen, sucrose was found to be the best for the former while the latter was most easily assimilated from asparagin, potassium nitrate or peptone and the optimum carbon: nitrogen ratio was 5:1. The shape and the size of spores were found determined to some extent by the nitrogen source.

PERFECT STAGE - ITS OCCURRENCE AND ROLE

Occurrence of perfect stage in India

In 1952, Chona and Bajaj (1953) observed the perfect state of sugarcane in India, for the first time, in an 18 months-old sugarcane crop at New Delhi and published it in 1953. They reported :

> "On a routine examination of an 18 months old sugarcane crop at New Delhi, some dried up old leaves of various cane varieties still attached to the plant, showed numerous, small black dots on the lamina, and to a lesser extent on leaf sheaths, resembling pycnidia or perithecia."

(The paper was accepted on October 4, 1953. 'Thanks' for the assistance were accorded to Mr. R. L. Munjal, Asstt. Mycologist). Whereas, Chona and Srivastava (1952) observed the occurrence of the perfect stage of *C. falcatum* in culture at a later date (the paper was accepted for publication on October 12, 1953! with no such thanks, indicating that identity of the perfect stage had already been established). It is not justified to report that Chona and Srivastava (1952) were the first to observe the perfect stage of *C. falcatum* in India as one finds in the writings of many authors. (Pers. comm. with Dr. R.L. Munjal).

Chona and Srivastava (1952) reported production of perithecia on dry old sugarcane leaves inoculated with the conidia and incubated under high humid conditions. Perithecia rarely developed on green autoclaved leaves and failed to develop on medium devoid of intact plant parts. Enrichment with carbon and nitrogen failed to elicit positive response for perithecial development (Chona *et al.*, 1960). Chona and his associates worked on various aspects of the perfect stage of the fungus (Dutta *et al.*, 1964 & 1965) but could not make much headway in elucidating the role of the ascosporic inoculum in the epidemiology of the red rot disease. Moreover, the contemporary workers in India were not able to detect the perfect stage in the field and became sceptical about its very presence in India, at least in sufficient numbers to become instrumental in generation of variability in the fungus. Duttamajumder (1996, IISR Annual Report) was able to show the presence of perfect stage of red rot pathogen in India in sufficient numbers throughout the length and breadth of the country. The abundant perithecia formation was observed in any area wherever a conscious search had been made. Perithecia appear on the dying/dead leaves during the monsoon and post monsoon period (August, September in Uttar Pradesh). Perithecia formation is also observed even in March when ambient environment is quite akin to that of August-September (Duttamajumder, unpublished). However, during this period perithecia are mostly restricted to the lower side of the dying mid-rib. In nature, nothing occurs in isolation. A succession of fungi occurs on the sugarcane leaf making the appearance of the perithecia obscure to the casual observer.

On the basis of observations on the behaviour of the mid-rib populations as well as the population that produces perithecia, the author has arrived at the conclusion that the red rot fungus, *C. falcatum* is basically a leaf parasite, and quite similar to *Colletotrichum graminicola* that occurs on sorghum (Duttamajumder, 2002). The pathogen completes its life cycle on the leaf of the sugarcane, and it does so, irrespective of the resistance and susceptibility of genotypes to stalk isolates. He further drew analogy from the human society, where most of the people are peace loving but only a handful, ranging from petty thieves, dacoits to the dreaded criminals create all sorts of nuisance in the society. Similarly, the population of *C. falcatum* in sugarcane agro-ecosystem is also peace loving and does not cause any significant harm to the cane crop as long as it restricts itself to the mid-rib. Essentially, most of the *C. falcatum* population thrives on the mid-rib without causing much harm to the yield potential of the cane. In fact, these isolates do not infect cane and develop resistant reaction on stalk inoculation. Occasionally an odd member acquires the audacity to invade the stalk and unleash an avalanche of terror. No doubt, erection of jails and enforcement of strict laws have contained the crime in human society but it remained short lived. With all the deterrents in place, criminals are still proliferating and causing nuisance. Long-term solution has always remained elusive to the society. In the same analogy, breeding of disease resistant sugarcane varieties provides a short-term solution to check the red rot menace. Time and again, new pathotypes/races of *C. falcatum* have arrived in the scene just to turn the apple cart down. A holistic system of sugarcane growing has to be developed that discourages the formation and proliferation of the aberrant type *C. falcatum* isolates. Like the spiritual upliftment of human society, it is well known that a beginning has to be made from the very young age to change the bent of mind of people. Proper management of the sugarcane crop from the very beginning (green manuring, land preparation, appropriate fertilizer application, sett treatment (MHAT, fungicide), roguing as well as use beneficial microorganisms like *Trichoderma*, etc.) is necessary to sustain the crop health. A healthy seed programme in a healthy field along with clean cultivation practices will go a long way to reduce the chance of *C. falcatum* attack and will improve the prospects of a better harvest.

7

VARIABILITY AND RACE IDENTIFICATION

Variability is the rule of nature

Variability is the rule of nature; it is the seminal mechanism available to an organism to maintain the diversity. Buxton (1959) wrote:

> "For evidence of variability in plant pathogenic fungi there is no need to look further than the many examples of crop varieties that have been bred for resistance to a given disease only to succumb to that disease after they have been brought into general cultivation."

The red rot pathogen is no exception to this general situation; rather it is more potent in changing the virulence pattern depending on the particular need to match the introduced host resistance. Both sexual and asexual mechanisms are operative for this purpose. However, one moot point that needs to be pondered at is the generation of variability in *C. falcatum*. What is the basis of frequent appearance of new virulences (races) that are able to start new outbreaks of disease both in new localities and in previously resistant host varieties?

Mechanism of variation

The variations in the asexual state of the fungus (*Colletotrichum* state) may originate through (a) heterokaryosis, (b) by recombination through parasexual mechanism, and (c) by the universal mechanism of mutation, selection and adaptation in response to the changes in the host environment. Heterokaryosis is the mechanism through which the fungus collects and consolidates two or more genetically different nuclei in the hypha and derives the benefit of the introduced genetic material. These newly gathered nuclei also multiply in tandem with the native nuclei. Anastomosis of hyphal cell is common **(Plate 32)** in *Colletorichum falcatum* (Duttamajumder *et al.*, 1990; Singh and Payak, 1968; Carvalho, 1968). Anastomoses may take place between hypha, hypha and conidium, conidium and conidium, conidium and appressorium **(Plates 33, 34, 35)**. The frequency of anastomosis increases with stress (physical as well as chemical) and the involvement of more than 500 conidia to form fusion aggregate has been noticed by the author **(Plate 34 d)**. The migration of nuclei takes place through the hyphal bridge formed due to anastomosis **(Plate 34)**. Through some undefined mechanism(s), fungus maintains a balance of native nuclei, and the nucleus gained through anastomosis and derives most of the advantages as necessitated by the changing environment of host or otherwise. Carvalho (1968) has shown the luxuriant growth of *C. falcatum* in the mixture of two nutrient deficient mutants (deficient in biotin and thiamine) in contrast to almost no growth of the parent mutants and explained that this change is due to the heterokaryotic condition resulting from anastomosis of hyphae and hetrokaryotic nuclei complementing/ supplementing each other to increase the fitness for survival. This may provide a reasonable explanation to saltation, which cannot be explained satisfactorily through the mechanism of mutation, as the rate of mutation is much lower to fit the situation in the artificial culture. Thus, heterokaryosis assumes a great significance in the development of races of *C. falcatum* whose population in nature constitutes a mixture of numerous variable biotypes occurring on the leaf causing innocuous laminar and mid-rib infections along with population capable of attacking the stalk.

Mutation vs adaptation

The ubiquitous occurrence of mutations and their subsequent selection to suit the available environment, gives an impression of a situation that most of the workers have accepted the change as adaptation. However, such an adaptation is mostly restricted to higher organisms where specific gene(s) is expressed depending on the environment it has to withstand. In microorganisms, a shift in population most suited to the new environment takes place. This indirectly implies the reason of higher genetic load that higher organisms have to carry as an insurance cover. In lower organisms like bacteria, due to their much shorter life-cycle, there is hardly any scope of adaptation, mutation rules the roost. Fungi, as a group are placed much higher in the evolutionary ladder (eukaryote) than bacteria (prokaryote) and have tackled this situation in a distinctive way. Some are coenocytic i.e., do not have any cross wall (septa) in the hypha and free flow of nuclei takes place in the thallus. The rest are with definite septa. However, presence of pore in the septa provides passage to the migration of nuclei in the hypahe. This is the group where *C. falcatum* belongs. The other (basidiomycetes) which is considered most evolved group among the fungi has also scuttled this situation by developing a clamp connection, which bypasses the septal apparatus and helps in the migration of nuclei. The advantages of heterokaryosis (multiplication and maintenance of different types of nuclei in the same thallus in addition to the native nuclei) and free migration of nuclei within the hypha gives an illusion (of adaptation) to this effect. However, it is to be noted that the adaptive enzymes also play a definitive role in most of the situations. The genes for these enzymes are present in the pathogen and these enzymes occur in a very minute quantity, often at a level beyond detection with common detection tools methods. However, the genes governing the production of these enzymes are activated when a particular situation arises especially when the fungus needs their activity to increase its fitness. The discovery of the operation of **lactose operon** supported the active operation of this phenomenon, which earlier was thought as an adaptation.

Parasexuality

The parasexuality, a mechanism of getting benefit of sexual mechanism (recombination of genes) without involving the mechanism of meiosis, is third most important means in the maintenance of variability to the desired level. In the parasexual process, occasionally two haploid nuclei fuse and give rise to diploid nucleus. Subsequently multiplication of diploid nucleus takes place. Thus, at a given point of time a thallus contains a mixture of haploid and diploid nuclei. Mitotic recombinations provide an opportunity of crossing over and development of new genetic recombinant at any part of the thallus. If this new hypha bearing the recombinant is more fit in the environment, it proliferates and signifies the improvement in the variability of the pathogen.

Sexual mechanism

Sexual reproduction (diplodization followed by meiosis- producing haploid ascospores in a typical 8 spored ascus) is operative in *Glomerella tucumanensis*. However, not much credence has been given to it for the generation of variability leading to the development of new pathogenic races responsible for the failure of the sugarcane genotypes. In the first, owing to the homothallic nature of the fungus, occurrence of high variability through sexual mechanism is not expected and the sexual state is mostly restricted to the leaf; and the leaf isolates are usually not capable of infecting the sugarcane stalk. However, this does not close the door for the development of new virulent races. It is assumed that occasionally an odd recombinant/mutant acquires the capability of infecting the stalk and gradually establishes itself in the alien environment of the stalk. This ultimately kills the stalk. Death of the cane plant reduces the availability of food to the leaf population, and thus poses imminent survival problem for the entire population. As observed by the author, *C. falcatum* isolates obtained from the stalk are shy in producing perithecia in artificial condition of the laboratory, whereas leaf/mid-rib isolates have no such qualms.

In spite of the inherent simplicity of homothallism, complexity of parasexuality, heterokaryosis and ubiquitous mutation provide the much-needed variability to *C. falcatum* to match the genetic complexity of sugarcane. Still it remained unexplained, how new resistant varieties suddenly succumb to the pathogen without any noticeable occurrence of disease in the crop. It is obvious that some kind of surreptitious spread of the pathogen is occurring in nature and effecting sudden failure of a resistant cane genotype in a larger area.

Race flora

For successful breeding of disease resistant genotypes and their deployment, a clear picture of race flora of the pathogen and their geographical distribution is essential. The disease resistant genotypes in turn are essential to contain the disease in more eco-friendly and effective manner. *Colletotrichum falcatum* has different forms varying in virulence and pathogenicity. This variation drew immediate attention of even the earliest worker on this disease. Earlier sugarcane planters also noticed that varieties differ not only in their growth characteristics but also in their behaviour regarding pests and diseases (Watt, 1893). This situation prompted frequent introduction of sugarcane varieties from different places (districts, states and country). The variation in the pathogen put pressure by curtailing the field life of a cane variety and thus, necessitated quick replacement. With the advancement in sugarcane breeding, emphasis was given for breeding resistant varieties against the pathogen. The development of interspecific hybrids (use of wild *Saccharum spontaneum* as the source of resistance) is the milestone in this regard. Indeed, the unsurpassed impact of sugarcane breeding in India paved the way in reviving sugar industry. This remained the first Green revolution (mostly unsung, except the Knighthood of Sir T. S. Venkatraman) in India. This spectacular success of breeding hybrid cane increased the sugarcane area tremendously, until it was hit badly by the red rot. More emphasis was laid to breed varieties resistant to this disease but often met with varied success. The erratic behaviour of *C. falcatum* in artificial culture, frequent loss of sporulation as well as pathogenicity always put the pathologists on the back foot. It became difficult to ascertain the virulence over crop seasons. Workers resorted to collect available variability each year and used the most virulent (aggressive) one from the collection for the screening purpose (Chona, 1954). At SBI, Coimbatore workers resorted to use only 'D' strain for long a time with the consideration that if a variety is resistant to 'D' strain it will have adequate resistance to other strains. In later years, however it was realised that resistance to 'D' strain is not uniform at all endemic locations and resistance of a genotype is dependent upon the composition of the local red rot flora used for testing (Alexander and Rao, 1976).

For 11 years (from 1992-2002), the author has studied the stability of pathotype Cf 01 in normal cultivation without putting any selection pressure. Each year, 8 rows (15 m long) were planted with healthy and naturally infected cane setts (Co 1148) without any selection, whatsoever. At the end of the crop year (March) the plot was harvested and these canes were used for planting. No selection of cane or cane sett was done. It was observed that the incidence of the disease hovered around 10-20%. Every year, isolation was made from the affected canes and virulence/aggressiveness was tested on the designated differentials. It was observed that aggressiveness of Cf 01 also varied in different years, but there was always an availability of isolates with high aggressiveness.

A perpetual question still haunts the minds of sugarcane pathologists. If *C. falcatum* looses its virulence so quickly, then how the stability of a race could be established, and will it be prudent to call it a race. Time and again, newly released resistant genotypes succumbed to the red rot, whenever any matching virulent isolate surfaced on the scene and knocked the genotype with impunity, leaving no option but to phase out the genotype from general cultivation. Thus, the average field life of a sugarcane variety hardly exceeds 8-10 years before the assault of red rot (Table 7.1). This intrinsic variability prompted earlier workers to group the *C. falcatum* isolates (variability) for better understanding and management. Initial grouping was based on growth and sporulation. Pathogen flora was classified into two broad groups, viz., a less virulent (less aggressive),

Table 7.1 Field life of a few popular sugarcane varieties in the subtropical belt of India

Variety	Year of release	Year of withdrawal / succumbed to red rot	Duration in years
Co 210	1918 (UP)	1932	14
Co 213	1918(UP)	1933	15
Co 312	1928(UP)	1946	18
Co 313	1928(UP)	1936	8
Co 419	1936(UP)	1948	12
Co 453	1947(UP)	1958	11
BO 3	1959(UP)	1967	8
BO 10	1958(UP)	1965	7
Co 997	1967(UP)	1969	2
Co 1148	1962(UP)	1980	18
Co 7717	1978 (Haryana)	1980	2
CoS 767	1979(UP)	1991	12
CoS 770	1977(UP)	1981	4
CoJ 64	1971(UP)	1979	8

dark race which was predominant in the pre-interspecific hybrid era (dark velvety mycelium, sparse sporulation) and the *light race*, the highly virulent type which was prevalent on the hybrid canes (white mycelium, heavy sporulation).

The concept of physiologic specialization in red rot pathogen dates back to 1935 when Abbott (1935) from Louisiana reported that red rot isolates form two morphologically distinct groups - one dark velvety non sporulating and the other, lighter, sporulating and usually capable of infection on inoculation. He developed an interesting theory in this regard, and believed that origin of the two types of races was the cause of failure of POJ 213 and CP 807. POJ 213 was affected by the light type badly, whereas CP 807 showed greater susceptibility to the dark type. He postulated that the dark type might be the original type, which thrived well on the old noble canes, and the light type has evolved due to the increase in area of the new hybrid POJ 213. Moreover, dark races were prevalent in the southern states (syrup districts) where old varieties (noble canes) were still being grown. However, peer groups did not support his hypothesis as indicated by Edgerton (1959)

> "From the evidence available, it would seem that this theory is not entirely acceptable. During the long period in which red rot was kept under close observation in Louisiana, epiphytotics varying in severity, occurred from time to time, and in general, cultures which were isolated and proved to be highly pathogenic, sporulated profusely. There is no evidence indicating that the organisms involved in the epiphytotics, such as occurred in 1923 and 1927 were morphologically different from those present in later years."

Similar observations were also made in India. Chona and Padwick (1942) indicated that the epidemic of 1938-39 resulted a new highly virulent light type isolate, which was responsible for the failure of Co 213. Rafay (1953), Chona (1954), Kirtikar (1960) and others also have reported the variation in the pathogen in terms of morphology and pathogenicity on different cane genotypes. Kirtikar (from Shahjahanpur) designated the variation in R series, like R 141, R 200 etc., whereas Rafay (from Lucknow), Chona (from New Delhi) and others grouped them as A, B, C, D, E, F, G, H, and I stains of *C. falcatum* (Table 7.2). However, there was no consensus among the workers in this regard. Due to the overlapping morphological groupings it was extremely difficult to put a *C. falcatum* isolate in the above categories. In spite of the problems, there had been only few instances where there was no difference of opinion in classifying *C. falcatum* isolates. This happened only with the more stable isolates of *C. falcatum*. During the past one hundred years of red rot research in India, only a few isolates could remain stable for a reasonable length of time retaining their pathogenicity in artificial culture. The prominent five strains were the 'D' strain (isolated from Co 453), 'I' strain (isotated from BO 11), strain 244 (isolated from Co 312), the Jagadhari isolate (from Co 1148, later designated as race Cf 01) and an isolate from Co 419 (later designated as race Cf 04). It is to be noted that the initial isolation characters of strain 'D', 'I' and Cf 01 were very similar - all were highly sporulating, covering the media with spore mass but with very sparse mycelium.

Table 7.2 Strain designation of *C. falcatum* prior to 1992

Strain	Character
A	Colony cottony to floccose, white in colour during first two weeks and assumes very light tint of grey with age. Slimy pink masses of conidia are absent.
B	Colony is loose and silky. Abundant spore masses are seen on aerial mycelium, while abundant salmon coloured slimy masses of conidia are produced on the surface of the medium. Old cultures exhibit a relatively more compact texture.
C	Colony is compact and velvety and is darker than type 'A'. Conidia are produced in pink masses sparingly with a tendency to confine themselves to the margins of the media, representing an intermediate stage between 'A' and 'B'.
D	The colony texture is loose with sparse aerial mycelium. The slimy masses of conidia are abundant, covering 50 per cent or more of the surface of the medium.
E	Mycelium is velvety in texture with sparse sporulation.
F	Mycelium is cottony, which turns grey with age, with moderate production of conidia.
G	Aerial mycelium is dark grey and profuse with very less conidial production.
H	Mycelium is loose and silky with pseudopycnidial masses of conidia on the surface.
I	Aerial mycelium is scanty, whole surface of the medium is covered with conidial mass.

Epidemic : arrival of a new race

The formation of new virulences of *C. falcatum* is always taking place in nature and their prevalence is guided by the sugarcane varieties in cultivation. In an epidemic, a new pathogenic isolate/strain prevails in a wide area affecting the cane variety. Even today, no satisfactory answer could be given regarding the origin of a new race and its wide coverage, affecting the prospect of a cane variety. It is often assumed that this new virulent pathotype is a mutant of existing strain(s). It is not clearly known whether highly pathogenic strains develop as a result of changes in the vegetative mycelium through anastomosis, conidial fusion or from ascosporic inoculum. However, ascosporic inoculum did not get much credence due to homothallic nature of the genus *Colletotrichum*.

It has been experienced by different workers repeatedly that whenever a popular sugarcane genotype succumbed to red rot in the field - invariably a new red rot pathotype has emerged having capability to breach the introduced resistance of that genotype. However, it is interesting to note that after the epidemic years, only one or two isolates have been obtained with the desired level of virulence required to knock down the variety in question. There was never a preponderance of variability, even at the flare up situation, which is contrary to the normal expectation that sheer population explosion of the pathogen will favour the generation of an array of variability. It is not necessary that the new pathotype is more capable to attack other genotypes as well. In several cases, a reduction in virulence was also observed (Chona, 1954). The occurrence of increased virulence may be due to the spectrum of varieties having same or similar background of genes governing resistance to red rot. It is on record that only a few of *S. spontaneum* clones have been utilized for the introduction of hardiness as well as resistance in sugarcane breeding and that too at the beginning of the interspefic hybrid-era. Subsequently, sugarcane breeders used the advantage of high heterozygosity of sugarcane, and resorted to an easier and more feasible approach of crossing between the hybrids and evaluating the selected progenies for their tolerance to red rot rather than following the difficult path of direct crossing with *S. spontaneum* with less chance of getting commercial hybrids. This approach, no doubt, has made significant progress in developing sugarcane varieties in quick successions but at the same time introduced an inborn weakness (narrow genetic base; permutations and combinations of same genes) to the varieties, which led to their quick phasing out of cultivation due to red rot. It has also resulted in a plateauing effect in sugarcane productivity in terms of both yield and quality. This situation has given birth to the apprehension that high sugar is associated with red rot susceptibility, overlooking the fact that *S. officinarum* contributing 90% of the chromosomes in the hybrid is susceptible to red rot. It is evident that simple reshufflings of genes governing resistance to red rot will not stand long with the enormous capability of the pathogen in developing variation in virulence.

Light type vs dark type

The other moot point that one is bound to consider is the frequent change of the light type isolates into the dark type. Changes are too common to be overlooked **(Plate 18c, d)**, and in fact, it is major problem in the maintenance of *C. falcatum* cultures in virulent form. Edgerton (1959) described it, as:

> "In old cultures, spots appear containing dark coloured mycelium, which sporulate sparingly. It is assumed that these result from mutation. Gradually, as transfers are made from these cultures, the dark coloured mutants become dominant and replace the original type. Generally, isolates which have been kept in culture for a year or more are dark coloured. Usually such cultures sporulate sparingly, or not at all and commonly show a low degree of pathogenicity."

Every worker, who has put his hand on red rot pathogen has experienced that a dark velvety mycelial patch often develops in the slant culture or petridish culture **(Plate 18c, d)**. The dark velvety area looses the sporulating ability and remains stable over subculturing. This is an indication of the doom of the isolate in terms of virulence. To circumvent this inherent changing nature of red rot pathogen in artificial culture, subculturing is done by picking up the spores from the slant culture or petridish culture, and religiously passaging through the host, as a sieve (possibly needs an impetus to retain genes for pathogenicity) to encourage the selective multiplication of the virulent form. Therefore, each year, inoculation and reisolation from the host, early in the season, remains one of the important activities to retain pathogen in virulent and sporulating form. Even with all the precautions, off and on, some isolates loose virulence. However, it has also been observed that some of the velvety cultures when inoculated also showed pathogenicity. The change of the pathogen to nonsporulating velvety form indicates a gloomy future, as without sporulation it looses its capability of secondary spread through the conidia. In the past, workers have used the most virulent isolate (indicated by the capability of knocking down maximum number of cane genotypes in the collection) for the screening of resistance types among the selections/ varieties (Chona, 1954).

The cultural characters of the isolates as described by Srinivasan (1962b) reflect the general situation of variability in *C. falcatum*. The cultures on their first isolation could be classified into four morphological types based on the spore bearing structures and occurrence of setae in the acervuli.

Type I cultures

Aerial mycelium white, scanty, appressed; conidia borne singly on vegetative hyphae, occasionally in a chain of two conidia, falcate, thin-walled 15-42 µm x 4.2-8 µm; conidial germination typical of the species viz., by forming a septum in the middle of the conidium and developing an appressorium at the end of a short germ tube. This type is rarely encountered.

Type II cultures

Aerial mycelium floccose, white in young cultures; conidial masses appearing in one to two weeks, salmon orange, slimy, acervuli devoid of setae; conidia falcate, 15-36 μm x 4.2-(7)-8 μm.

Type III cultures

Aerial mycelium floccose, white in young cultures; sporulation in one to two weeks; acervuli appearing black at first, surrounded by setae, later developing conidial masses of vinaceous colour; conidia falcate, 17-33 μm x 2-6 μm.

Type IV cultures

Aerial mycelium scanty, colourless; acervuli borne on black stromata of thick walled polygonal brown cell, 0.2-0.8 mm dia and up to 1 mm in height; acervuli surrounded by setae and developing conidial masses of vinaceous colour; conidia falcate, 17-33 μm x 2-6 μm. This type is also rarely encountered.

Except type I, which is less virulent (aggressive), all the types are virulent.

The type I and type IV type did not change in cultural types even after 15 transfers whereas within 3 transfers type II and type III cultures gave rise to cultures with a felt-like, smoky grey mycelium, almost devoid of acervuli with poor sporulation (Table 7.3). The other (light type) continued to give rise to cultures resembling the parental type. Dark sectors did occasionally arise in the cultures, but monoconidial transfers from the light area resulted in continuance of the latter without change in morphological type.

Table 7.3 Stability of pathogenicity and virulence of *C. falcatum* after repeated subculturings

Virulence reaction	Variety Co 213				
	R	MR	MS	S	HS
At the time of isolation (type II)				**2**	**4**
After 10 subculturings	3	1	-	1	1
At the time of isolation (type III)				**14**	**19**
After 10 subculturings	20	5	-	3	5
Total after 10 subculturings	**23**	**6**	**-**	**4**	**6**
	59%	**15.4%**	**-**	**10.2%**	**15.4%**

(Adapted from Srinivasan, 1962)

RACE IDENTIFICATION

The development of red rot isolates capable of knocking down the dominant genotype Co 1148 in Haryana and western Uttar Pradesh, and succumbing of early high sugar genotype, Co 997 in Andhra Pradesh forced several workers (Satyanaraya and Satynarayana from Anakapalle, Satyavir and co-workers from Haryana and Alexander from SBI, Coimbatore) to look into the issue of pathogen variability. Satyanaraya and Satynarayana (1976) observed that red rot isolate from Co 997 did not infect CoC 671 and vice versa, so was the case with Co 1148 and Co 7717 as observed by Satyavir and his co-workers. Alexander and his co-workers identified five major pathotypes of *C. falcatum* prevalent in India, viz., MP 1 to MP 5. MP 1 and MP 2 prevalent Tamil Nadu and Andhra Pradesh, respectively; MP3 in eastern U.P. and Bihar; MP 4 in western U.P., Haryana and Punjab and MP 5 was prevalent in Kerala. This type of classification of pathotypes based on cane growing zones did not hold for long, as variation of the pathogen was not limited by the geographical location, rather it was dependent on the spectrum of sugarcane varieties in cultivation.

Alexander and his associates in SBI, Coimbatore and Vijay Singh at IISR, Lucknow also tried serology to establish any possible relationship between serological groups and pathological groups of *C. falcatum*. However, no conclusive results emanated from their studies. It is obvious that existence of a direct relationship betwen food habit and surface proteins is not generally available in nature.

Breeding for stable resistance needs efficient screening against the probable virulence of the pathogen that the new genotype is going to face in its field life. *C. falcatum* is also notorious in losing (shedding) virulence in artificial culture. This intrinsic variable nature of the pathogen always introduced certain level of uncertainty in the mind of pathologists about the outcome of inoculation with cultures that have undergone several subculturings. Obviously, this guided the development of standard procedure of revival of virulence (aggressiveness) by passaging the culture through inoculation early in the season (each year). Unfortunately, the intrinsic variable nature of *C. falcatum* did not allow developing a consensus among the breeders and pathologists in this regard.

The work on identification of pathotypes of *C. falcatum*, at all India level was initiated under the umbrella of AICRP (Sugarcane) in 1984-85 and Dr. K. C. Alexander, as Principal investigator of the pathology group, took the lead. Data on pathogenic variability on a set of designated cane genotypes were collected and it was discussed thoroughly in the subsequent AICRP (S) meetings.

Table 7.4. Tentative key for characterising *Colletotrichum falcatum* strains/ pathotypes characterised on differentials (1992, AICRP(S))

Sources/location of isolates		Designated differentials												
		1	2	3	4	5	6	7	1	2	3	4	5	6
Designated race	Source/ Location	Baragua	Khakai	SES 594	Co 975	Co 62399	CoS 767	BO 91	Co 1148	Co 7717	CoJ 64	Co 419	Co 997	CoC 671
Cf 01	Co 1148 NWZ	R	**S**	R -	**S**	S	R	R	S	R	S		S	S
Cf 02	Co 7717 NWZ	R	**R -**	R	**R -**	**S -**	R	R	R-	S	S -		S	S
Cf 03	CoJ 64 NWZ	R	R -	R -	R	**R-**	R	R	R-	R	S		S	R
Cf 04	Co 419 ECR	R-	**S**	**R**	S	R-	R-	R	S	**S**	S+	S+	S -	S
Cf 05	Co 997 ECR	**R**	S -	R	**S+**	R-	R-	R	S -	**R-**	S -	R	S+	S+
Cf 06	CoC671 ECR	**S**	S -	R	S -	R-	R	R	S -	R-	S	R-	S+	S+

NWZ = North western zone, ECR = East Coastal region, Cf – stands for *Colletotrichum falcatum*
Reactions are grouped into two main categories for the reasons of convenience. However, finer details of reactions are designated by using symbols as "+" and "-". For example, MR is recorded here as R- and MS reaction as S- and HS as S+ etc. This indicates if the reactions are on the higher or lower side of the designation. R and S are features of the isolate (resistance/ susceptible).
Cf01 = Differentiated on Khakai, Co 975/Co 7717
Cf02 = Differentiated on Khakai, Co 975, Co 1148/CoC 671, Co 7717
Cf03 = Differentiated on Khakai, Co 975, Co 997, Co 1148/CoC 671, Co 7717
Cf04 = Differentiated on Co 7717, Co 419
Cf05 = Differentiated on Co 7717, Co 419
Cf06 = Differentiated on Baragua

In 1992, in the meeting of AICRP (Sugarcane) at IISR, Lucknow under the Chairmanship of the then ADG (Crop Protection) Dr. S. Nagarajan, a workable system of virulence mapping for *C. falcatum* was chalked out. Obviously, the basic input was from Dr. K. C. Alexander (PI, Pathology) and the scientists from different centres contributed to it. The group finalised a set of designated differentials and designated races for validating the pathogenicity data from different locations and environments (Table 7.4). Accordingly, three north Indian isolates and three south Indian isolates were selected for testing at all India level for stability in different environments. The isolate from Co 1148 was named as Cf 01, isolate from Co 7717 got the designation Cf 02 and so an isolate from CoJ 64 was designated Cf 03. These three designated races were for subtropical region. The red rot isolate from Co 419, the wonder cane of India, received the designation Cf 04; similarly an isolate obtained from Co 997 was termed Cf 05 and the isolate from early high sugar genotype CoC 671 got the designation Cf 06. The six designated pathotypes typically produced HS (S^{+}) reaction with the matching genotype. These six representative cane genotypes were taken as the maintainers of virulence. For the differentials three species representative viz., SES 594 (*S. spontaneum*), Khakai (*S. sinense*) and Baragua (*S. officinarum*) were taken. For the differentiation of north Indian isolates Co 975 and for south Indian isolates Co 62399 were also taken. BO 91 and CoS 767 were added as the next two vulnerable genotypes, as any occurrence of red rot will jeopardize their cultivation affecting major area in the subtropical belt and will force a modification in the designated race identification system. The identification of designated races was felt necessary to systematize the disease resistance breeding and the deployment of the cane genotypes in any particular area depending on the non availability/prevalence of the matching designated race.

Single window testing

The other important aspect was to explore the possibility of single window testing of the cane genotypes to avoid any variation regarding their resistance behaviour, as it is often influenced by various conditions prevailing in different locations. This indiscriminate behaviour of this host-pathogen combination often became bone of contention and bickering between pathologists and breeders; as in most of the cases, an excellent genotype often failed to get a berth due to non-possession of appreciable resistance against the prevalent races of *C. falcatum*. Breeders often felt like 'sinking of a ship at the shore, after crossing all the rough waters' and always tried to push a genotype to scale the red rot wall. Breeders, many a times tested the integrity of pathologists, by providing a clone with different numbers in the same year or in different years for evaluation of the resistance against red rot. And many a times, discrepancies did arise putting a big question regarding the accuracy of evaluation.

As observed earlier races of red rot are mostly variety specific with geographic limitations. They are distinctly different so far as tropical and sub-tropical agro-climates are concerned. Under the aegis of AICRP(S), trials were conducted for this purpose. Designated differentials were raised at the participating centres and differentials were inoculated with the same designated races using plug method. As expected, there was hardly any variation in the disease reaction due to the variation in environment. Response of any individual genotype, irrespective of the designated race used for inoculation, behaved in same fashion and variation was limited to fluctuation of 1 unit in 0-9 point scale. This has opened up a new possibility of single window testing of cane genotypes at the pre-release stage. Two central institutes viz., IISR, Lucknow for the subtropical region and SBI, Coimbatore for the tropical region may act as the testing window for red rot. In the meeting of AICRP(S) of 1992, it was also emphasized to maintain a repository of *C. falcatum* pathotypes at IISR, Lucknow and its exact replica at SBI, Coimbatore to facilitate uniform testing of cane genotypes. In a prelude to this, working cultures of *C. falcatum* pathotypes are now being exchanged among the participating centres of the zone for the purpose of evaluation (with strict vigilance, so that no such pathotypes get a chance to escape from the research stations). Now the onus rests on the breeders and pathologists to work in tandem and to streamline the system of varietal testing against red rot, varietal identification as well as deployment of cane genotypes according to the prevalence of pathotypes in the specific zone. It is now necessary to give more emphasis on survey and surveillance to monitor the change in the variability pattern across the country and make interim corrections in the composition of pathotypes to be used for screening in the sugarcane varietal development programme.

Of late, five more races have been designated (Cf 07, Cf 08 and Cf 11 from CoJ 64 ; Cf 09 from CoS 767, and Cf 10 from the clone 85A261). The isolation and establishment of four pathotypes (Cf 03, Cf 07, Cf 08 and Cf 11) having varied virulence (CoJ 64) have put a question on the development of new virulence in nature. Have these pathotypes originated independently or shared the same origin with minor modifications in virulence pattern or all these CoJ 64 isolates owe their origin to the isolates of Co 1148? Alexander and Rao (1976) also made similar observations with different collections of so called 'D' isolate from different places. The instability of the isolates on artificial culture often poses a question regarding the tenability of races, but so far, no satisfactory answer could be given to explain the phenomenon. There is an urgent need to develop the differentials, preferably gene based and to strengthen the system of race identification so that the number of races for which any breeder will target the resistance breeding, is attainble.

On a closer look at the designated race grid (Table 7.4) it is evident that the highly virulent isolate of Co 419 (Cf 04) has paved the way for the less virulent ones (Cf 05 and Cf 06). So is the case with north Indian isolates Cf 01 (Co 1148) isolate paved the way for the less virulent Cf 02 (Co7717 isolate) and Cf 03 (CoJ 64 isolate). It is obvious that fitness of a particular isolate in the target environment counts most, and gets precedence over virulence. Duttamajumder and Verma (1994) with cotton-*Xanthomonas malvacearum* system, have shown that in a given location where an array of resistant and susceptible host genotypes are available, less virulent types of the pathogen predominates the race picture on *barbadense* cotton (with less number of genes for resistance) whereas, more virulent types survived better on *hirsutum* cotton (more number of genes for resistance). Due to the limitation in the fitness, highly virulent types, available in the same location could not get much foothold on the *barbadense* cotton, whereas less virulent ones having more fitness on *barbendense* cotton got established.

In the past, most virulent strains of *C. falcatum* available in the collection of the pathologists were used for the screening purpose (Chona, 1954); over the years it has become apparent that targeting resistance, only for the highly virulent type will not suffice. A broad based resistance is essential to sustain the field life of a variety. It is obvious that all the highly susceptible *S. officinarum* clones, available in the germplasm collection are probably due to slow red rot development in the field condition (as none of the *officinarum* type was found in the wild). It is also apparent that the prevalent practice of screening of only the selected progenies (good agronomic quality) for resistance will not solve the problem. A good amount of progenies with appreciable resistance is always left out due to their moderate performance in yield and quality. Again, there is no guarantee that the new resistant genotype bred against the prevalent races will survive longer in the field. There is every possibility that the newer genotype may go down with even a pathotype having very low relative virulence but is highly virulent in specific gene combination(s) matching the new genotype. Time and again, the epidemics of red rot have shown the appearance of a specific kind of variation in the pathogen; the variation, that is most detrimental to the genotype. Each epidemic yielded a new virulent type and that too a stable one! However, at the same time it is also on record that in spite of the explosive population increase during the epidemic, usually one virulent pathotype has been encountered. Obviously, epidemiological mutations are rare, and this provides an opportunity to manage the disease through other means. It indicates the necessity of deployment of a large number of cane genotypes varying in resistance, in a particular cane growing area to form a mosaic pattern in the fields and a quick replacement of any genotype showing any attack of red rot. The deployment of cane genotype, clean cultivations in addition to the broadening of the genetic base of red rot resistance will go a long way to contain the perpetual flare up of red rot in India. It may be relevant to mention that there have been instances where a variety declared susceptible (CoJ 64, Co 86032) to red rot and recommended as 'not suitable' for commercial release survived in the field for a quite long time. The success of CoJ 64 rests much on the clean seed programme, which is often misinterpreted as operation of field resistance. In any perennial crop, disease resistance has evolved in the direction of horizontal resistance (field resistance) as necessitated for its long time survival. However, both types of resistance coexist in a fashion that provides greater stability of a genotype against the pathogens prevalent in specific geographical location. The race specific susceptibility is always vertical. For that matter, whether the variety in question is CoJ 64, Co 86032 or Co 1148, horizontal resistance is in built and they all ward off other races except the matching ones.

8

DISEASE RESISTANCE

Disease resistance: the rule of nature

In nature, resistance against the pathogen is the rule and susceptibility is the exception. In the living world, those who cannot produce food on their own are dependent on the green plant (directly or indirectly) for their food requirement. A majority of the microorganisms live as saprophytes, leading their life on the dead plants or plant refuse. Only a few, which have acquired the capability of attacking the green plants cause disease and inflict damage. In general, plants have in-built resistance against hoards of microorganisms and only in specific conditions favouring the pathogen, disease occurs.

The contribution of disease resistant varieties in present-day agriculture is immense. Improvement in yield and quality usually accompanied the resistant variety. Better and normal growth of crop requires intrinsic improvement, irrespective of the factors of resistance. Farmers provide better care to a crop that shows promise. Certainly, the varieties that have gained ready acceptance invariably have improved commercial quality and high yield potential along with disease resistance. Disease resistance always provides sustainability to any crop production system. It has not only increased the food security but also has provided an opportunity in planning the crop with greater confidence. With the resistant varieties, even if the crop is somewhat less yielding, there is certainly something to harvest in contrast to the susceptible varieties. Sugarcane is no exception in this regard and disease resistance has served sugar industry of different countries and many a times has infused a new lease of life. Notably, the resistance to '*Sereh*' saved the sugar industry of Java in 1890's. So was the case with sugar industry of United States of America, where resistance to mosaic and root rot saved the sugar industry in 1920's. Similarly, in India, resistance to red rot has changed the face of Indian sugarcane; from perpetuating epidemic and crop failure, the country has entered into a realm of stability in sugarcane production.

The primary prerequisite for breeding disease resistance for any crop is the availability of sources of resistance and ease in the transfer of genes in the resultant hybrids. The success of such gene transfer depends on (i) source of resistance; (ii) amenability of genes governing resistance to such transfer as well as their expression in the new background, (iii) heritability of such genes, (iv) reliable method of assessment of resistance and finally (v) how frequently introduced resistance is breached by the pathogen (durability of resistance).

Unfortunately, resistance sources against red rot in the collection of sugarcane germplasm are limited and high degree of resistance nearing immunity is not available. The flow of new resistant cane varieties is not able to keep pace with the development of new variations in the red rot pathogen leading to frequent shortage of cane genotypes for replacement. The development of varieties fortified with durable resistance is cumbersome due to the involvement of a large number of genes for resistance. Further, high ploidy level and herterozygosity of the cane crop have complicated the breeding for red rot resistance. Being of semi-perennial nature, sugarcane has an in-built system of horizontal resistance that helps it to tide over the adverse conditions from season to season and from year to year.

Information on the inheritance of red rot resistance in sugarcane is rather sketchy. There is no clear-cut pattern of inheritance that one can embark upon. The inheritance, as known today, is indiscriminate as crosses between resistant and susceptible clones in all possible combinations (RxR, SxR, SxS and reciprocals) give rise to some resistant plants (Azab and Chilton, 1952, 1959). Degree of resistance of parents makes little difference to the percentage of resistant progenies, however, variations do occur in certain cross combinations. In sugarcane, the best-known source of resistance to red rot is *S. spontaneum* and most of the present day commercial hybrids owe their red rot resistance to it. However, except the initial involvement of a few *spontaneum* clones in the crossing programme, very little has been done to tap this potential resistant donor. Of late, a programme (ISH development) to broaden the genetic base of the parental pool was run at SBI, Coimbatore and several clones with new gene/ gene combinations have come up from this effort. It is hoped that a structured programme at the national level be started to introduce an array of genes from the wild relatives like *S. spontaneum, S. robustum* and *Erianthus* sp. to tackle the problems of pests and diseases as well as the problems of drought and waterlogging.

Horizontal vs vertical resistance

The concept of 'horizontal' and 'vertical' resistance needs a little elaboration to dispel the misconception prevailing among the sugarcane breeders and pathologists. A resistance gene in the host is identified by the corresponding virulence allele, *not the avirulence allele*, at a corresponding locus in the pathogen. That is, it is identified in a state of susceptibility, not resistance. An erroneous belief in specific resistance, as distinct from specific resistance genes, has led to much "barking up the wrong tree". Among the examples of misdirected "barking" are the phytoalexin and hypersensitivity theories of the host-pathogen specificity (Vanderplank, 1984).

The more the involvement of resistance genes, the clearer it becomes that susceptibility is specific and resistance unspecific i.e. there is race-specific susceptibility but not race specific resistance. In a modest 10-gene system, the possibility of races is $2^{10} = 1024$. Is it possible to target all the races in resistance breeding? In fact, a plant having all the 10-resistance genes is susceptible only to race 1 to 10 and is resistant to rest of the possible races (provided there is no interaction among these resistance genes). This remains the main target at the resistance breeding to counteract the virulence of the pathogen; not to increase the level of resistance of host.

Flor (1953, 1956) documented contradictions between observations in the field and greenhouse. In the field, unnecessary virulence reduced the fitness of the pathogen. He analysed field surveys (USA) of the flax rust fungus *Melampsora lini* from 1931 to 1951 and arrived at the conclusion that with the introduction of more resistant flax cultivars more virulent races of *M. lini* appeared. Moreover, the most prevalent races were those possessing the least number of virulence genes compatible with survival, i.e. compatible with the ability to attack the popular cultivars. Throughout the period of his analysis the percentage of isolates of *M. lini* carrying unnecessary genes for virulence, i.e. gene enabling them to attack cultivars not grown commercially, decreased. Greenhouse and laboratory experiments are dangerous in matters relating to the comparative fitness of phenotypes to survive in populations of pathogens in a fluctuating environment as available in nature.

Flor worked with autocious rust- an obligate parasite, and there was no question of artificial culturing. (It would be of interest to note that Flor in the early part of his career worked on sugarcane). In case of *Colletotrichum falcatum*, it sheds its virulence at different generations of subculturing on different substrates like the most commonly used media oatmeal agar - obviously the gene(s) governing pathogenicity are no longer needed for fitness in the saprophytic environment of petridish, wherein abundant food is available and that too without any competition.

The boom and bust cycle is well known in disease management. When a new gene governing resistance is introduced through a new variety, the acreage of the new variety increases but at a certain point, pathogen makes inroads in this new genotype and the introduced resistance is breached. Mutation is the key for the change in the pathogen and is responsible for increasing the fitness of the pathogen in that particular environment developed due to the introduced resistance. Without preference for any allele, allele frequency is maintained in the population to the tune of mutation rate. If the virulence allele is more fit than the avirulence allele, its frequency in the wild-type population will exceed the mutation rate. If it is better fitted in the new situation, over generations the virulence allele would eventually cover the entire population (an epidemic situation).

Ideally a successful disease resistant breeding programme is one whose outcome i.e., a new resistant cultivar, is accepted by the farmers and commands substantial acreage, giving the pathogen little scope for survival. It does so without incurring epidemiological mutations in the pathogen. Of the kinds of mutation, epidemiological mutation is the only one in which the frequency of mutation per unit of the pathogen, e.g. per million or billion of spores, depends on the area of cultivation of the host. The pathogen population always remains in a state of flux. Recurrent mutations with local flares of virulence in different loci remain the key in the survival of the pathogen. Most of the flares simply die out like matchstick. Occasionally some flares catch on to become the conflagration called 'epidemic'.

Vanderplank (1963) used illustration for the two possible types of resistance that a host plant can have. Horizontal resistance is expressed uniformly against races of the pathogen; and vertical resistance is expressed differentially. He explained the situation with potato varieties Kennebec and Maritta, and late blight pathogen *Phytophthora infestans*. Both these varieties possess resistance gene R1 (resistance gene is identified on the basis of susceptibility) which confers resistance to race no. 0,2,3,4 and combinations of (2,3), (2,4), (3,4) and (2,3,4). It does not confer resistance to race 1, or any combination with race 1 i.e., (1,2), (1,3), (1,4), (1,2,3), (1,2,4) and (1,2,3,4). The resistance conferred by the gene R1 is expressed differentially against the races and is vertical. Against the races to which the gene R1 does not confer resistance, Kennebec and Maritta behave differently. Kennebec succumbed faster to blight than Maritta. Grown side by side, Kennebec is blighted brown while Maritta is still mainly green. Maritta has more "field resistance" or "horizontal resistance". In the graphical representation where there is no vertical resistance against a race, the horizontal resistance will be higher for Maritta (as it remains green longer) than for the Kennebec. Both forms of resistance coexist, as pathogen does not come alone, comes as an appropriate mixture of races.

Vertical resistance is effective only against the initial inoculum reaching the field. It effectively screens the races that are not able to attack (a hit or miss situation). Thus, only those that can survive in the new environment thrive. Obviously, the second cycle goes with the race that can only thrive at low level and that's where horizontal resistance plays the decisive role in retarding the rate of growth and sporulation. Thus, horizontal resistance is the rule for survival. Plant has to have horizontal resistance; vertical resistance is like icing on the cake.

Based on the evolutionary pattern of crops, ranging from annual to perennial, the resistance pattern varies. The component of vertical resistance available in the annual crops increases, like in cereals, against the pathogens for which inoculum comes from outside (like rust in wheat). On the contrary, in perennial crops, the component of horizontal resistance increases so that the plant can thrive well over the years and is able to face the odd pathogen attack during its lifetime. In fact, it is the environment that plays the crucial role in the predisposition of the plant as well as determines the availability of inoculum in sufficient quantity that can make inroad in the resistance system of the plant.

Types of resistance

Host resistance falls into two broad categories viz. (i) resistance offered by the host at the time of the entry of the pathogen and (ii) subsequent resistance, that the pathogen encounters during the invasion and establishment in the host tissue. In the first category, thickness of the cuticle and epidermal layer (epidermis), cuticular wax, relative abundance of the sclerenchymatous tissue in the stalk, presence of stomata and other openings (like coverings on root primordia, suberization of leaf scar, etc.) determine the level of resistance. The second category comes in the way of the pathogen, retarding its march in the stalk/internodal tissues, through the relative abundance of vascular tissues in the internode or hard, lignified sclerenchymatous tissues in the node, which the pathogen has to overcome for a successful entry and establishment in the cane.

(i) Morphological resistance

The morphological resistance apparently holds some promise for prolonging the life of any cane genotype. Taking the evolutionary ways of development of *Saccharum barberi* and *S. sinense,* (the two natural hybrids of *S. officinarum* and *S. spontaneum* that flourished in India and China over the millennium) it appears that nature has favoured that type of plant architecture, which has certain degree of entry point resistance to red rot pathogen. From the beginning, both *barberi* and *sinense* types are taken as susceptible to red rot on artificial inoculation (Abbott, 1938; Chona, 1954) but they provided much needed support for sugar manufacture in India before the advent of man made interspecific hybrids in 1918, starting with Co 205. In spite of the spectacular success of hybrid canes, these thin indigenous canes commanded considerable area in 1930s and 1940s. Cultivation of these canes was mostly abandoned in 1950s in favour of hybrid canes. In a study, Duttamajumder and Misra (unpublished) have shown that certain pathotypes of *C. falcatum,* when inoculated breaching the entry point barrier, produced susceptible to highly susceptible reaction in Khakai (*S. sinense*); whereas, the same isolates failed to infect the cane stalk in nodal inoculations (where entry point resistance was operative). Obviously, this type of entry point resistance has evolutionary genesis in the prevailing environment of red rot ravaged subtropical India. Physical resistance is a long lasting one, and forms a significant part of the host resistance. Daniels and Daniels (1976) were of the opinion that the noble cane, *S. officinarum* has indeed originated in Indo-Burma region. It could not establish itself in the centre of its origin due to red rot, and drifted towards more congenial environment (south east Asia) where it could flourish well. On the contrary, the tolerant ones viz., *S. barberi* and *S. sinense* types could adapt in this centre of origin due to their resistance to red rot and these have adapted to this environment pretty well.

So far, no breeding programme has been targeted in India and elsewhere to improve this type of resistance (entry point resistance) in cane genotype to increase their field life. Obviously, current knowledge in this regard is limited to harness the benefit of this resistance potential. It is usually assumed by different workers that this type of resistance plays significant role also in sugarcane but so far, no definitive work has been carried out in sugarcane - *C. falcatum* system.

(ii) Physiological resistance/biochemical resistance

The physiological resistance is activated at the initiation of infection process and subsequent development of the pathogen in the host tissue. This is also called active resistance and directly related with the activity of the host. This type of resistance, offered by the standing cane against red rot pathogen is well known. Resistance is also offered by cut canes but with a decrease in efficiency; grades of resistance go down and level of susceptibility increases by 1-2 units in the 0-9 scale of disease rating. The resistance in host-pathogen combination is often ascribed by the higher content of polyphenols, total phenols, gum deposition, PAL and TAL activities and activities of specific compounds like chlorogenic acid, flavonoids, chitinase, etc.

Several host enzymes namely chitinase, β 1-3 glucanase, poly phenol oxidase (PPO), Phenylalanine ammonia lyase (PAL) and Tyrosine ammonia lyase (TAL) have been identified as the key enzymes that govern the production of defence related compounds in sugarcane. PAL and TAL are the key enzymes involved in the biosynthesis of phenolics, phytoalexins and lignins. These enzymes lead to the formation of activated cinnamic acid, which is the precursor of many defence related compounds. Madan *et al.* (1992) were of the opinion that red rot resistance in sugarcane is due to the high specific activity of PAL and TAL. Polyphenol oxidase (PPO) quickly oxidises phenols to highly toxic quinines that arrest the growth of invading organism. The increase in PPO activity is often taken as an indication of resistance - higher the activity, greater is the degree of resistance. Similarly, higher activity of peroxidase (important enzyme in the bio-synthesis of lignin, oxidative degradation of IAA and formation of quinones) was also taken as the indication of higher degree of resistance.

Several workers have reported that resistance against red rot in sugarcane may be linked with the amount of phenols produced by a variety. However, Singh *et al.* (1976) have shown that there is no positive correlation between the total phenol content and the degree of resistance against *C. falcatum*. However, after infection, the quantity of the total phenol gradually goes down over time in susceptible genotypes, as against the maintenance of same level in resistant genotype.

Conflicting reports in literature have marred the explanation of resistance against *C. falcatum* in sugarcane. Most of the popular works on biochemical resistance were undertaken with Co 1148 as the reference variety as it was backed by its commercial success over the years. Co 1148 remained resistant for a considerable period against several pathotypes of *C. falcatum* and eventually fell prey to it with the development of the matching race Cf 01. With the arrival of matching race, all the resistance (both morphological and physiological) fell like a pack of cards. Co1148 could not stand erect for long. This has put a severe rider on the actual role of the biochemical/physiological parameters for ascertaining resistance against red rot. However, these biochemicals are essential for the survival of the plant in the backdrop myriad of microorganisms that it has to ward off, so their role cannot be undermined. Moreover, no directed breeding has been made so far, to increase the phenol content or in the augmentation of biochemical activities leading to the resistant behaviour of a cane genotype.

However, all said and done, it should not be forgotten that Co 1148 has not lost its resistance to any other race for which it was resistant earlier. The understanding of resistance in cane is like the observations on elephant by the blind pupils (in Panchtantra). One is too preoccupied to analyse the things in hosts' perspective to find an explanation, leaving pathogen in the wilderness. That's why only one-sided story is always available in the literature and it is happily served for consumption. The pathogens have evolved as thieves; as they do not remain contained with the plant refuse to lead a saprophytic life. They have learnt the art of stealing the food from living plant cells. Obviously, for survival, they have to steal food (as they do not have the ability to manufacture food), and they cannot compete with the saprobes for the host refuse. Their survival depends on how best they can overcome or breach the introduced barrier (resistance) in the host; it is apparent that they have to modify certain armoury in the quiver to retain the steady flow of food supply by breaching the introduced resistance. In this context, the pathologists and breeders should develop ways and means to restrict the generation of highly aggressive strains (similar to proliferation of nuclear bomb in human society) and develop suitable measures in arresting their damage potential within a reasonable period of time so that any catastrophe may be averted.

Of late, involvement of toxins in pathogenesis has been indicated by a group of workers. The typical red rot symptom (white spot in the earthy red background with sourly alcoholic smell) could not be reproduced either with the crude culture filtrate or by the partially purified toxin. The toxin theory in the pathogenesis involving *C. falcatum* has little bearing with the actual disease. It is the damage of conducting tissues due to nodal rotting that leads to the yellowing of crown and that's why the yellowing is often confused with the damage due to termite, rodent or Gurdaspur borer. The metabolites produced by *C. falcatum* induce a reserved zone in the cane tissues, thus preventing other microorganisms to take over. In fact, the internal parenchymatous tissues of cane stalk that act as the storehouse of sucrose do not offer much resistance except the production of phenolic compounds in response to infection. The defence reaction is quick due to the obvious reason. Nature has stored carbohydrate in sugarcane as sugar (saccharose) for future use, deviating from the ubiquitous polysaccharide starch, the storage carbohydrate of choice. Guarding simple form of sugar is more difficult than the complex starch and therefore, it warrants a mechanism of quick defence. The high activity of defence enzymes in response to any invasion or damage to the cane clearly suggests their active role. Obviously, *C. falcatum* has developed some mechanism to withstand the level of phenolics that hinders the growth of other microorganisms in cane tissues. Similarly, work is also going on for inducing/ boosting resistance of cane plants through the application of bio-inducers like *Trichoderma*, fluorescent Pseudomonads or the application of salicylic acid, jasmonic acid and other chitinase inducers in plants. This approach may come handy in future to combat red rot in varieties that are showing certain degrees of susceptibility to the secondary infection.

However, the moot question remains as to whether this type of biochemical resistance is breedable. Will it be possible to incorporate this type of resistance in the desired combination (genotype)? So far, such type of breeding or selection has not taken place or has not received the desired level of success in sugarcane. It is obvious that the present understanding is inadequate and more in depth study is needed in host-pathogen recognitions and interactions at the molecular level to make a new dent in this arena.

Do the size and type of xylem tissues provide resistance?

Atkinson (1938) came out with an observation that variety Co 290, showing susceptibility to red rot has more number of continuous vessels (vascular bundles) through the nodes as compared to that of CP 29-116, which was resistant. Abbott (1938) also supported the view of Atkinson. The work of Varma and Mital (1949) indicated that the genotypes with more open vascular bundles through the node provide more susceptibility. They thought that like the *Fusarium* conidia, conidia of *C. falcatum* quickly travel through the vascular bundles and cause infection higher up in the stalk. Satyavir and his co-workers and others have also expressed similar opinion in this regard. Unfortunately, these authors failed to appreciate the fact that in the xylem of the standing cane, *C. falcatum* does not sporulate at all. From where will the xylem get conidia for transporting? Even the sporulation in the infected cane starts, that too in the internodal cavities, when the cane has already been damaged severely. Moreover, no genotype is susceptible to all the pathotypes and thus, put a big question mark on the role of running vascular bundles to the susceptiblity of a genotype. The question has no satisfactory answer. As indicated earlier, almost all the genotypes, irrespective of their resistance behaviour at the time of initial evaluation, have fallen prey to red rot later on. Co 1148, a highly resistant genotype of 1970's, with all its morphological and biochemical resistance, succumbed to red rot with the development of a matching pathotype of *C. falcatum*. The toxin theory or the theory of continuous conducting vessels helping passage of conidia is based on the mere circumstantial evidence with no direct relationship.

Breeding for disease resistance in red rot - some reminiscence

In the symposium on red rot in 1954 Sri K.L. Khanna regretted that:

> "... The promises made in 1939-40 to keep down the menace of red rot were short-lived on account of the recurrence of the epidemics. The gradation of varieties with regard to their susceptibility to red rot did not hold long… With the development of new virulent races they could not win the confidence of the Industry, which was not interested in the explanation but the positive results".

Over the years, the painstaking efforts of breeders and pathologists have given definite answer to the 'promise' in this regard. A varietal saturation with a large number of resistant genotypes has reduced the red rot incidence to a great extent and today we have come out of the shadow of epidemic that was looming large on the Indian sugarcane.

In the same symposium, Sri N. L. Dutt, the then Director of SBI, Coimbatore made it clear that:

> "... Actually no breeding for disease resistance was yet under way at Coimbatore and only the seedling progenies were being tested against the disease to determine which parents led to large proportion of disease resistant progenies."

He cited Dr. Brandes, according to whom *" the game was not worth the candle"* as sugarcane is a highly heterozygous polyploid in nature and the inheritance is not Mendelian. The economic crosses effected were being tested for disease resistance before final selection.

Inheritance

The breeding for red rot resistance in sugarcane is like '*catching a black cat in the dark cellar*'. It is more of an art (as well as chance) than the actual planning (science). Sugarcane is a highly complex polyploid crop plant where basic chromosome number (x) is 10 (chromosome numbers of commercial genotypes are usually in the range of 2n=112-128). Sugarcane genome is heavily loaded with redundant (?) genetic load. Due to this inherent complexity, loss or gain of some chromosomes (genetic material) may not always reflect in the phenotype. Moreover, it is on record that different cells of the cane may have different number of chromosomes, thus forming a mosaic of genetic material.

Inheritance of genes for red rot resistance also drew attention of earlier workers in USA and Abbott (1938) furnished the information generated in a preliminary investigation (Table 8.1). It is to be noted that percentage of susceptible progenies was more than 60% for any cross combinations. In the backcrossing, (CP 30/23 is the progeny of Co 281 x US 1964) there was some change in the disease reaction pattern of progenies. In comparison to the parents, nearly 25% share of MR category improved the resistance rating whereas nearly same per cent lost its resistance and became susceptible. However, the inheritance pattern remained inconclusive.

Azab and Chilton (1952, 1959) tried to look into this vexed question of red rot resistance. Cross combinations viz., RxR, RxS, SxS produced resistance genotypes in the progeny. The percentage of susceptible progenies in all cross combinations was above 50% as earlier observed by Abbott (1938). No definite patterns of inheritance emerged out (Table 8.2). The high ploidy level in conjunction with heterozygosity has made sugarcane an enigma to the geneticists, and even after 100 years' of cane breeding very little could be fished out from the virtually unexplored realms of sugarcane genome. Of late, through EST approach (expressed sequence tag) an endeavour is underway in different world centres to unravel at least the expressed portion of the sugarcane genome.

Table 8.1 Behaviour of sugarcane parent and progenies with respect to red rot resistance

Cross	Red rot reaction of parents	No. of plants tested	% R	%MR	%S	%HS
Co 281 x US 1964	R X R	390	4.4	26.2	45.6	23.8
Co 281 X POJ 2878	R X MR	316	21.5	25.0	37.3	16.1
POJ 2725 X CP 1161	S X R	105	16.2	14.3	26.7	42.9
Co 281 X CP 30/23 (Back cross)	R X S	101	8.9	15.8	19.8	55.4

(Adapted from Abbott, 1938)

Table 8.2 Behaviour of sugarcane parent and progenies with respect to red rot resistance

Cross	Red rot reaction of parents	No. of plants tested	% R	%MR	%MS	%S
CP 29-116 x CP 36-105	R x R	110	29.9	24.5	25.5	29.1
CP 36-13 x CP 36-105	R x R	103	10.7	15.5	24.3	49.5
F 31-962 x CP 36-105	R x R	117	40.2	21.4	18.8	19.6
		330	**24.5**	**20.6**	**22.7**	**32.1**
CP 38-34 x Co 356	S x S	127	22.8	13.4	16.5	47.2
CP 38-34 X CP 36-138	S x S	106	26.4	16.0	18.9	38.7
		233	**24.5**	**14.6**	**17.6**	**43.3**
CP 34-120 x CP 36-105	S x R	99	33.3	28.3	22.2	16.2
CP 43-64 x Co 356	R x S	117	23.9	17.9	17.9	40.2
CP 43-64 x CP 33-372	R x S	113	33.6	15.9	18.6	31.9
		329	**30.0**	**20.4**	**19.5**	**30.1**

(Adapted from Azab and Chilton, 1959)

In India, from time-to-time different workers have tried their hand to elucidate the inheritance pattern for the resistance against red rot but with no definite results that may improve upon the present understanding. Alexander and his co-workers from SBI, Coimbatore thought of the introduction of horizontal resistance against red rot involving different resistant parents through conventional breeding. Unfortunately, Alexander and his group met with the same fate – with no clear-cut inheritance pattern or successful cane genotype fortified with horizontal resistance. As discussed earlier, sugarcane for all-purposes, behaves like a perennial crop and horizontal resistance is the rule for its survival. The variety specific pathotype of *C. falcatum* only indicates the success and failure of vertical resistance. Breeders rely on this type of resistance to fortify the new variety (at least as a short term goal) with the hope that it will last much longer in the field. The boom and bust cycle will continue as a recurrent feature. This essentially puts a lot of pressure on the breeder to stay a step ahead of the pathogen in developing resistant cultivars to sustain sugarcane productivity and production. This situation certainly increases the cost of production of new varieties and the cost involved in subsequent replacement and adoption by the farmers and millers.

Chona (1956) was of the opinion that with the complex hybrids used as parents in present-day cane breeding programmes, reaction to red rot is not an important factor in the selection of desirable parents as the degree of resistance of the parents has made hardly any difference in the percentage of resistant seedlings.

Robinson (1962) in the 11^{th} meeting of International Society of Sugarcane Technologists has put forward the apprehension and plight of the breeder and policy planner in this regard. He wrote:

> "Those responsible for the release of new sugarcane varieties are sometimes faced with the decision of what to do with the variety which fall, after an acceptable period of testing, into the category which is neither resistant nor susceptible to a major disease, the control of which is being attempted.
>
> The variety will of course have other favourable agronomic qualities making its extension most desirable and in this case, there is a temptation to exploit these favourable qualities until disease overtakes the variety in question. The conservative view of discarding a troublesome variety outright to forestall disease spread and safeguard other valuable varieties is also put forward for consideration. Most frequently, economic considerations, sometimes justifiable, overrule the latter policy and a further setback is administered to disease control."

Most actions with regard to this type of variety must be judged solely in relation to prevailing circumstances but the wisdom of releasing such a variety has always appeared doubtful to the writer unless the new cane represents a highly significant advance on those, which it will replace.

A close look at the data supplied by Alexander (1995) (Table 8.3) and clubbing the *robustum*, *sinense* and *barberi* with hybrid canes as per the new findings of Irvine (1999) about the non-tenability of these species reveals an interesting picture. It became apparent (Table 8.4) that out of 104 old hybrids (*robustum, barberi* and *sinense*) only 5 (4.81%) were R, 17 (16.34%) were MR, 15 (14.42%) were MS, 16 (15.38%)were S and 51 (49.04%) were HS. The susceptibility picture is also no different in freshly developed ISH hybrids. The corresponding figures are quite similar. However, there is a definite shift in resistance reaction in the time tested old hybrids (recognised as *barberi* and *sinense*) that withstood the rigour of red rot in the subtropical India over the millennium. More of MR type was encountered, with the significant reduction of MS type. It should be noted that the pattern of high susceptibility was around 50% in both the time tested and the new ISH hybrids. It indicated the role of natural selection pressure in shifting the population towards more of tolerance in the intermediate category (MS and MR) rather than imparting any major change in resistance or susceptible pattern (R and S category). Probably physical resistance/entry point resistance has played a decisive role in the stabilization of cane genotypes over the millennium. It is quite likely that due to this entry point resistance Khakai survived in the subtropical belt, in spite of its susceptibility to red rot as indicated by Duttamajumder and Misra in their studies (unpublished). When Khakai was inoculated, some of the *C. falcatum* isolates that produced HS reaction on plug method of inoculation failed completely to get an entry in the cane through nodal method.

Table 8.3 Red rot reaction in different species of *Saccharum*

Group	No. of clones evaluated	Reaction of red rot				
		R	MR	MS	S	HS
Saccharum officinarum	208	6	14	6	23	159
Saccharum robustum	65	2	8	9	9	37
Saccharum sinense	21	2	6	3	1	9
Saccharum barberi	18	1	3	3	6	5
Saccharum spontaneum	315	109	114	26	54	12
Inter-specific hybrids (ISH)						
For major pathotypes of East- Coastal Zone	144	-	1	7	70	66
For major pathotypes of North-Central Zone	234	1	14	48	62	109
For major pathotypes of North-Eastern Zone	238	1	23	51	29	134

(After Alexander, 1995)

In fact, due to this complexity of sugarcane, even today one is not sure about the genes that are governing the accumulation of sugar, the prized product for which one grows sugarcane! So is the case with red rot resistance. Indeed Flor (1942) had shown that a simple inheritance of host resistance genes also makes the inheritance of the pathogenicity genes simple. Complexity in any complimentary system forces the other system to follow suit. A complex inheritance pattern of host forces the pathogen to adapt a complex system to match that of the host, so that it remains a step ahead. After all, the pathogen has to drive nourishment from the host for its own survival. Now the arena is open to the molecular biologists to make a breakthrough in this direction.

Table 8.4 Red rot reaction in hybrids and species of *Saccharum*

Group	No. of clones evaluated	Reaction of red rot				
		R	MR	MS	S	HS
Saccharum officinarum	208	6	14	6	23	159
Saccharum spontaneum	315	109	114	26	54	12
Total	523	115	128	32	77	171
Percentage (species)	**100**	**21.99**	**24.47**	**6.12**	**14.72**	**32.70**
Old hybrids						
Saccharum robustum	65	2	8	9	9	37
Saccharum sinense	21	2	6	3	1	9
Saccharum barberi	18	1	3	3	6	5
Total	104	5	17	15	16	51
Percentage (old hybrids)	**100**	**4.81**	**16.34**	**14.42**	**15.38**	**49.04**
Inter-specific hybrids (ISH)						
For major pathotypes of sub-tropical India	472	2	37	99	91	243
Percentage (new hybrids)	**100**	**0.42**	**7.84**	**20.97**	**19.28**	**51.48**

Disease resistant varieties

> "The horse which wins the derby automatically increases his value as a sire. On a similar thinking, it is often assumed that the best commercial sugar cane varieties should prove most valuable as breeding stock, and the majority of breeding programmes do, in fact, include a high proportion of crosses in which one or both of the parents are successful commercial varieties. A study of breeding records shows, however, that commercial excellence, as a sugar producer is little indication of potential value as a parent. On the contrary, many of the best varieties are bred from parents unsuitable for commercial use."

Thus wrote Stevenson (1965) in his famous book '*Genetics and Breeding of Sugar Cane*'.

The reason for this state of affairs is fairly clear in the light of the modern knowledge of genetics. It is known that varieties best suited for commercial cultivation owe their success in a large measure to heterosis, dependent upon counterpoised genes rather than the parental contributions *per se*. When such varieties are used as parents, these gene combinations are upset by chromosome segregation and recombination at meiosis, and derivatives heterotic to a comparable magnitude as that of the parents may not occur. In fact, in the past

100 years of cane breeding only two *S. spontaneum* forms (Java and Coimbatore) and two *S. barberi* forms (*Chunnee* and *Kansar*) have been effectively used. This awareness of limited genetic base prompted breeders to broaden the genetic base to infuse more variability in the commercial hybrids. Thus, to achieve this goal a large number of ISH clones (inter-specific hybrids) have been developed from the previously untapped gene pool involving different species of *Saccharum* and allied genera.

As mentioned earlier, Dr. Barber emphasised the need of breeding resistant varieties to check the menace of red rot. Over the years, several excellent varieties were bred carrying resistance to the prevalent race flora and were released for general cultivation. However, with the passage of time, most of these varieties have fallen prey to red rot. The variety Co 213, released in 1920s, occupied a large area in subtropical India but the 1938-39 epidemic pushed it out of cultivation. Similarly, boom and bust cycle continued and many excellent sugarcane varieties like Co 312, Co 419, Co 453, Co 997, Co 1148, CoC 671, CoJ 64, CoS 767, etc. succumbed to the red rot pathogen and were phased out of cultivation. In sugarcane, the average field life of a variety hardly goes beyond 10 years. Breeding a cane variety takes up almost equal number of years! The breeders are a bit tensed in this regard. In spite of rigorous testing of clones during the evaluation, and in pre-release stage, it is on record that some sugarcane genotypes caught red rot infection even in the very first year of field life or release for general cultivation!

The pursuit for resistance against red rot in sugarcane is a never-ending one. The task of breeding sugarcane genotypes with a good degree of resistance is cumbersome due to the non-availability of high degree of resistance or immunity in donor parents (Srinivasan, 1965, Stevenson, 1965), occurrence of large number of races and screening a large population is too arduous to embark upon.

The discussion on disease resistance will not be complete without deliberation on methods employed to evaluate the resistance/susciptibility of progenies. A technique suitable to evaluate the progenies satisfactorily for resistance, irrespective of environmental fluctuation, will be appropriate in this regard. Moreover, the technique has to be less demanding on resources and energy and has to be simple enough so that it can be used easily. Like the specific evaluation methods available for different crop-disease combinations, sugarcane-red rot system has the 'plug method', which is mostly engaged in the evaluation of resistance against red rot.

INOCULATION TECHNIQUES

Inoculation technique does not make a major dent in the understanding of any disease, as any technique producing more than 80% disease in favourable environment will suffice for evaluation. Specific inoculation techniques sometimes play a very crucial role in identifying the resistant progeny accurately in any host-pathogen combination. The techniques of inoculation as used for *C. falcatum* need some elaboration. These artificial techniques always remained a cause of disagreement among pathologists and breeders due to variations in the results. Depending on the part of the plant, age and stage of the crop at the time of inoculation, inoculation techniques may be grouped into five categories, viz., *nodal inoculation, internodal, foliage, seedling* and *soil inoculation*.

Nodal method

In nodal inoculation, inoculum is placed on the surface of a node and *C. falcatum* has to overcome the nodal resistance (partially or fully) for entry as well as successful establishment in the cane. This nodal inoculation may be effected with or without removal of the leaf. (i) When leaf remains intact, node corresponding to LTM leaf or 1st /2nd node just below the LTM leaf is used for this purpose. The targeted leaf is pulled a bit to loosen the leaf sheath so that a gap is formed between stalk and leaf sheath, in which the inoculum is placed. One millilitre of conidial suspension in distilled water (10^6 viable conidia/ml) is placed, usually with a pipette, in the space available between stalk and leaf sheath. (ii) When the leaf is removed, cotton swab soaked in conidial suspension is placed on the node and loss of moisture is prevented with coverings. The primary prerequisite of nodal inoculation is that it should be completed at the onset of monsoon. IISR nodal method of Singh and Budhraja (1964) and Parafilm method **(Plate 38)** of Duttamajumder and Misra (2004) are the two important nodal methods to reckon with. In the IISR nodal method, 3rd or 4th node from the ground (basal node) is targeted and after inoculation, a plastic tubing is inserted from the crown to maintain the humidity. In the Parafilm method, 5th or 6th node from the top (crown node), usually coincides with LTM leaf or 1st /2nd node just below (more mature node) is carefully cleaned by removing the leaf. On the freshly cleaned node, a cotton swab soaked in inoculum is placed around the node, which is then wrapped with 1½ inch broad Parafilm strip to avoid any moisture loss. The Parafilm method scores over the other nodal methods by fixing the growth status (tissue condition) irrespective of the age or maturity of cane and by providing adequate moisture for conidial germination and successful infection. The Parafilm covering on inoculum impregnated cotton swab around the node helps to prevent moisture loss. Due to the young tissue, the delineation of susceptibility is very fast, and any cane genotype may be scored for susceptibility in 10-15 days of inoculation, as compared to 60 days, in all other inoculation methods. Moreover, the topple down effect of inoculated canes of susceptible genotypes from the node of inoculation makes the identification of susceptibility very easy even by a layman.

Internodal method

The most popular method of inoculation, even today, is the much maligned 'plug method'. Incidentally Butler and Khan (1913) introduced this method of inoculation. In this method, a hole is made in the stalk (internode), inoculum is placed inside the hole and the opening is sealed with modelling clay. Usually 2nd or 3rd internode above the ground (basal node) is taken for inoculation (to avoid any contamination due to stagnation of water during the rains) as the inoculation is mostly carried out in August-September. This method is too harsh and does not account for any entry point resistance to the pathogen. However, grading of genotypes, assessed through this method, is more consistent. For making hole in the stalk a plunger-type inoculator was developed by Menon and Singh (1960), which became popular and is popularly known as IISR inoculator **(Plate 39 b)**. In Haryana, Virk introduced modified hypodermic syringe for this purpose. To reduce the drudgery of using IISR inoculator, Duttamajumder and Dwivedi (1995) introduced a portable (rechargeable battery operated) hand drill **(Plate 39 c)** to make a hole in the cane. With the availability of different size bits for making hole, inoculation in thin *spontaneum* canes like SES 594 or hard canes like CoLk 8002 became easy and damage or cracking of the cane internode was reduced to the bare minimum.

Plug method remained the method of choice. Accordingly, different workers have tried to improve the efficacy of this method. Due to the problems of inadequate sporulation in different strains/pathotypes of *C. falcatum*, homogenized slurry of mycelial growth along with the media (OMA) was often used as inoculum. Thus, there was no uniformity in the inoculum. Prakasham and his group from Anakapalle tried to standardize the inoculum with definite quantity of conidia sans any mycelium or extraneous material. They used different concentration of spore load and found that a concentration of 10^6 conidia/ml is adequate for evaluating resistance of cane genotypes against red rot (Table 8.9). The use of a pure conidial suspension has been now adopted as a standard practice. In spite of the standardized inoculum load of 10^6 conidia/ml, results often fluctuate. To avoid such fluctuations, an inoculation and evaluation of a minimum 50 canes is recommended in the plug method. Duttamajumder and Misra (unpublished) tried to eleucidate the nature of variation that necessitated inoculation and evaluation of at least 50 canes to arrive at the resistance of a cane genotype. They inoculated

1000 canes of Co 1148 with Cf 01 at one go, so that variations in other parameters are kept at bare minimum. After 60 days of inoculation they observed (Table 8.5) that 85 per cent canes were showing HS reaction, 4 per cent were showing S reaction and 4 per cent were classified in MS category. It came to light that in 5 per cent cane *C. falcatum* could not made any significant progress in producing disease and were classified as MR type and in rest 2 per cent infection was restricted to the inoculated internode and was classified as R type. However, out of these 7 per cent cane showing resistant behaviour also showed the sign of some secondary invaders. These secondary invaders probably checked the proliferation of *C. falcatum* in the cane tissue.

Table 8.5 Variation of disease reaction in plug method (1000 cane inoculation, in Co 1148 and Cf 01 combination)

	Disease reaction				
	HS	S	MS	MR	R
Per cent cane	85	4	4	5	2

Chona (1954) who popularised this method commented that the 'plug method' though harsh will continue, so long as any other effective method of testing of the cane genotypes is developed. Though breaching of morphological resistance is the major drawback, it hardly plays a significant role, as the physiological resistance is considered most important to the cane to fight against red rot. An active involvement of an intact and live host determines the resistance/ susceptibility. Currently AICRP(S) follows two methods of inoculation viz., plug and nodal simultaneously to evaluate resistance of cane genotypes. However, the result of plug method still rules the roost for the recommendation of a variety for general cultivation.

Foliage inoculation

Foliage inoculation was developed by Srinivasan (1962a) in which seedlings of six weeks age were sprayed with conidial suspension in high relative humidity (RH) and favourable temperature. This inoculation killed majority of the seedlings leaving a very few for selection. Similar observation was also made by Duttamajumder (unpublished) by placing *C. falcatum* inoculum grown on Sorghum seed and scattered in the seedling bed at the age of one month. Though it appeared good in reducing high initial population to a manageable level, chance of getting desired varietal material did not improve. Thus, it put a serious question on the killing of seedlings at a very early stage of their life. In fact, Prasadarao (1985) commented that the seedling method of screening was illogical, and he even titled the paper as "*Throwing out the babies with bath water*" to indicate its fallacy. Singh *et al*. (1978) modified this method by increasing the age of seedlings and carried out inoculation in 5-6 months old seedlings in field condition (coinciding with monsoon rains in July-August). Essentially, this inoculation also reduced the number of cane genotypes for evaluation by the breeders and eventually, all the selected genotypes were tested again using plug method. However, it did not improve the chance of getting the desired varietal material. This method also could not gain support, as leaf-resistance and stalk-resistance do not corroborate.

Cut cane technique

The controlled condition testing (CCT) was developed and used by the workers at SBI, Coimbatore in 1992. The limitation of field screening imposed by the germplasm collection at Coimbatore forced them to search for an alternative method of screening that can be carried out in the glass house. In search of a suitable method, this CCT method was developed. It is essentially a cut cane method. Canes of 6-8 months age are cut from field and brought to the glasshouse. These cut canes are placed in poly-bags filled with pot mixture for rooting. Inoculation is carried out using nodal method and inoculated cut canes were kept under controlled temperature and humidity (in the glasshouse). A good correlation with field testing was claimed. Being a cut cane method, the resistance is not always comparable with that of field-testing, as the degree of susceptibility increases by one to two units (0-9 scale) in cut canes.

Grading of resistance against *C. falcatum*

Srinivasan and Bhat (1961) advanced the red rot resistance breeding significantly. They rationalised the grading system for the evaluation of red rot resistance following the plug method of inoculation (Table 8.6). During the Platinum Jubilee Celebration of SBI, Coimbatore in his reminiscences, N. R. Bhat (1987) wrote :

> "On studying the technique employed by the pathologists for testing canes for red rot reaction, it appeared to be unsatisfactory as results of tests were often inconsistent. The inoculum used was a mixture of isolates collected from diseased canes. The criterion used for judging grades of susceptibility was only length. At my instance, both these were changed. Firstly, a pure culture of "D" strain, the most virulent available at that time, was used for inoculation. Secondly, in addition to lesion-length, observations were recorded also on striking symptoms developing in diseased canes of various grades of susceptibility. The study of the data collected on the revised basis revealed that, lesion-width, development of white spots on lesions, and drying of cane tops served as a better guide in judging the reaction. The results were so consistent and convincing that they could be definitely used in formulating a reliable key for grading red rot reaction as indicating resistance, tolerance and susceptibility."

Srinivasan and Bhat (1961) came out with a system of grading sugarcane genotypes (Table 8.6) based on lesion length (node transgressed), lesion width, condition of top and the presence of white spots. They advanced the system from lesion length to well defined criteria with their respective weightages for evaluating resistance. The success of this system is judged by its popularity. Sugarcane pathologists engaged in red rot research use this scale (modified) and it has been adopted as the key system of grading for red rot evaluation (Table 8.7) in All India Co-ordinated Research Project (Sugarcane).

Table 8.6. Suggested key for resistance categories by Srinivasan & Bhat, 1961

Disease reaction	**Characteristics**
Highly resistant (HR)	Tops green; lesion confined to inoculated internode. Serial spots and pith lesions absent.
Resistant (R)	Tops green; lesion crossing one node and tending to remain restricted in width with a sharp, dark red margin (1); white spots absent (0). Serial spots and pith lesions absent.
Moderately resistant (MR)	Tops green; lesion crossing one node and tending to remain restricted in width with white spots circumscribed. Serial spots and pith lesions, when present, non-progressive; nodal necrosis, when present, not tending to taper.
Moderately Susceptible (S)	Tops green; lesion crossing more than one node and tending to spread laterally to a greater or lesser extent (2); white spots circumscribed, or prominent (2) and typically running in a transverse direction. Serial spots and pith lesions, when present, progressive; rarely, nodal transgression nil, but serial spots and/or pith lesions progressive; nodal necrosis, when present, not tending to taper.
Highly susceptible (HS)	Tops yellow or dry. (Lesion crossing a few nodes or more often the greater part or the entire cane and covering the greater part of the entire width of cane; white spots prominent, transverse. Serial spots and pith lesions, when present, progressive, but often not distinguishable due to coalescence of lesions; nodal necrosis, when present, covering entire node; cavities present, in the growth ring of certain varieties.)

(After Srinivasan and Bhat, 1961)

Explanation :
Lesion width : 0 = Lesion confined to inoculated internode, 1 = Lesion sharply restricted in width, 2 = Lesion tending to spread laterally
White spot : 0 = Absent, 1 = Circumscribed, 2 = Prominent
No. of nodes crossed : 0 = 0, one = 1, two to four = 2, more than four = 2
Condition of top : 0 = Green, 1 = Yellow or dry

Observations on lesion width, nodal transgression and white spots are based on the symptoms/reactions expressed in the internode above the inoculated internode.

The top condition is either yellow or green, white spot may be absent or present. When present it may be restricted in development and may be spreading.

Table 8.7. Modified grading system of Srinivasan and Bhat (1961), as accepted by AICRP (Sugarcane)**

1.	Condition of top	:	0 (green), 1(yellow, dry)
2.	Presence of white spot	:	0 (absent), 1 (restricted), 2 (enlarging)
3.	Nodal transgression	:	0 (restricted to inoculated internode), 1 (one node crossed), 2 (two nodes crossed), 3 (three or more nodes crossed)
4.	Lesion width	:	0 (absent), up to 1/3rd (1), up to 2/3rd (2), >2/3rd (3).

*** (0-2.0 =R, 2.1-4.0=MR, 4.1-6.0=MS, 6.1-8.0=S, >8=HS)*
(Minimum number of canes =50)

The spread of infection up to three nodes above the inoculated internode is counted for grading. Similarly, spread of the disease in the cane i.e, maximum width of the lesion in the internodes above the inoculated internode for which nodal transgression was recorded is taken into account (Anon., 1973-2005).

Prasadarao and his group from Anakapalle further advanced the grading system of Srinivasan and Bhat (1961). They delimited lesion width as 1/3rd, 2/3rd and >2/3rd and assigned 1, 2, 3 points, respectively. Similarly, nodal transgression was delimited to three nodes and assigned one point for each node crossed. Thus, they implemented the 0-9 scale. Duttamajumder and Singh (1999) have modified the grading system of Srinivasan and Bhat (1961) to the effect that the nodal rotting was taken as the measure of spread.

Prasadarao and coworkers also proposed a new system of grading based on discriminant function. Prasadarao *et al.* (1978) argued that maximum weightage should be given to white spots as this symptom invariably appears in the susceptible genotypes on inoculation with red rot pathogen. On this basis, they developed an equation/discriminant function.

$$Z = 3.5X_1 + 36.7X_2 + X_3$$

X_1 = Lesion width; X_2 = Occurrence and nature of white spots; X_3 = No. of nodes penetrated

If $Z > 89$ it is taken as susceptible, a value between 42 and 89 places it in the intermediate category and $Z < 42$ is taken as resistant.

Scoring

X_1	*Lesion width < 1/3rd of internode = 1*
	Lesion width > 1/3rd of internode = 2
X_2	*White spot absent = 1*
	White spots surround by necrotic ground tissue of dark colour = 2
	White spots large and running in a transverse direction = 3
X3	*The no. of nodes trangressed by the pathogen.*

This system mainly considers white spots, giving it a weightage of 36.7 with no weightage to the nodal transgression and only 3.5 to lesion width. White spots appear only in susceptible genotypes. Obviously, it becomes a scale to test the susceptibility, not the test of resistance, as evidenced by its skewed nature to white spots. In any disease resistance breeding programme, one is obviously interested in the degree of resistance, not the degree of susceptibility.

Physical barriers are restricted to the entry of the pathogen to the host tissues only, the actual establishment of the pathogen in the cane is determined by the physiological behaviour of the internal tissues. Obviously, the pathogen has to overcome the physiological barrier(s) at the first instance to be successful in leading a parasitic life. Moreover, after the entry into the cane through the nodal region pathogen faces the hostile environment of sclerenchymatous tissues

Table 8.8. Grading system for evaluation of cane inoculated with the plug method for red rot resistance

Criteria for grading	Score (0-9 scale) **			
1) Condition of the top	Green (0)	Yellow/dry (1)		
2) White spot	Nil (0)	Restricted (1)	Enlarging (2)	
3) Nodal transgression (Average of three internode above the inoculated internode)	Nil (0)	Up to 25% area (1)	Up to 50% area (2)	>50% area (3)
4) Nodal necrosis (Node just above the inoculated internode)	Nil (0)	Up to 25% area (1)	Up to 50% area (2)	>50% area (3)

(After Duttamajumder and Singh,1999)
** (0-2.0 = R, 2.1-4.0 = MR, 4.1-6.0 = MS, 6.1-8.0 = S, >8 = HS)
(Minimum number of canes = 20)

Explanation:

1. Condition of top : 0 (green), 1 (yellow, dry)
2. Presence of white spot : 0 (absent), 1 (restricted), 2 (enlarging)
3. Nodal transgression : Average of number of node crossed above the inoculation point (0, 1, 2, 3) x lesion width in the internode above the inoculated internode (0, 1, 2, 3): Here the two characters are clubbed into one to give an area concept i.e. % of area/tissue affected.

$$\frac{\text{Node transgressed X Lesion width}}{3}$$

4. Nodal necrosis : 0 (absent), up to 25% (1), up to 50% (2), >50% (3): This character is added in this method of screening. Observations on lesion width (LW) and nodal necrosis (NN) are based on the symptoms/reactions expressed in the node and internode just above the inoculated internode.

of the node. Until and unless the pathogen is able to overcome this very hardy tissue, it stands no chance to go down the most favoured parenchymatous tissue of the internode and to cause any significant damage to the cane. Thus, the nodal rotting (necrosis) of the cane is the most important criterion on which testing/ evaluation of resistance should be based. Taking nodal rotting as the main criterion, one can explain and minimise the discrepancies arising out of different inoculation techniques, especially the plug and nodal methods. To overcome this dual problem, a grading system within the main frame of Srinivasan and Bhat (1961), was proposed by Duttamajumder and Singh (1999) to take out much of the harshness of the plug method and to provide more realistic grading of the sugarcane genotypes (Table 8.8). In this modified system, horizontal spread of the disease within the cane should be taken on node, not in the internode as nodal rotting is characteristic of red rot, as described earlier. (Top condition =1, White spot = 2, vertical spread =3, horizontal spread at node (nodal rotting) =3). The degree of nodal rotting when the cane was subjected to plug method of inoculation was found to be the indicator of field resistance. This grading system helps to take out much of the harshness of plug method in the sense that many MS grade genotype improve their performance on the count of resistance, which is more realistic.

Inoculum load

It is apparent that even in the most congenial environment of the host tissue certain amount of inoculum load is necessary to initiate successful infection. Theoretically, one conidium can initiate successful infection whereas, in practical terms even placement of 6000 conidia/ml could only produce 8.7% HS category whereas, 6 million conidia (1000 times of 6000) produced 90.7% HS symptom in Co 419 (Table 8.9). It is also to be noted that genotypes differ in this requirement of inoculum load. In case of Co 997 an inoculum load of 6000 conidia successfully produced 74.2% HS symptoms. However, infiltration of setts with different number of conidia gave altogether different picture. Infiltration of conidia through cut ends of two-bud setts revealed that 1 million conidia (10^6), 0.1 million (10^5) registered complete failure of germination

Table 8.9 Relative concentration of spores in the inoculum used in plug method and development of disease

Spore concentration	Per cent cane showing HS category (top drying)	
	Variety	
	Co 419	Co 997
$6x10^6$	90.7	100.0
$6x10^5$	89.8	100.0
$6x10^4$	29.1	88.3
$6x10^3$	8.7	74.2

(Source: 75 years' research at Anakapalle)

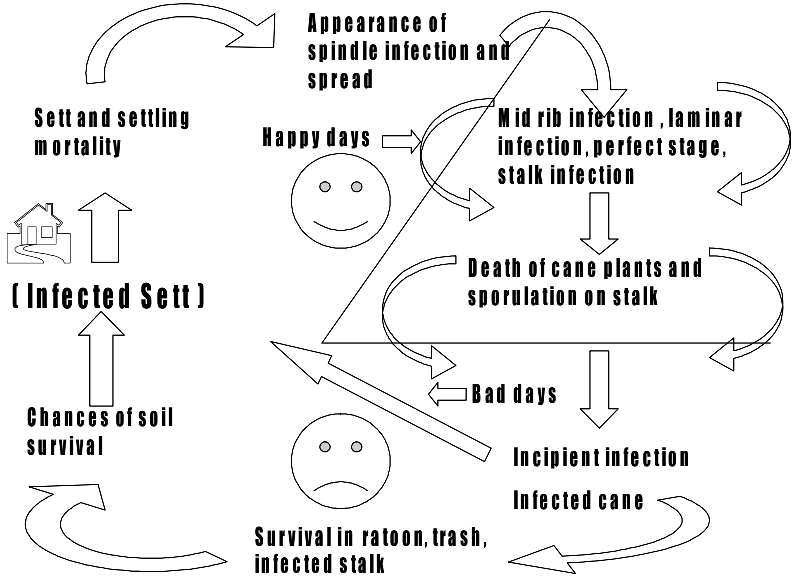
9

EPIDEMIOLOGY

Life cycle and disease cycle

Manifestation of red rot varies depending on the nature of infection, time of the season and the prevailing environment. If sufficient inoculum is present in/on the sett in active form and if adequate moisture is available, it causes both pre- and post-emergence mortality of the sprouts during April-May in the subtropical region. With the advent of pre-monsoon showers, symptoms of the disease start appearing, and with the onset of the monsoon when the weather is most suitable, full manifestation of the disease takes place. Later on, infected plants turn yellow and finally dry up. At the grand growth phase, when sufficient stalk and sugar have formed, typical stalk rot phase (red rot phase) appears. Red rot is not easily identified from the external appearance of the cane unless it has caused irreparable damage by rotting the internal tissues. In such cases, the rind loses its natural bright colour, turns violet or displays dark reddish tinge, and becomes dull in appearance. The affected plants appear sick; usually yellowing of the 3rd or 4th leaf, these being the prominent leaves of the crown, draw the attention of the observer from a distance (The yellowing may start from any leaf of the crown - there is no hard and fast rule). Yellowing of leaf starts from the tip and proceeds down along the leaf margin. *C. falcatum* proliferates within the cane stalk happily, and usually comes out through the root primordia for secondary spread when sufficient rotting of the internal tissues has taken place. The fungus totally transforms the root primordia into black acervuli bearing abundant conidia (the black colour is due to the dark setae). At this stage, if the cane is split open longitudinally, typical symptoms of red rot, viz., reddening of the internal tissues with interrupted red and white patches (white spots) in the affected internodes, rotten or damaged node, and the presence of typical sourly alcoholic smell may

Fig. 9.1. Life cycle and disease cycle of *C. falcatum*



be observed (Butler, 1906; Abbott, 1938; Edgerton, 1959; Singh and Singh, 1989; Agnihotri, 1990). At the final stage of the disease, interior of the stalk darkens and the tissues shrink, leaving a cavity which may be filled with the mycelium, conidia, setae, appressoria, etc. Gradually the affected cane looses moisture and dries up. On the surface of the cane rind, numerous acervuli laden with conidia are formed; these help in the dissemination of the pathogen (Singh and Singh, 1989; Agnihotri, 1990; Duttamajumder, 2002).

The pathogen is a transient visitor of soil and remains viable as long as the infected debris lie undecomposed in the soil (Singh *et al.*, 1977; Agnihotri *et al.*, 1979). Due to its high sensitivity to microbial antagonism, *C. falcatum* cannot withstand the stiff competition with the saprophytes in soil (Butler, 1906; Chona and Nariani, 1952). The role of soil in the epidemiology of red rot is not apparent, as sugarcane is available round the year. Rather, it is the environment that modulates the growth of the pathogen, both in terms of incidence and intensity of the disease. As far as the need of survival goes, *C. falcatum* does not face any situation of non-availability of host in nature, like most of the pathogens of perennial crops. This eventually does not put any pressure on *C. falcatum* to be competitive in the soil environment. Thus, like most of the transient visitors, it also survives in the soil as long as the infected debris is not fully captured by the soil microflora for degradation. Moreover, the inoculum present in the debris is always surrounded (as an outer layer) by the antagonistic microflora which are actively involved in the decomposition of the debris, and thus, do not leave any room for *C. falcatum* to come out and proliferate. Experiments with infected debris by various workers have indicated only limited success of infection from debris and that too when the debris is directly in contact with the cane.

Host-pathogen and environment: the trio of epidemiology and management

Considering the limited role of soil as the reservoir of inoculum, due to stiff microbial antagonism, the role of host becomes far more important in the survival of the pathogen. Obviously, *C. falcatum* has tried to evolve with the host and often behaves like a semi-obligate parasite. In fact, the leaf form (prevalent on mid-rib and leaf lamina) does not cause any noticeable harm in the overall productivity of the cane, nor does it threaten the life of individual cane. Obviously, the major factors that determine the disease appearance and its progress are the environmental factors like relative humidity (RH), rainfall, sunshine hours, dew, temperature, wind, etc. It has been noticed by various workers that the linear spread of red rot in the cane stalk gradually gets restricted with the fall in ambient temperature and the success of infection dwindles during the peak winter, even with the plug method of inoculation. It is obvious that the major role is played by the environmental factors that modulate the infection like any other perennial crop.

In certain period of the year (season) when maximum air temperature lies between 30-35^{0}C and minimum air temperature is around 25^{0}C, *C. falcatum* grows happily and damages the cane rapidly. (Fig. 9.2-9.5). In spite of the crucial role of pathogen-environment combination in the epidemiology of this disease, there is a big gap in the present understanding. A comprehensive study is warranted in this direction. Similarly, the role played by the host in the same environmental fluctuations needs critical evaluation from the standpoint of epidemiology and spread of the pathogen. Still it is not clear how red rot suddenly flares up from nowhere affecting the so-called resistant varieties in a large area. Does the pathogen spread with the seed cane surreptitiously? How does it increases its population for the big day or the type of environment needed to activate the pathogen from the slumber and engineer an epidemic – there are many such questions that need a satisfactory answer.

Sugarcane, being available round the year, the pathogen gets easy access to the major component of host-pathogen-environment system for meeting its food requirement. Therefore, it is left to the vagaries of the environment to modulate the disease incidence and intensity. However, the dominant role is played by the resistance of the host and virulence of the pathogen. Time and again, it has been noticed that red rot is quite variety specific, i.e., only a specific

Fig. 9.2. Weather parameters at IISR farm, Lucknow- year 1995

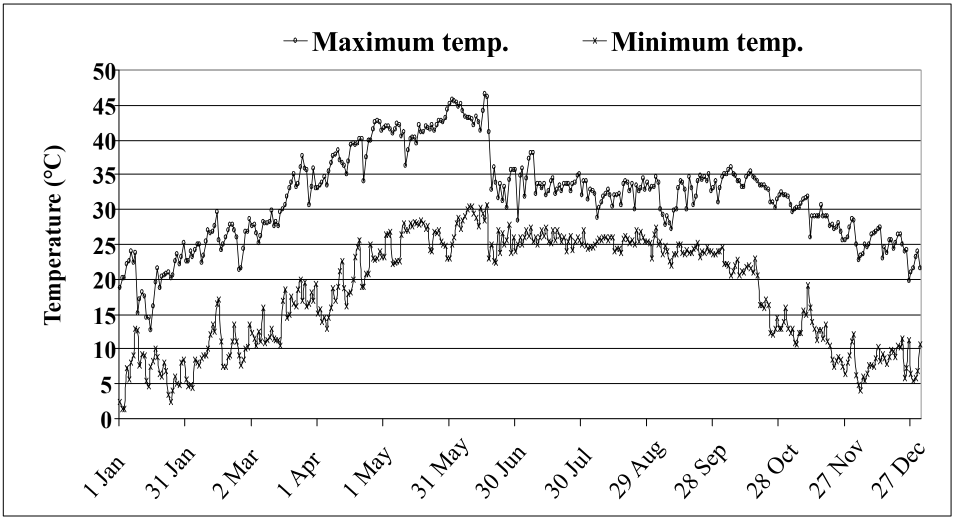


Fig. 9.3. Weather parameters at IISR farm, from July to October 1995

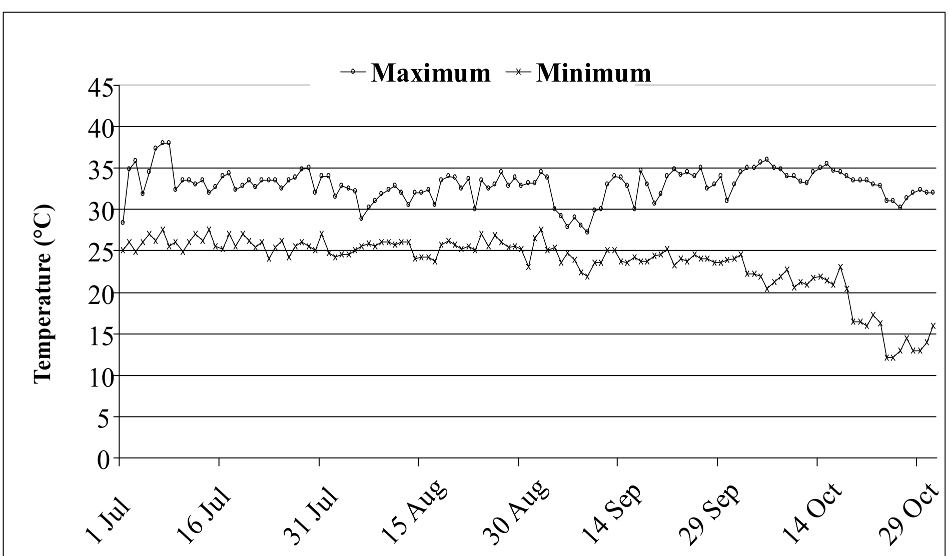


Fig. 9.4. Weather parameters at IISR farm, Lucknow - year 1999

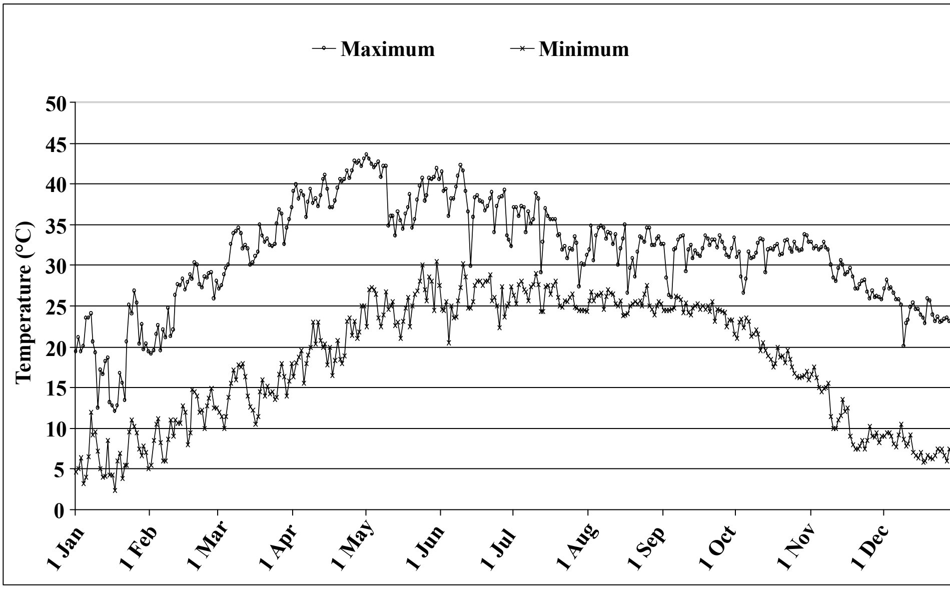
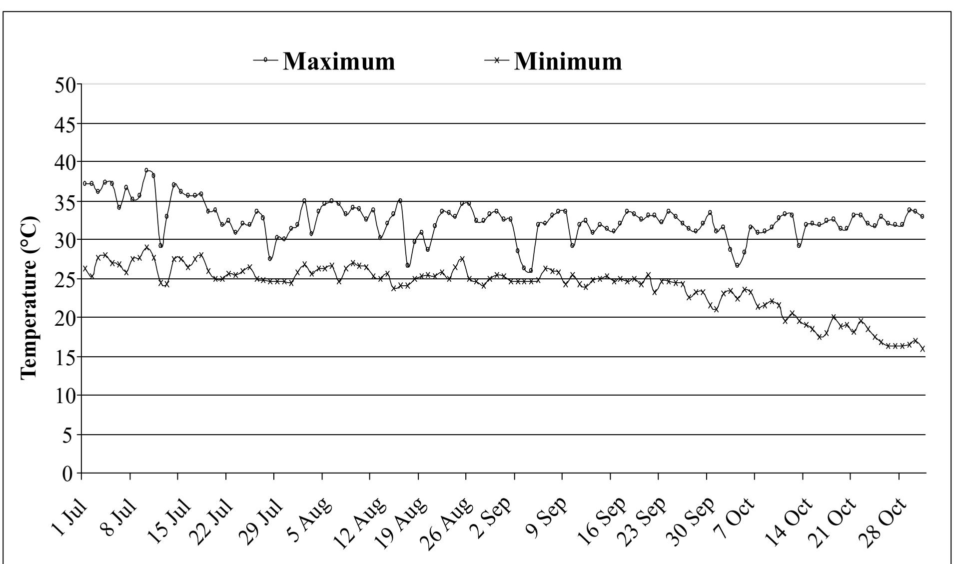


Fig. 9.5. Weather parameters at IISR farm from July to October 1999



variety is affected at any given point of time. Rest of the varieties usually remain free from disease in commercial cultivation. This indirectly helped to formulate deployment strategy of cane varieties as well as reducing the share of acreage for any given variety so that a reasonable cane supply to the sugar mill was ensured even in a bad year.

Varietal resistance against the prevalent races/ pathotypes in a specific geographical area is the primary prerequisite to raise a healthy crop. Secondly, sugarcane is commercially propagated through stem cuttings called "setts"; these setts should be free from red rot infection to prevent the ingress of the initial inoculum through the planting material. Thirdly, fields having immediate history of red rot should be avoided for cane planting. Although the soil inoculum does not pose any immediate threat to the crop, it surely produces contaminated planting material, which shows the effect later.

Spread of the pathogen in the crop

Sett, the planting material, remains the focal point of carrying infection to the field as the primary inoculum. The latent infections are the key infections that defy detection at any level, reach the field with sett, and express later on. If the planting material is heavily infected, it rarely germinates, as *C. falcatum* damages the nodes badly, which in turn kills the bud and root primordia. The data of an experiment conducted in 1950's by Rafay and Singh (1959) are given in Table 9.1 to highlight the situation. Maximum number of diseased cane was developed from the apparently healthy cane (cane with no reddening at the cut ends). A recent observation by the author on the outbreak of red rot in CoJ 82 in Barabanki (U.P.) has reinforced the observations of earlier worker on two counts

Table 9.1 Appearance of disease in the crop planted with apparently healthy, partially diseased and diseased setts

Condition of setts (two bud)	Year	Buds			No. of millable canes	No. of diseased cane	Per cent diseased
		Planted	Germinated	Germination (%)			
Both end diseased	I	40	4	10	6	3	50
	II	40	7	18	10	7	70
One end diseased	I	40	19	48	21	3	14
	II	40	17	43	24	9	38
Apparently healthy from diseased plot (incipient infection)	I	40	28	70	36	12	33
	II	40	25	63	39	6	15
Control (from healthy plot)	I	40	30	75	36	-	-
	II	40	29	73	44	-	-

(Adapted from Rafay and Singh, 1959, work carried out at Pusa, Bihar on Co 299)

viz., (i) the importance of seed-borne infection (in the introduction of inoculum from other region as well as its effectiveness in spreading the disease), and (ii) early appearance as well as flare up of the disease in the congenial microclimate. On tracing the source of the seed-cane, about 10-12 per cent incidence of red rot was noticed in the plots contributed by the seed-cane (now a ratoon crop) in August. The seed cane was initially brought from western Uttar Pradesh and was multiplied for a year. From this multiplication plot, seed-cane was distributed to five farmers spanning a distance of more than 50 kilometres. In the farm of one progressive farmer the disease flared up and the entire crop was affected. On the other hand, in fields of rest of the four farmers, disease incidence hardly exceeded 10-20 per cent. Lack of adequate irrigation was noticed in three locations and incidence of termite was observed in the other. In the termite affected crop the disease incidence was minimum (around 10 per cent). It was apparent that the crop that got better irrigation and agronomic management in the early phase of growth favoured the secondary spread of the disease. It may not be out of place to put on record the experience that the author has gained from various cane disease surveys conducted in the past 22 years. The red rot was frequently noticed in the fields of poor farmers who did not have enough resources for seed replacement and continued to grow the old susceptible varieties that had long been denotified by the State. Similarly, red rot was also noticed in the fields of progressive farmers who took good care of the crop and introduced different cane genotypes from several locations for evaluation. Red rot is less common in the fields of average farmers who are intermediate between these extremes.

In order to assess the significance of secondary infection, Mr. Rafay laid out an experiment in March 1954 by planting setts from red rot affected canes of Co 453 in a row (as infector row). The two adjoining rows had the healthy setts of Co 453. In the first week of July 1954, three clumps in the infector row had the spindles affected with red rot. In September, stalk infection was noticed in both the diseased and adjoining healthy rows. Until harvest, primary infection (directly developed from infected sett) was observed in 23 cases and secondary infection (where such direct connection was not apparent) was noticed in 198 stalks (Anon. 1953-2005). Similar observations have been made by later workers with different variety-pathotype combinations in quite different geographic locations. In a similar experiment, Duttamajumder and Misra (unpublished) observed that the inoculum developed as a result of spindle infection accounted for majority of the secondary spread. Such a spread usually covered a distance of 1.50-1.80 metres from the cane with spindle infection. It will not be out of place to mention that the young shoots bearing spindle infection die out quickly, and these often become the favourite food of *Trichoderma*, basidiomyceteous fungi, bacteria and termite.

Mr. Rafay also monitored the mid-rib lesions during this period. The mid-rib infection was noticed in June as minute dots on the upper side of the mid-rib of leaves. From July to October, the lesions increased and developed a straw coloured central region with innumerable dark dots consisting of acervuli of *C. falcatum*. They were formed on the mid-ribs of almost all the 305 varieties used in the experiment. Cultures taken from leaves of all varieties having red rot lesions on the mid-rib produced 'dark type' cultures except Co 956 and Co 995, which produced the 'light type' (probably spindle infection was taken as mid-rib infection). It was significant that no stalk infection was recorded in any variety despite the incidence of red rot lesions in leaves of a large number of varieties. The two varieties, i.e., Co 956 and Co 995 which yielded light type isolate also had stalk infection (Anon. 1953-2005).

Limited information is available on the role of environmental factors aiding the spread of the red rot pathogen in the crop and its role in the development of an epidemic. In 1970's, Sreeramulu (1973) presented a lone paper on sugarcane disease epidemiology regarding the dynamics of spread of conidia. He wrote that:

> "... conidia of *Colletotrichum falcatum* were not caught in appreciable numbers in the Hirst trap (located at heights both within and above the crop) on most days even when the occurrence of innumerable number of profusely sporulating acervuli were present. Very few conidia were caught on the slides, only on few days, and an analysis of these catches indicated the possibility of the existence of a 'nocturnal pattern' in the daily rhythm with the maximum about at 8 PM".

The 'nil catch' records on most of the days imply that there is no conidial presence of the *C. falcatum* in the air over the sugarcane field. On the contrary, Singh and Lal (1996) thought that the spread of the conidia and dried acervuli freely took place in the winter months, and played a significant role in the epidemiology of the disease as numerous acervuli appeared on the surface of dried red rot affected canes during the winter months. These authors have failed to appreciate the fact that the conditions favourable to cause infection are not available during the winter months, and lying on the soil surface for the favourable weather to come is not the cup of tea for *C. falcatum*. Even if the dry conidial matrix spreads in the crop canopy, it does not pose any immediate threat to infect the cane crop, even to the newly emerging shoots.

Satyavir and co-workers worked on some aspects of the disease development in relation to weather parameters in Haryana. They have reported a minimum temperature of 25.5 to 26.6^{0}C and an evening relative humidity (RH) of >60 per cent as the two important parameters in disease development. Initiation of infection was negatively correlated with maximum temperature (r= -0.65 to -0.81) and morning RH; rainfall had a direct bearing on the successful initiation of infection. Similarly, the rainfall was found mainly responsible for the secondary spread of the pathogen.

The most damaging phase of the crop occurs only during the late monsoon, emphasising the importance of rain, cloudy weather, moderate temperature, etc. in the development and spread of the disease. A look at the temperature of any particular year reveals that the period of July-August-September, is the critical three months period when red rot has its field day, when the gap between maximum and minimum temperature is narrow, and relative humidity (RH) for greater part of the day remains 80% or above in most of the period (Fig. 9.2-9.5).

Again, during these three months, a cloud cover for a considerable period provides needed ambience in the microclimate; a common feature in the subtropical India. Red rot is a nagging problem in eastern Uttar Pradesh and Bihar where flooding or waterlogging is common. The accentuated susceptibility of cane plant is due to the predisposition in the above situation. Waterlogging forces the production of aerial roots, and as early as 1906, Butler had pointed out that these emerging roots act as the conduit and catch the infection of *C. falcatum* easily.

Irrigation and rain/flood water play a very important role in the dissemination of the red rot pathogen. Rain splash on the spore mass developed on spindle leaves, mid-rib and the nodal region including rind takes the aerial route and spreads with micro-droplets of water carrying conidia (Duttamajumder, 2002). The excess water also runs down taking the spores in its course and reach the ground. At this time, any running water helps the spread of the spores from one field to the other. Late in the season, enormous sporulation also appears on the surface of dead plants, and if weather favours dispersal of this inoculum, it often initiates infection which mostly becomes latent. Latency is the hallmark of the genus *Colletotrichum*. Due to this reason, in autumn planting (October), there is every chance that red rot infected cane (latent infection) will be planted, as there is no chance of detecting the latent infections. In fact, it has been proved beyond doubt by Dr. Kishan Singh and co-workers that in red rot prone area, October planting favours more incidence of the disease in comparison to the spring planting (February-March). Due to this reason, the avoidance of autumn planting is often recommended in the red rot prone areas.

Collateral/Alternative hosts

Collateral hosts play a significant role in the epidemiology of any disease and in many cases these act as the stepping-stone for the pathogen. Sorghum, whether cultivated or wild, readily supports the growth of *Colletotrichum falcatum*.

Moreover, *Sorghum halepense*, a common weed is often encountered in the cane fields and the adjoining area. Different species of *Saccharum* also harbour the pathogen. *Saccharum spontaneum*, though serves as the resistant donor for red rot resistance, also harbours leaf infection of red rot pathogen in good number. Similarly, *C. falcatum* can also survive on other grasses like *Leptochloa*, *Miscanthus*, etc., and thus maintains the innoculum in active form. Red rot inoculum from these grasses reaches the sugarcane crop through wind, wind-borne rain, splattering rain-drops, floodwater and running water and may initiate infection in the crop. However, these sources remain more cosmetic in contribution of inoculum and rarely play any significant role in this regard.

Soil survival

As mentioned earlier, Butler tried to study the development of infection of the cane through the soil route. He could not produce an infection when inoculum was placed in soil. *C. falcatum* could not last long in the moist soil environment. Before the epidemic of 1938-39, the role played by the soil in assisting *C. falcatum* for perpetuation and causing new infection was not taken seriously. Dr. Chona was of the opinion that in some unexplainable situation, soil may be acting as the source of inoculum (Chona, 1980).

The role of *C. falcatum*, perpetuating in soil and initiating new infection is yet to be elucidated unequivocally. Longevity of *C. falcatum* in soil, inoculated artificially in October and kept in the open at Pusa, was for over seven months, as against eight months in the laboratory (obviously 4 months i.e. Nov., Dec., Jan., and Feb. are the key winter months). On the other hand, during summer its mycelium could not live longer than two months in the open and 2½ months in the laboratory. Spore suspension in distilled water, stored in daylight at room temperature was viable for 2½ months. In oatmeal agar medium (OMA), mycelium survived under similar conditions for 6-7 months, while in a refrigerator (5^0C), it survived for 7-8 months (Khanna, 1943). Chona and Nariani (1952) found mycelium and spores to be viable for over 3 months in unsterilized soil and

Table 9.2. Survival of *Colletotrichum falcatum* in the affected stalk pieces buried in soil

Burial depth (cm)	Survival in days	
	Winter (1973)	Autumn (1974)
5	63	34
10	63	16
15	63	34

(After Singh et al., 1977)

soil-compost mixture and for 5 months in the former and 6 months in the latter when they were sterilized. In compost, the fungus was not found to survive for more than one month. This shows that the mycelium is unable to hold on its own against other microorganisms, which offer a keen competition (Chona and Nariani, 1954).

Another curious point one needs to ponder on is the very low survival of *C. falcatum* in tissues of cane showing HS reaction, after the death and drying of cane. After two weeks of harvesting of such canes, recovery of the fungus gradually went down (correspondingly with the gradual loss of moisture) and after one month, it became exceedingly difficult to isolate the pathogen. This situation was encountered both in late and early part of the season. In the early part, *Trichoderma* sp., *Fusarium* sp., and a species of basidiomyceteous fungus follow *C. falcatum* from behind and overtake the infected area very quickly, leaving no room for the red rot pathogen. Similarly, late in the season, *Fusarium*, *Nigrospora, Chaetomium,* the rind disease pathogen (*Phaeocytostroma*) and fungi belonging to basidiomycetes overtake the red rot damaged cane (harvested cane and kept under laboratory condition in December-January-February).

10

DISEASE MANAGEMENT

Before any attempt is made to manage a disease, it is imperative to go for the SWOT analysis to understand the Strength and Weakness of any host-pathogen system and to find out where the Opportunity lies and how to guard against the Threat. The management of red rot is no exception to this general rule. The pathogen and host both have their respective strengths and weaknesses which one has to account for in developing any management strategy. These may be outlined as follows:

Strength

1. It is essentially a sett-borne disease.
2. Only during the grand growth phase the disease is highly damaging.
3. Secondary spread is not very efficient and it hardly completes one or two disease cycles in the season. Red rot is primarily a compound interest disease, but mostly behaves as a simple interest one, in case of stalk infection (sporulation on cane stalk appears after the death of the cane, late in the season and available temperature and humidity in the winter months do not allow establishment of successful infection). Secondary spread from the inoculum developed in the spindle infection during the early part of the crop growth is efficient but the new infection rarely results in the development of another round of spindle infection and sporulation.
4. The pathogen is a transient visitor of soil and survives in soil as long as the debris remains undecomposed on the soil surface.
5. Spores and mycelium of *C. falcatum* are favourite food of mites and do not survive long in nature.

6. Young affected shoots are often killed/devoured up by termites, *Trichoderma*, bacteria, basidiomycetes, etc., thus put a natural check on the sporulation and spread.

Weakness

1. Non-availability of immunity/ high degree of resistance in the available sugarcane germplasm.
2. So far, no chemical has been identified that can effectively eradicate the infection from the cane or cane sett.
3. Sett treatment is difficult due to bulky nature of seed-cane and impervious nature of rind prevents the entry of fungitoxicants in sufficient quantity to the target site.
4. Spraying in standing sugarcane crop is not a feasible option.
5. Monoculturing, ratooning, round the year availability of the cane crop as well as often growing of the crop in a large contiguous area under a single variety in factory command area.
6. Poor knowledge on genetics of sugarcane and basis of red rot resistance.
7. High variability of *C. falcatum*.
8. Difficulty in introgressing varied sources of resistance from the wild species and quickly incorporating it in the commercial varieities.

Opportunity

Farmers, millers and policy planners are aware of the problem and are responsive to any measures to contain the disease.

Threat

1. Complacency — at the level of farmers, millers and policy planners.
2. Inadequate supply of good quality seed-cane and lack of strong seed programme.
3. Practice of ratooning, a built in mechanism that aids in increasing the disease severity and spread.
4. Once affected, the crop cannot be saved.
5. Unrestricted movement of cane.

Management of the disease: available options

At present, control measures in plant pathology are predominantly preventive. In sugarcane, heat therapy is quite effective, and in many cases, it is the only recommendation that can put a number of diseases under check (Singh and Agnihotri, 1987; Duttamajumder and Agnihotri, 1992, Duttamajumder, 2002). In sugarcane, due to the industrial nature of the crop, it is difficult to practice the principles of avoidance for all practical purposes, as sugarcane has to be grown in the mill-reserved zone to run the sugar mill. Avoidance, in some situations, should be forced to prevent the spread of red rot e.g., in low lying and waterlogged areas. Moreover, it has been observed that stress in the form of submergence favours *C. falcatum* to consolidate available variability through fusion of conidia (Duttamajumder *et al.,* 1990). Appearance of spindle infection is concomitant with early rains (May-June) and water stagnation helps in quick spread of the disease.

Quarantine

The unrestricted movement of the seed-cane from one district or state to another has largely been responsible for the migration of important disease pathogens like *C. falcatum* in the past, and such spread is continuing surreptitiously. Earlier, pathogens had spread to the new areas and now virulent pathotypes are moving to new pastures (Duttamajumder and Agnihotri, 1992; Agnihotri and Duttamajumder, 1994; Duttamajumder, 2002). Therefore, legislation as well as its strict compliance is essential to curb spread of the disease. The sugar mills have to take necessary precautions in the letter and spirit before introduction of any genotype from other locations. This unrestricted introduction/movement of cane has been responsible for the spread of red rot. To cite a recent example is the introduction of CoJ 82 from western part to eastern part of U.P. (Barabanki). Red rot flared up in the new environment and caused a wholesale damage **(Plate 2)** or for that matter the indiscriminate introduction of seed-cane of CoC 671 to Gujarat from Andhra Pradesh and elsewhere was responsible for the colossal loss due to red rot.

Seed selection and field sanitation

Clean cultivation is the primary pre-requisite for growing a healthy sugarcane crop. This starts with the selection and planting of healthy seed-cane in a healthy field. This practice reduces the load of initial inoculum and helps in checking the red rot. Timely removal of diseased plants in the early part of the season (May, June, spindle infection), as and when detected, also helps in reducing the inoculum available for the secondary spread. Removal of weeds also helps in this direction, as this practice checks the growth of alternative hosts supporting the active growth of *C. falcatum*. Moreover, a weed free field boosts crop growth and ensures better harvest.

At the Samalkota farm (in Andhra Pradesh) sett selection was regularly carried out by Barber to check the red rot. Any sett with running red vascular strand or red internode was rejected outright. Barber was perhaps the first worker in sugarcane, to advocate this method of sett selection for fighting red rot menace (Butler and Khan, 1913). Sett selection reduced the incidence of red rot substantially but was unable to check the disease completely (Butler and Khan, 1913) and the disease continued until the variety was replaced with more resistant one. This practice of sett selection should be religiously followed from the very beginning of the field life of a new genotype, if one really desires to check the damage from red rot. Many a times it has been observed that whenever seed consciousness took a back seat due to some reason or the other, everybody from the tillers to the millers, had to pay a heavy price due to red rot. Several red rot epidemics that hit the Indian sugarcane, from time to time, bear this out.

Physical treatment: the viable option

Sugarcane, a vegetatively propagated crop, responds well to the use of physical energy in the form of heat therapy/treatment. In this treatment, seed-cane is exposed to a critical temperature that is lethal to the pathogen but not to the sugarcane buds, and thus it effectuates a differential killing of the pathogen. Incidentally, Kobus introduced heat treatment in sugarcane to get rid of the '*Sereh*', in Java. Use of heat energy has been tried in various forms (hot water, hot air, moist hot air or aerated steam) with varying degrees of success against red rot (Singh, 1973; Singh *et al.*, 1980; Natarajan and Muthuswamy, 1981). Heat treatment can root out fresh and superficial infections very well, but it becomes largely ineffective against the deep-seated and dormant infections present in the nodal region of seed cane (Srinivasan, 1971; Singh and Agnihotri, 1987). Complete elimination of red rot pathogen was not achieved by the MHAT treatment at 54°C for 2½ hours (actual treatment time), however it significantly reduced the number of surviving propagules/inocula causing infections later in the season (Singh *et al.*, 1980). Curative property of hot water treatment may be increased by adding the desired fungicide which can tolerate the temperature of the water or by using a post-fungicidal treatment (Singh and Agnihotri, 1987).

It will not be out of place to mention the controversy that engulfed the red rot researchers in the 1970s regarding the effectiveness of heat treatment in checking the sett-borne infections of *C. falcatum*. Under the guidance of Dr. D. N Srivastava, the then ADG (PP), ICAR trials were conducted under the AICRP(S) at Pusa, Shahjahanpur and Lucknow (for subtropical environment), Anakapalle and Coimbatore (for tropical environment) to resolve the matter. For heat treatment, both MHAT and HWT were used. From the outcome of these trials, it was concluded that both the heat treatment procedures were effective in checking the red rot, but cent per cent eradication was not possible.

However, it should be borne in mind that these trials were conducted with full blown red rot affected canes and full blown diseased canes never get a chance in planting. In a low level of infection, efficiency of heat treatment increases substantially and thus, it was recommended to continue with the practice of heat treatment of seed-cane. Moreover, heat treatment has other benefits also; it checks sett-borne infections of smut, GSD, RSD and leaf scald. Nowadays, it is mandatory to provide heat treatment to any parental material to be used for seed production and distribution programmes in sugarcane.

Chemical treatment: a piquant situation

The primary aim to study any disease is to devise ways and means to keep the pathogen under check so that its incidence and intensity as well as the resultant effects remain below the economic injury level. The success of some agrochemicals in controlling plant diseases provided necessary impetus and opportunity for a better management of the crop. However, it should not be taken as the panacea for all ills. The success of chemicals in other crop diseases has given an impression that the plant protection chemicals will also prove effective in controlling red rot of sugarcane. This is not so. Red rot is one such disease where the pathogen has defied present day fungicides very well. Though these fungicides are effective in checking the fungus *in vitro*, they fail miserably in checking the pathogen in the cane tissues. This piquant situation is often attributed to the lack of inhibitory concentration of fungicides at the target site to effect any killing and eradication of the pathogen. Different workers have tried their hand to workout a suitable chemical control with available fungicides, but with no appreciable success. From a mixture of limewater to dithiocarbamates, through organo-mercurials to the systemic fungicides – everything, that was available or that the workers could lay their hands on, were tried to achieve a control of this disease in the field but with not so happy results.

Kirtikar and Verma (1963) reported that dipping healthy setts of red rot susceptible variety Co 331 in mercurial, sulphur and copper fungicides of different strengths for 10 min did not provide protection against red rot infection. Chand *et al*. (1974) reported that one-hour treatment of setts of Co 312 with Benomyl (0.25%) increased the germination over control by 50% and reduced the incidence of red rot. Anzalone (1970) reported control of red rot with Benomyl (100-200 ppm) when treated under pressure.

Table 10.1 Effect of different fungicides on the recovery of *Colletotrichum falcatum* after soaking of artificially infected single bud setts in fungicide solution/suspension for 24 hours

Fungicides	No. of sett germinated					No. of setts produced dis. plant
	Healthy sett (25)		Diseased sett (25)		Recovery of fungus (%)	
Bavistin (0.2%)	24	96%	10	40%	81.3	2
Topsin M (0.1%)	21	84%	4	16%	75.0	1
Bayleton (0.2%)	20	80%	7	28%	68.8	1

(Adapted from Rao and Satyanarayana, 1995)

Pandey and Agnihotri (1996); Rao and Satyanarayana (1995) tried to combine the effectiveness of heat therapy (hot water treatment) with fungicides. The former authors indicated almost complete control of red rot with fungicides like Bavistin, Benomyl, Topsin M, Aretan at 0.1% level when added to hot water tank and treated at 50^0C for 2 hours. Rao and Satyanarayana (1995) reported effective control by treating single bud setts both at 50^0C for 2 hours or at 52^0C for 30 min along with the treatment of Bavistin 0.1%. However, fungicidal treatment alone was only partially effective in controlling sett-borne infection of red rot, indicating effectiveness of this combination of both the treatments. As a standard practice, hot water treated setts are further treated with fungicides to guard against the ingress of opportunist microorganisms in the sett through the

Table 10.2 Efficacy of hot water treatment and fungicidal treatment in control of sett-borne infection of red rot in single bud setts

Treatment	Per cent tissues yielding red rot fungus			Percent germination	Per cent germinants exhibiting red rot
	Internal tissue	Leaf scar	Bud		
Hot water 50°C 2 hours	0.0	0.0	0.0	16.0	0.0
Hot water 50°C 2 hours + Bavistin 0.1%	0.0	0.0	0.0	15.0	0.0
Soaking setts in 0.1% Bavistin for 2 hours	26.6	13.3	8.0	28.0	28.5
Check (setts soaked in cold water)	80.0	53.3	66.6	8.0	100.0
Hot water 52°C 30 min	0.0	0.0	0.0	40.0	0.0
Hot water 52°C 30 min + Bavistin 0.1%	0.0	0.0	0.0	48.0	0.0
Soaking setts in 0.1% Bavistin for 30 min	40.0	26.6	33.3	25.0	60.0

(Adapted from Rao and Satyanarayana, 1995)

cut ends. This treatment is essential as the natural microflora of the sugarcane, which put a check on the various other microorganisms, also get badly affected by the heat treatment and thus, the hot water treated setts need extra protection in this regard.

Several fungicides, both systemic and non-systemic (e.g., Bavistin, Vitavax, Topsin M, Aretan, Blitox, Dithane Z-78, Dithane M-45, etc. are effective, *in vitro,* in killing *C. falcatum* but they do not provide satisfactory control of the red rot disease under field conditions (Kirtikar and Singh, 1964; Chand *et al.*, 1974, Narasimharao and Satyanarayana, 1974).

Divergent views are found in abundance, both in support and against the effectiveness of the chemical fungicides in controlling red rot in seed pieces and standing cane. Better crop stand was achieved in many cases by treating the setts with benomyl, carboxin, carbendazim, agallol, etc. and probably this was due to better germination and establishment of the crop but not because of the actual control of the pathogen. So is the case with the application of fungicides in the standing crop. Different fungicides were able to check the mid-rib infection well but failed miserably in checking the stalk infection. Thus, chemical control was forced to take a back seat in the management of red rot. Of late, certain chemicals (like salicylic acid, jasmonic acid, etc.) have come to the forefront for inducing systemic resistance in cane plant. However, these are now at the preliminary stage of experimentation to prove their worth against red rot.

The possible reasons for the ineffectiveness of the fungicides, as suggested by Duttamajumder and Agnihotri (1992) are as follows:

i) Impervious nature of the cane rind prevents the entry of fugitoxicants.
ii) In case of limited entry, only sub-lethal amount of fungitoxicant reaches the target site. This makes the fungicide ineffective in eradicating the red rot pathogen from cane tissue.
iii) Lack of effective water-soluble systemic fungicides against *C. falcatum* that can easily enter the cane sett and preferentially kill the fungus at the target site.
iv) The toxic effect of the fungicide is effectively annulled by the abundant nutrient present in the sett.

Three-tier seed programme: a sound option

Based on the success of MHAT (moist hot air treatment, at 54^0C for 2½ hours; actual treatment time) in checking sett-borne pathogens of sugarcane, including red rot, a three-tier seed programme was launched for production and distribution of healthy seed-cane to the sugarcane farmers (Singh, 1983). This heat treatment essentially eliminates/reduces the level of initial inoculum present in/on the sett. This programme has three well defined stages, viz., (i) Breeder seed, (ii) Foundation seed, and (iii) Certified seed. It starts with the production of breeder seed that gives rise to foundation seed and in turn certified seed is produced. The crop raised from heat-treated cane may be quickly reinfected by the pathogens. Therefore, a strict vigil is necessary to guard against the possibilities of reinfection and to maintain freedom from diseases.

The research carried out at the Indian Institute of Sugarcane Research, Lucknow and elsewhere suggests that there is no single method available that can manage red rot to one's complete satisfaction. Due to this reason, red rot is often termed as a '*managerial disease*', and in pathologists parlance it is often remarked that '*what cannot be cured has to be endured*'. In fact, from the very beginning of red rot research in India, it responded well to various management practices. The disease surfaced in a big way, only when major lapses (knowingly or unknowingly) occurred. An integrated approach is a must to contain this pathogen. Resistant varieties remain the most practical option to combat this disease, even though *C. falcatum* is breaching the introduced resistance with impunity in a regular fashion. The integrated approach currently recommended to achieve this goal is as follows:

i) Selection of the field having no immediate history of red rot. Preferably, the field should be well drained/ upland type.

ii) Selection and use of healthy seed-cane.

iii) Where ever possible, the seed-cane should be subjected to MHAT.

iv) Chemical treatment of seed-cane with a fungicide like Dithane M-45, Bavistin, etc. to protect against seed-borne as well as soil-borne inoculum and also to protect the setts from secondary infections after MHAT.

v) Application of bio-agents like *Trichoderma,* as sett- as well as soil- treatment, at the time of planting and also on the trash mulch, used in ratoon crop for moisture conservation, helps in faster degradation as well as production and release of chemicals inhibitory to *C. falcatum*.

vi) Roguing of infected stools (entire clump) and spot application of fungicide as and when disease is detected (pre monsoon) is necessary to reduce the level of available inoculum in the field and to remove the focus of infection.

vii) Discouraging the practice of ratooning, if any incidence of red rot is observed in the plant crop.

11

EPILOGUE

No doubt, impressive work has been done on many aspects of the red rot disease in India, yet there are many facets in the epidemiology and management of the disease that need critical studies.

In epidemiology, the role of perfect stage remains a pathologist's enigma. How is the sexual reproduction in *C. falcatum* contributing to the generation of new variability or development of new pathotypes and in the aerial spread of the pathogen covering a large area ? Where does the inoculum come from to cause the mid-rib infection in the pre-monsoon period? In addition, how does this mid-rib infection contribute in the generation of new virulence in the pathogen? No satisfactory answer is available even today. The exact role played by the environmental factors like temperature, humidity, rainfall, light intensity, wind, etc. need to be worked out in detail. The contribution of alternative hosts in perpetuating virulent races of red rot pathogen is yet to be determined. Dispersal mechanisms of infective propagules like conidia, ascospores, chlamydospores, acervuli, etc. formed on cane plant during and after the rainy season need critical evaluation. Available information is inadequate to explain the reason of large-scale flare up of red rot covering hundreds of acres. The exact role of perpetuating structures of the fungus like chlamydospores/ appressoria, thick walled hyphae and setae in the epidemiology of the red rot, and the part played by *Fusarium* and other associated microorganisms in accentuating the damage need immediate attention of researchers. The role of associated bacteria needs critical study, as *C. falcatum* when grown in sucrose medium or cane juice does not produce typical sourly alcoholic smell, which is quite characteristically observed in diseased canes of susceptible genotypes.

The nature of variability and existence of physiological races in *C. falcatum* *vis a vis* sources of resistance in host have to be catalogued using modern molecular biological techniques. Although a modest beginning has been made in this direction (Khirbat *et al.*, 1980; Anon., 1973-2005) – the task is formidable and serious attempt has to be made to resolve the race picture. Preliminary RAPD and ISSR analysis of the designated races did no good. The stability of *C. falcatum* pathotypes in terms of virulence, growth and sporulation has always remained a puzzle to those engaged in the research on red rot. Time and again, pathologists of different genre have questioned the prudence of identification of races in red rot pathogen. This apprehension has stemmed from the very instability of the pathogen in retaining the gene governing pathogenicity. *C. falcatum* is too notorious to lose virulence in culture. A critical molecular analysis of such cultures may help to unravel this riddle – whether it is a case of mutation or gene silencing or any other novel possibility. In the past hundred years of red rot research in India, only six or seven pathotypes remained stable for a reasonable period of time, even though loss of virulence was also noticed in the artificial culturing of these pathotypes. The loss of sporulation in artificial culture is another moot point in this regard. Without the production of infective propagules (conidia), the fungus cannot stand a chance of spread and survival in nature.

Red rot, even today continues to threaten sugarcane cultivation in India. So far, no fungicide is available that can effectively eradicate internal infection of red rot from seed-cane (setts). The challenge before research workers is to make chemotherapy a viable proposition. Of late, induced systemic resistance through the application of chemical inducers and bio-agents is opening up new avenues to strengthen the existing resistance of the cane plant. It may hold promise in increasing field life as well as stability of a variety. A concerted effort by all concerned in this regard will definitely be rewarding and it will go a long way in containing the menace of red rot.

Breeding and utilisation of resistant sugarcane variety/genotype has become the mainstay in the perpetual fight against red rot in India. In spite of the immense importance of resistant varieties, a directed breeding effort against red rot has hardly been targeted to this end due to the complex inheritance of resistance. The predominant breeding strategy still widely followed is the screening of elite clones (genotype with good agronomic attributes) against red rot. This screening of selected elite population has created an aura of false impression that high sugar content in cane is associated with red rot susceptibility. The sugarcane ideotype is essentially the noble cane (*S. officinarum* contributes 90% of the genome of hybrid cane), which is fortified with desired level of tolerance to different stresses including that caused by the pathogens. Obviously, progenies selected for better agronomic qualities are skewed towards *S. officinarum,* which is susceptible. An analysis of the data furnished by Azab and Chilton (1952) and

Alexander (1995) clearly shows that 50 per cent of the hybrid progenies are susceptible. The inheritance pattern of the genes governing red rot needs to be elucidated critically, leaving little or no room for speculation. So far, very limited work has been carried out in this direction and in many cases, limited progeny size did not allow proper analysis of inheritance pattern.

The success of *S. spontaneum* in imparting resistance to the hybrid progenies however did not improve its chance as parent. Over a hundred-year of cane breeding only a few *S. spontaneum* have been used for this purpose. A look in the parentage of the present day commercial hybrids immediately reflects the use of *S. spontaneum* (for resistance) only at the early part of the development of interspecific hybrids. There is hardly any fresh gene flow for resistance to red rot in the new commercial hybrids. This situation eventually put a rider on the genetic base of red rot resistance in the present day varieties. Thus, the red rot resistance of the present day varieties stems from the permutations and combinations of the existing genes obtained much earlier from a very limited number of accessions of *S. spontaneum*. Pathogen population is no longer under high pressure of survival due to non-introduction of new genes for resistance. This, in turn is reflected in the very short field life span of a new cane genotype as *C. falcatum* quickly overcomes the permutation-combination of the existing genes.

The inherent difficulty in sugarcane breeding system, in crossing and introgression of *spontaneum* complement, in producing the commercial cane hybrids (lack of synchronization in flowering, development of hardly a few clones of commercial merit at the first hybrid generation, etc.) along with the urgencies in breeding commercial genotypes prompted the cane breeders to take the more feasible route i.e., crossing between the hybrids. This indirectly has enforced inherent weakness in the system by narrowing the genetic base. It is high time to broaden the genetic base for red rot resistance. The problem of pyramiding the resistance is accentuated by the non-availability of immunity/near immunity / high degree of resistance against red rot in the parent population. Due to the highly heterozygous nature of the crop and high ploidy level, the cane genome is sufficiently buffered to take up such adventure with acceptable outcome.

The loss caused by red rot, at times is annihilating. In future, a forecasting approach, based on analysis of weather parameters, may improve management strategies to avoid serious losses. Obviously, healthy seed in a healthy field should be the motto in controlling red rot. An effective quarantine and its strict compliance is necessary at both macro-and micro-level to contain its spread in new area. Sugar mills have to stop the prevalent practice of introduction of canes from different regions without quarantine check. The fate of CoC 671 in Gujarat should be taken as the warning signal in this regard. Once a variety becomes susceptible to a pathotype in the field, it is difficult to prolong its field life without incurring significant losses.

Time has come to change the mindset to achieve cent per cent control of red rot in sugarcane. It is high time to learn to live with red rot amicably. Introduction of resistance in host population is also not a permanent solution, as has been experienced in the past. Introduced genes for resistance create a situation to starve the pathogen and force the pathogen to find out new avenues for survival (Duttamajumder and Verma, 1994). One is fully aware that hunger is the mother of all discord and discontentment in human society; so is the case with the fungal society. The food scarcity/forced hunger due to introduction of new genes for resistance pushes the pathogen population to tap all its available resources to devise a way to overcome this inflicted situation for survival. A happy situation could be that when a majority of the cane varieties have moderate resistance and thereby allow the pathogen to co-exist in a more peaceful way.

In fact, every organism has its own birthright to survive in this world. It will be quite unjustified to try to wipe it out completely from the face of the earth. *C. falcatum* has specialized itself to have sugarcane as its food and when food supply is limited, it tries to adapt to the new food situation, not happily, but for survival. This force-feeding often leads to the accumulation of matching genes in the pathogen to counter the resistant action of the genotype. Whenever the pathogen could muster the matching virulence gene(s), it attacked with vengeance (high knocking intensity). It has been observed in many other crop-pathogen combinations that when the matching gene becomes available in the pathogen to counter the introduced resistance, the disease reaction is often much severe, even greater than what a highly susceptible genotype displays with no or very limited genes for resistance. Obviously any race specific resistance is always vertical, whereas horizontal resistance is the rule of nature that is needed for basic survival.

It will be unwise to overlook the fact that agriculture is hardly 200 generations old, and modern agriculture is only of 10 generations! Throughout the millennium, nature has painstakingly maintained all the so-called susceptible genotypes and that's the legacy which has been inherited from the past. Today, within the span of 10 generations the situation has been pushed to such an extent that, at times, it becomes extremely difficult even to maintain a so-called modern genotype in normal cultivation. If left alone, these genotypes may not survive in nature at all. It is obvious that we are not in line with nature and it is high time that we mend our ways for a better century.

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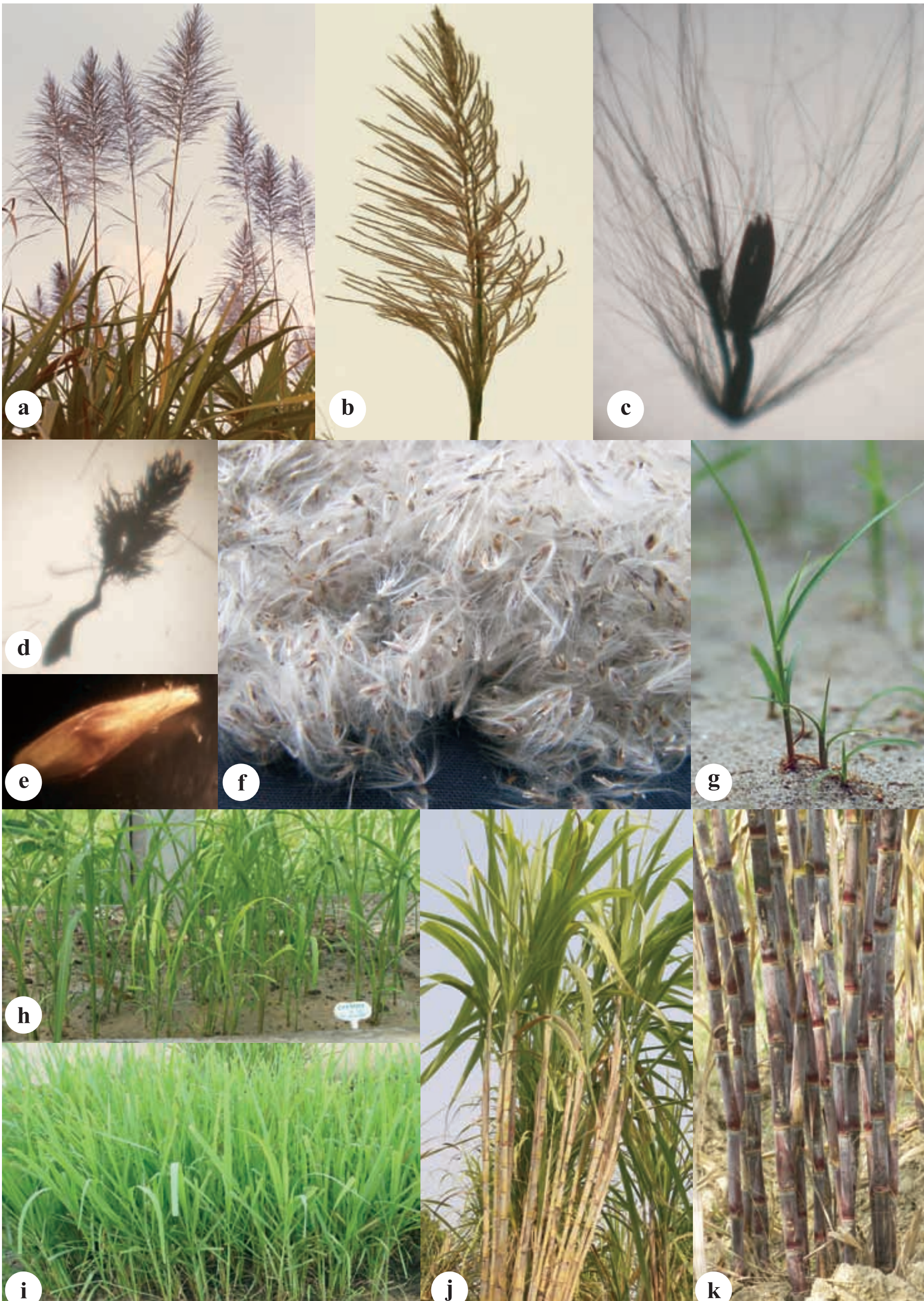

Plate 1 : **Life cycle of sugarcane**
(a) Flowering/arrowing is sugarcane (b) An arrow (c) Spikelet (d) Feathery stigma (e) True seed (defuzzed, magnified) (f) Fluff (g) Seedling emergence (h) Young seedlings (i) Seedlings ready for transplanting (j)&(k) Cane formation from single seedling.

Plate 2 : The damage potential of *C. falcatum*
Red rot decimated the susceptible variety completely (in August)

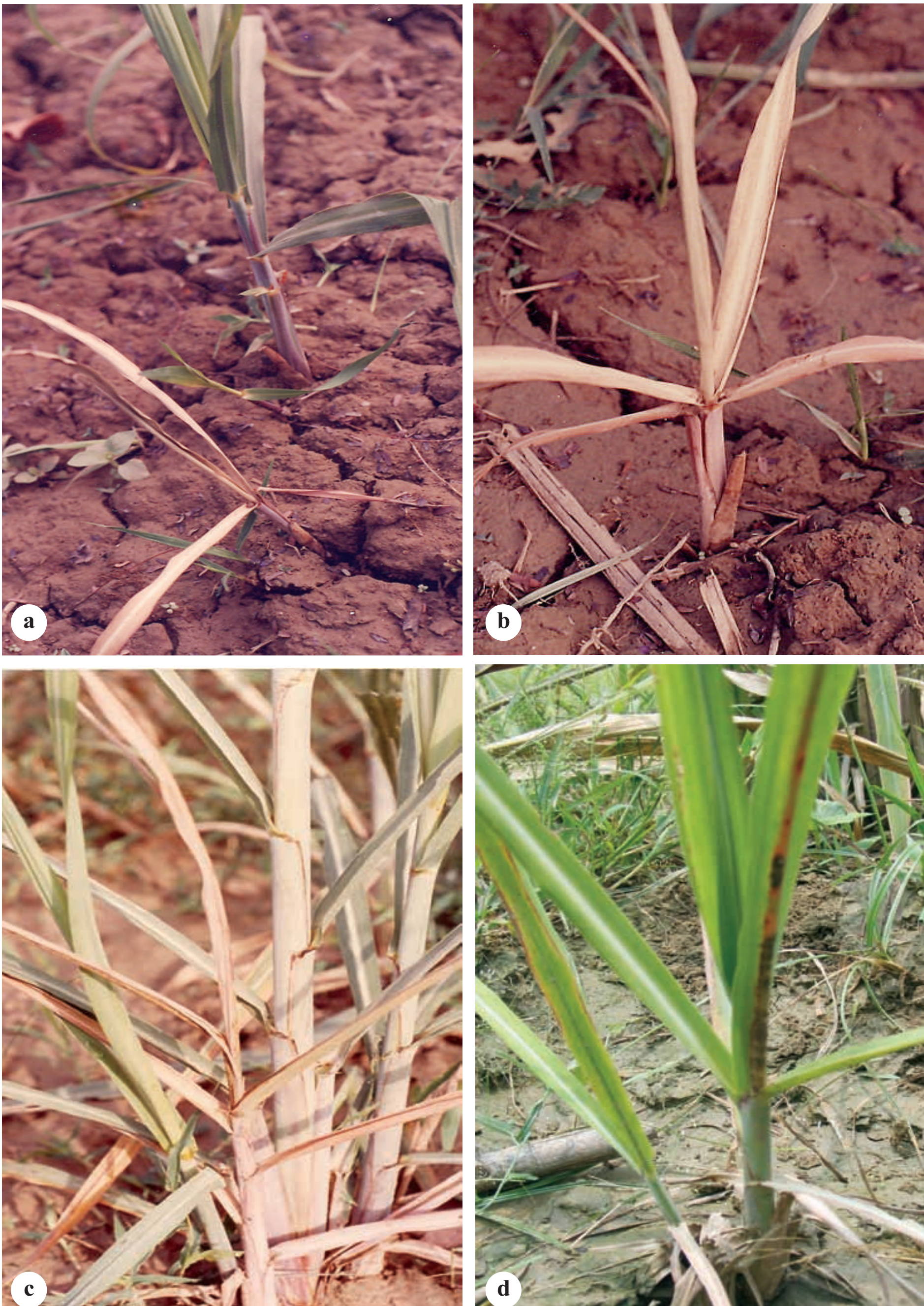

Plate 3 : Mortality of young shoot due to red rot
(a) Death of emerging shoot (germination failure) (b) Death of young shoot (c) Tiller mortality (d) Early spindle infection (at ratoon initiation).

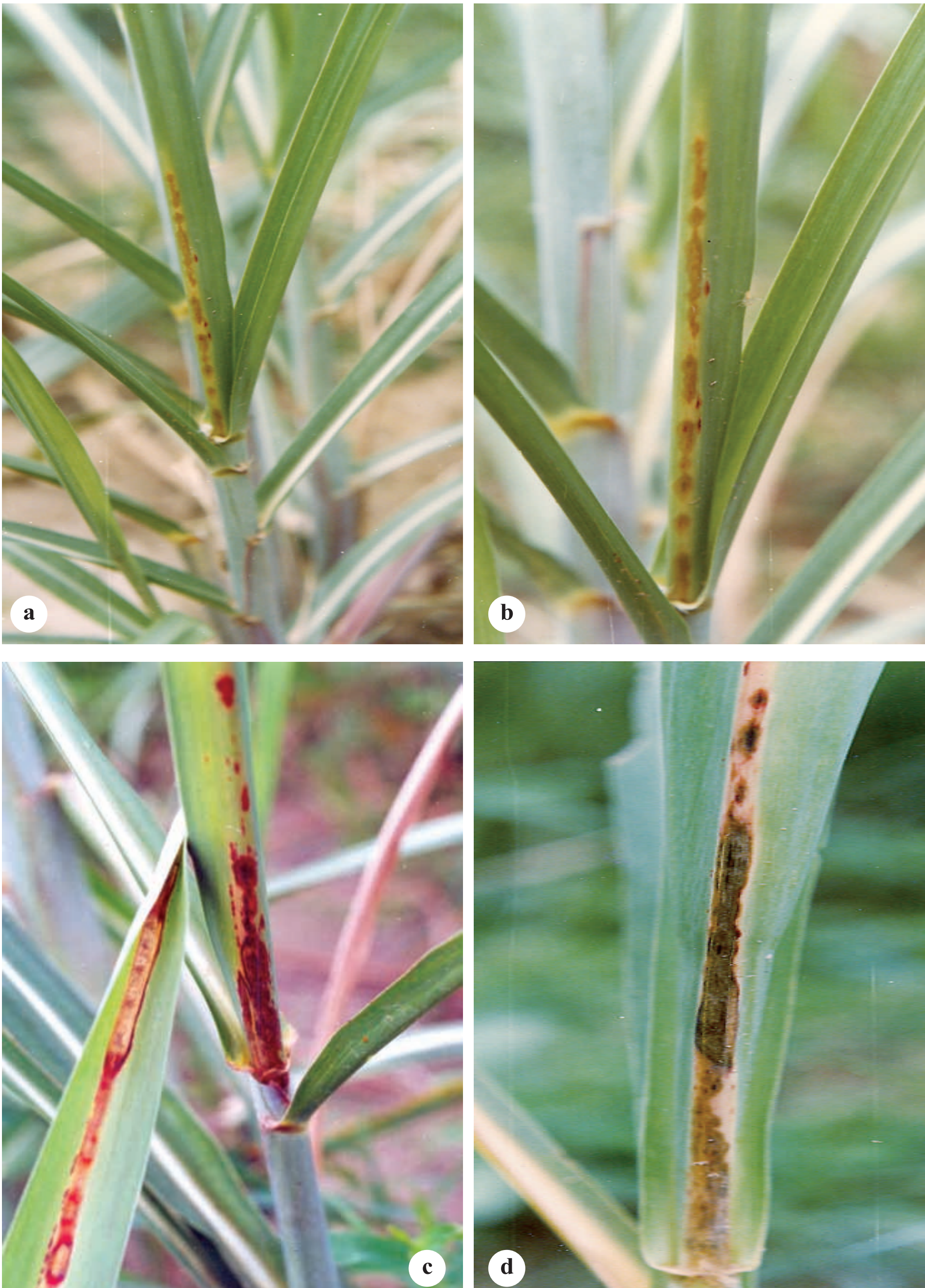

Plate 4 : Spindle infection or red rot

(a)&(b) Appearance of water-soaked spots on the abaxial side (back side) of the mid-rib. Infection progresses from the base of the unfolding leaf. (c)&(d) Quick and heavy sporulation appears on the spots turning them black. Sporulation also appears on the adaxial side of mid-rib.

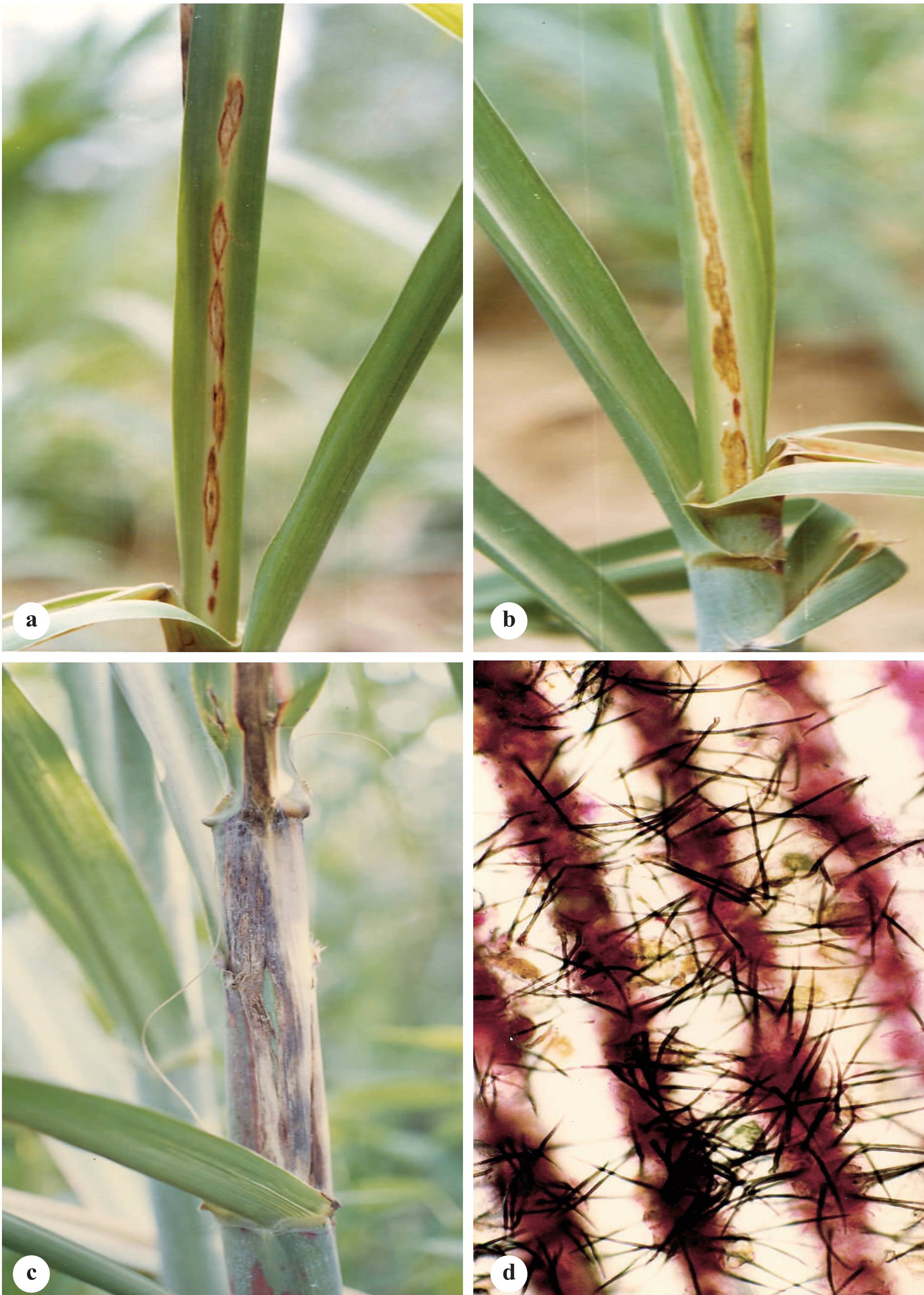

Plate 5 : **Spindle infection (contd.)**
(a) On the unfolded leaf, eye-shaped spots surrounded by yellow halo develop (b) Sporulation starts at the centre of the spot (c) Occasional damage of leaf sheath (on the way of lamina) (d) A photomicrograph showing abundance formation of setae.

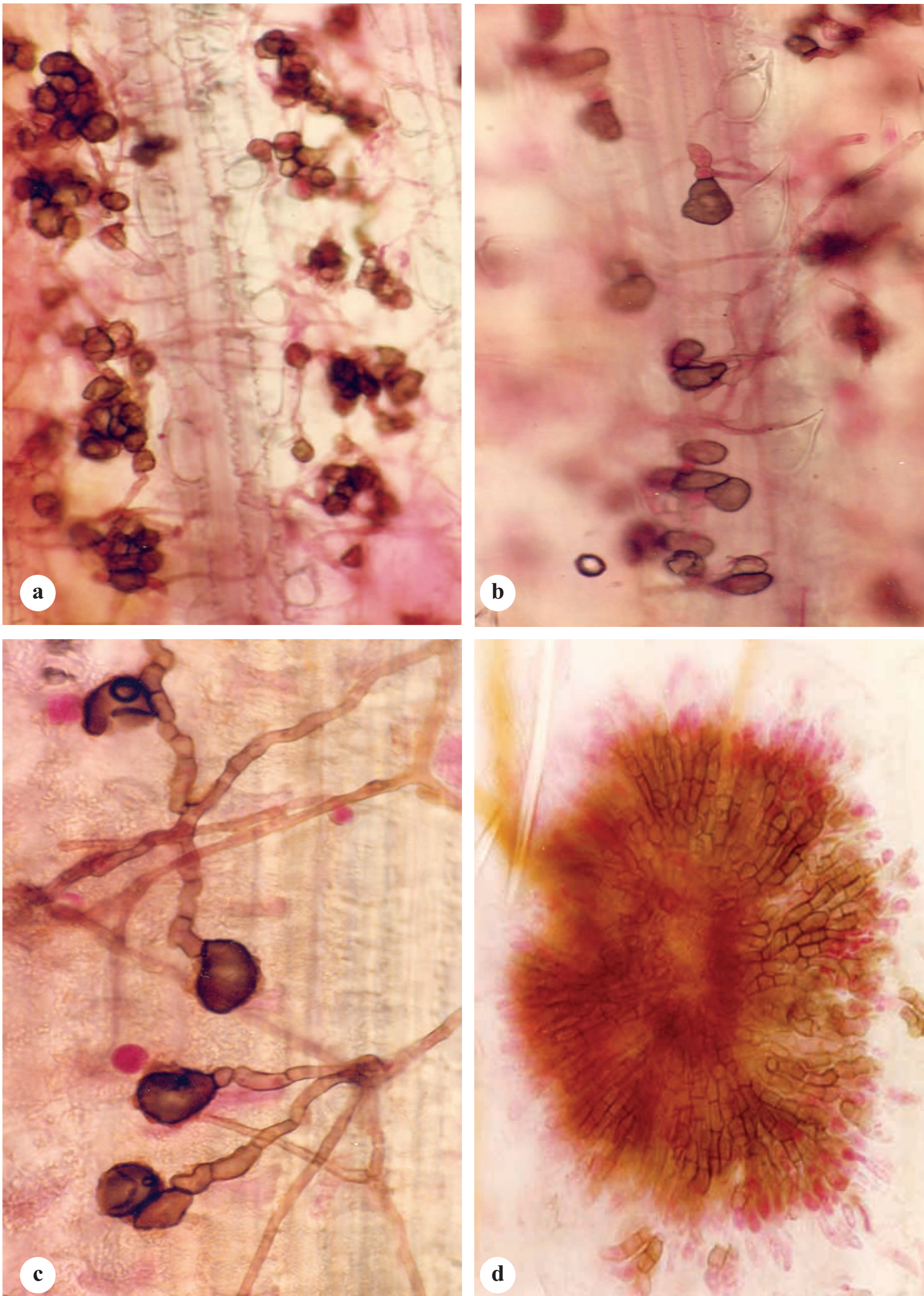

Plate 6 : **Development of appressoria on leaf surface of sugarcane**
(a), (b) and (c) Appressoria/chlamydospore developed on leaf (d) Development of appressed acervulus in the leaf fold.

Plate 7 : **Yellowing of crown leaves (first symptom in mature cane)**
(a) Yellowing more visible in the prominent leaves (b) Entire crown turns yellow as the disease progresses.

Plate 8 : **Symptoms at advanced stage of red rot**
(a) & (b) Yellowing of top and death of plant.

Plate 9 : **External symptoms on cane** (Discolouration of nodal & internodal tissues)
(a) Discolouration of internode (epidermis) (b) Internal appearance of cane after removal of epidermis (c) Damage of node and development of depressions in the internode (in advanced stage).

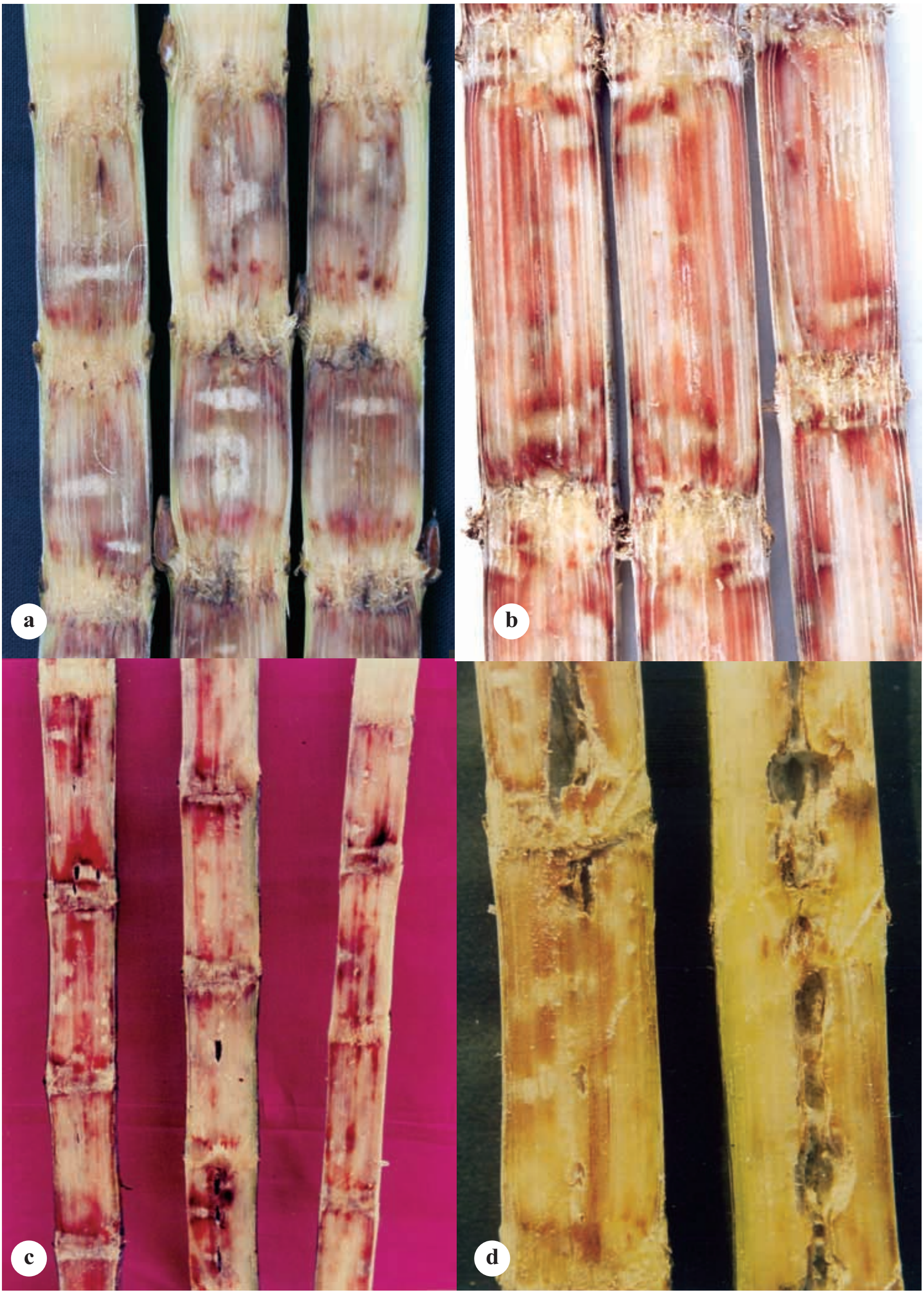

Plate 10 : Internal symptoms of red rot
(a) Development of white-spots (b)&(c) Damage of nodes and rotting of internode (d) Development of internal cavity.

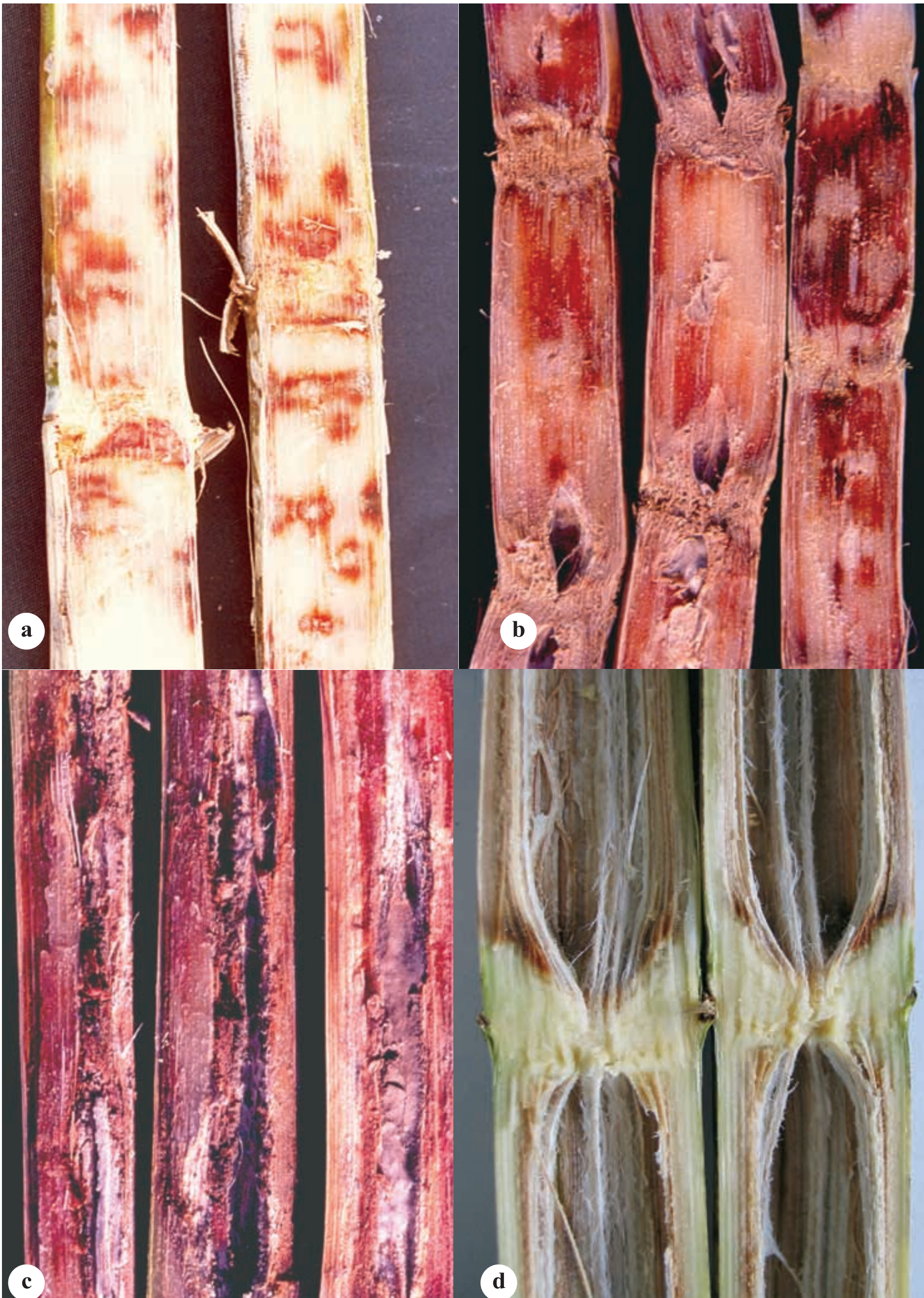

Plate 11 : **Internal symptoms (contd.)**
(a) Delimited white-spots (b)&(c) Cavity formation, fungal growth and sporulation in the cavity (d) Internal symptom of wilt (node unaffected, dry & larger cavity)

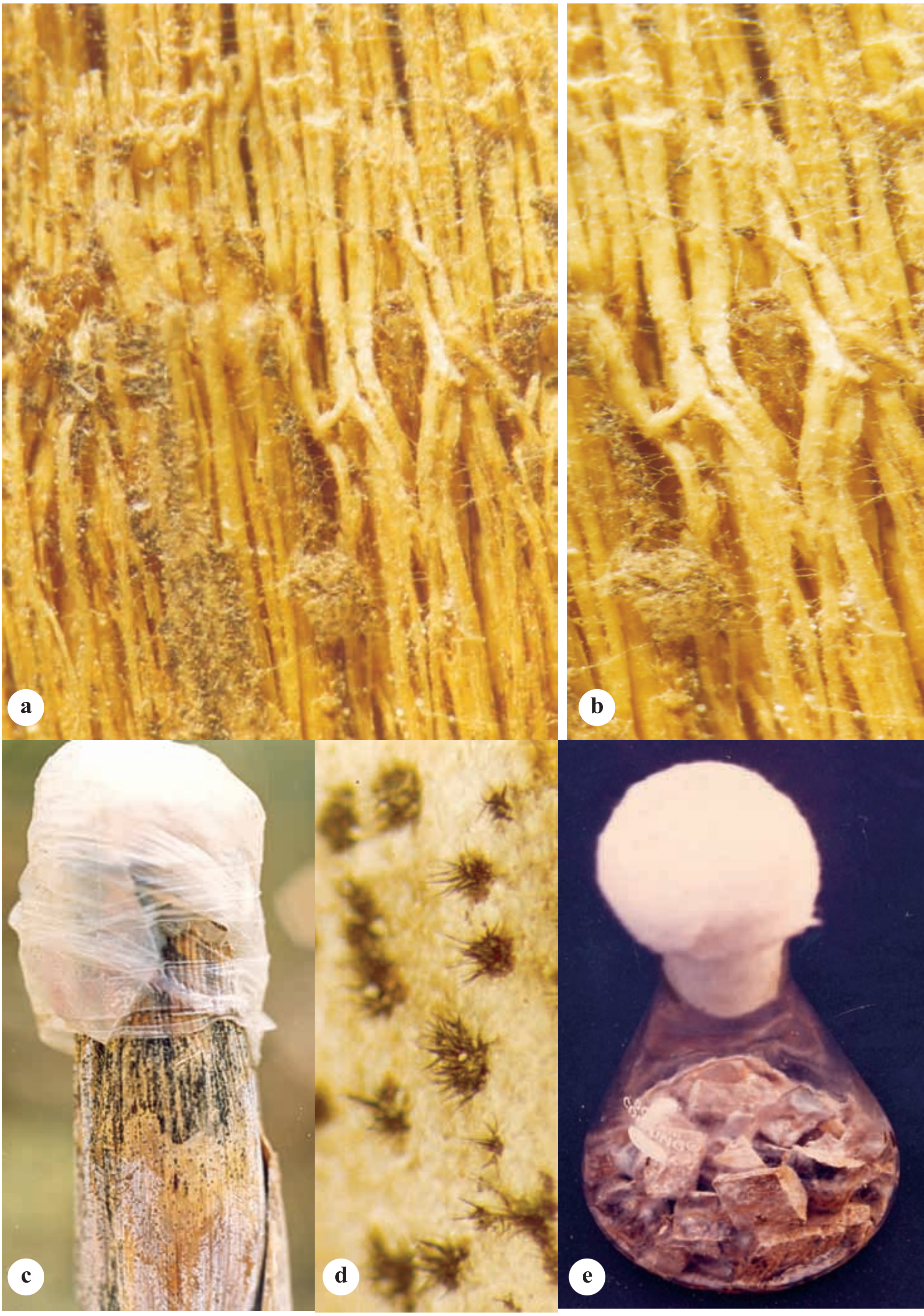

Plate 12 : Nodal rotting and sporulation

(a)&(b) *C. falcatum* preferentially damages the nodal tissues leaving vascular bundles unaffected (c) Sporulation on damaged cane (d) Acervulus formation (e) Sporulation on sterilized stalk pieces (more spores, less setae).

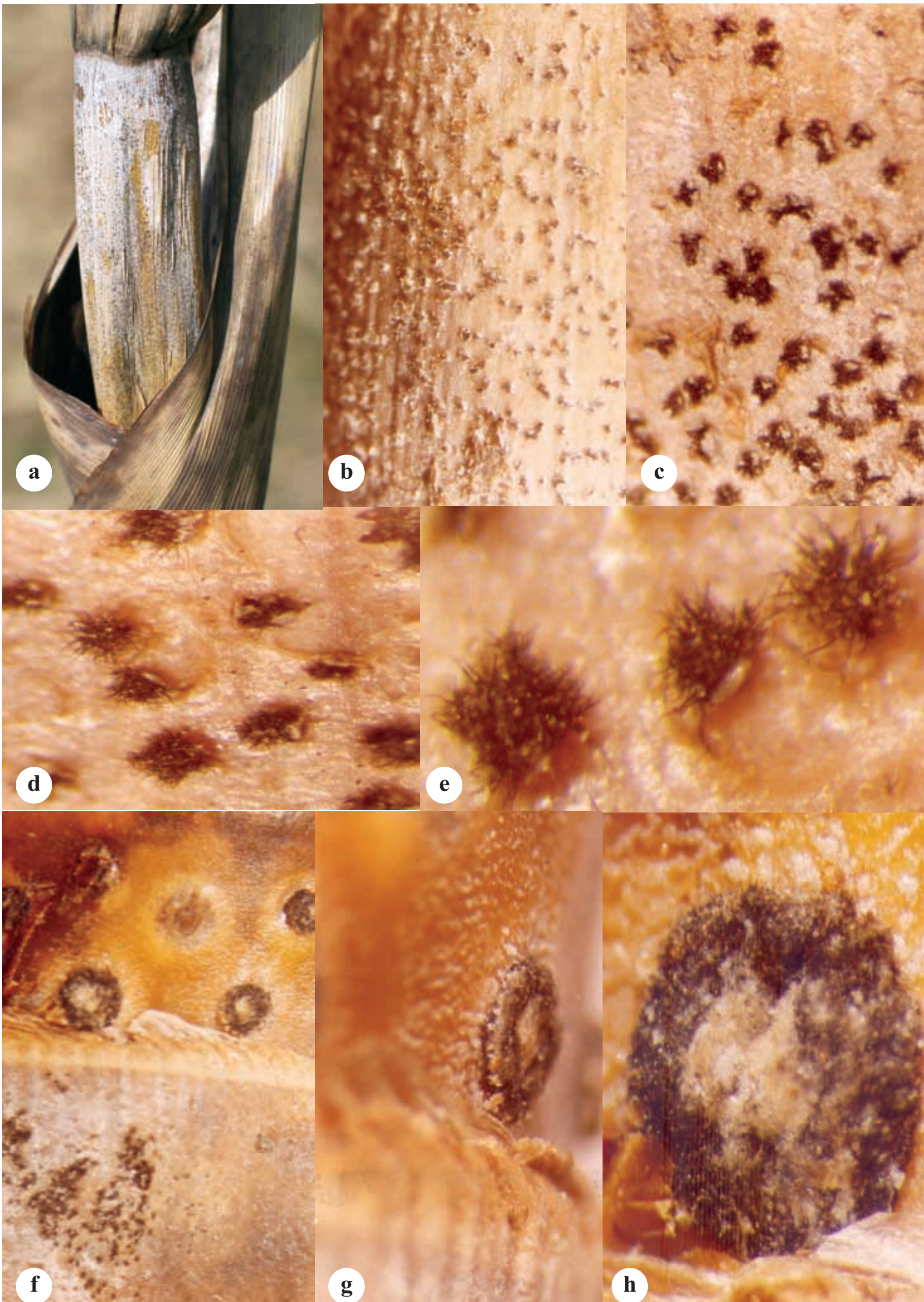

Plate 13 : Sporulation on the stalk

(a) Sporulation on diseased cane (drying) (b)&(c) Acervuli formation; fungus coming out breaking the cuticular/epidermal wax (d)&(c) close-up of the acervulus (f), (g) & (h) Damage of root band. Root primordia completely transformed into acervulus.

Plate 14 : Mid-rib infection of red rot

(a)&(b) Development of eye/spindle shaped spot on the mid-rib; initially blood red, later on centre turns ashy (c)&(d) Sporulation (development acervuli, setae & conidia) on the ashy area.

Plate 15 : Mid-rib infection covering entire leaf
(a) Large size mid-rib infection (b) Mid-rib infection covering entire length of leaf. Yellowing and drying starts tip downwards.

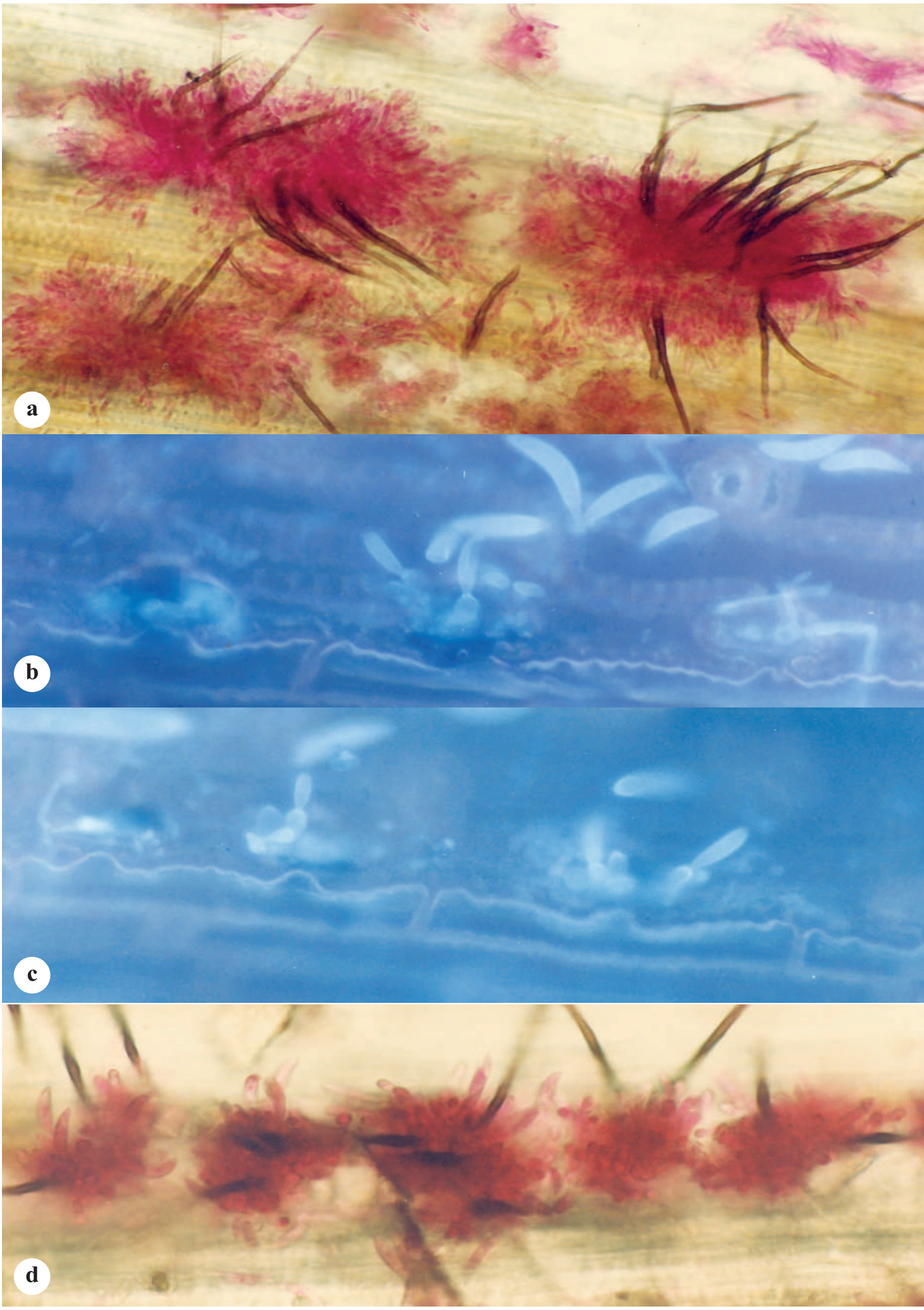

Plate 16 : Development of acervulus in *C. falcatum*
(a) Acervulus formation on the surface of leaf (b)&(c) Conidiophore bearing conidium coming out through stomata (d) Formation of acervuli on the stomata.

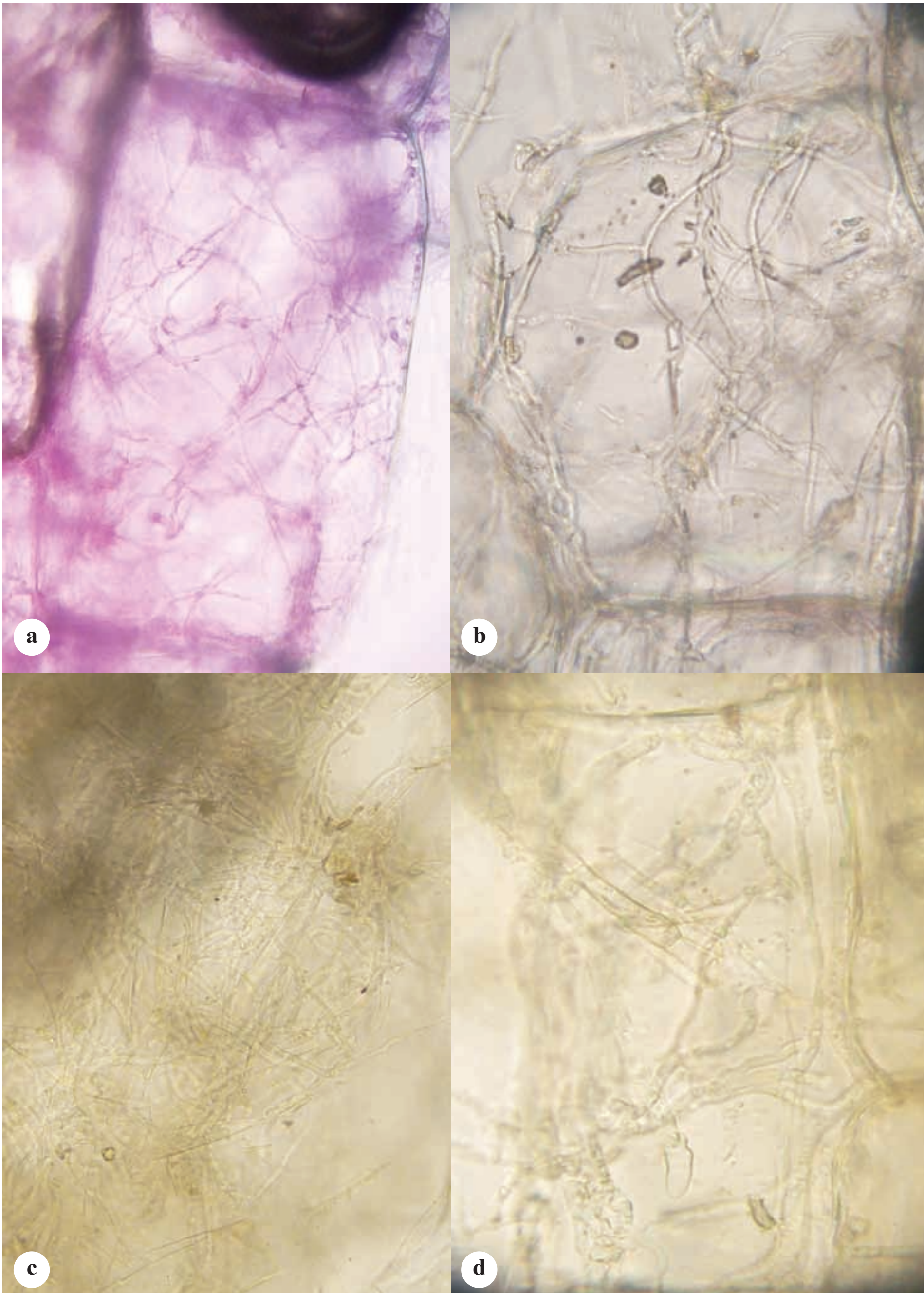

Plate 17 : **Proliferation of hyphae in stalk (white spot)**
(a), (b), (c) & (d) Stalk cell is filled with *C. falcatum* hyphae.

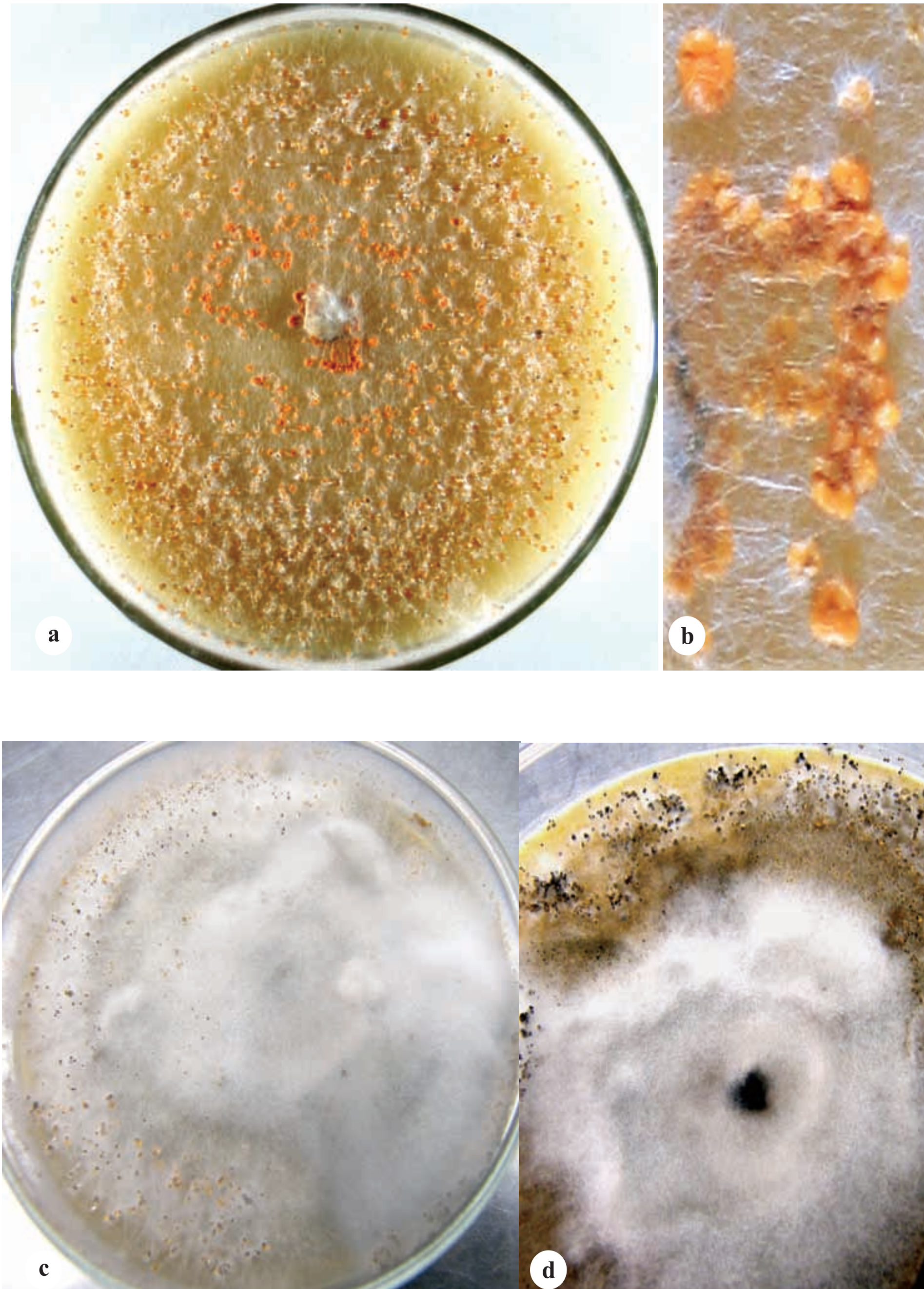

Plate 18 : Growth of *C. falcatum* on oatmeal agar (OMA)
(a) Culture on OMA - Cf 01 (b) Close-up of sporemass (c) & (d) Development of greyish white hyphal growth and sparse sporulation in Cf 01.

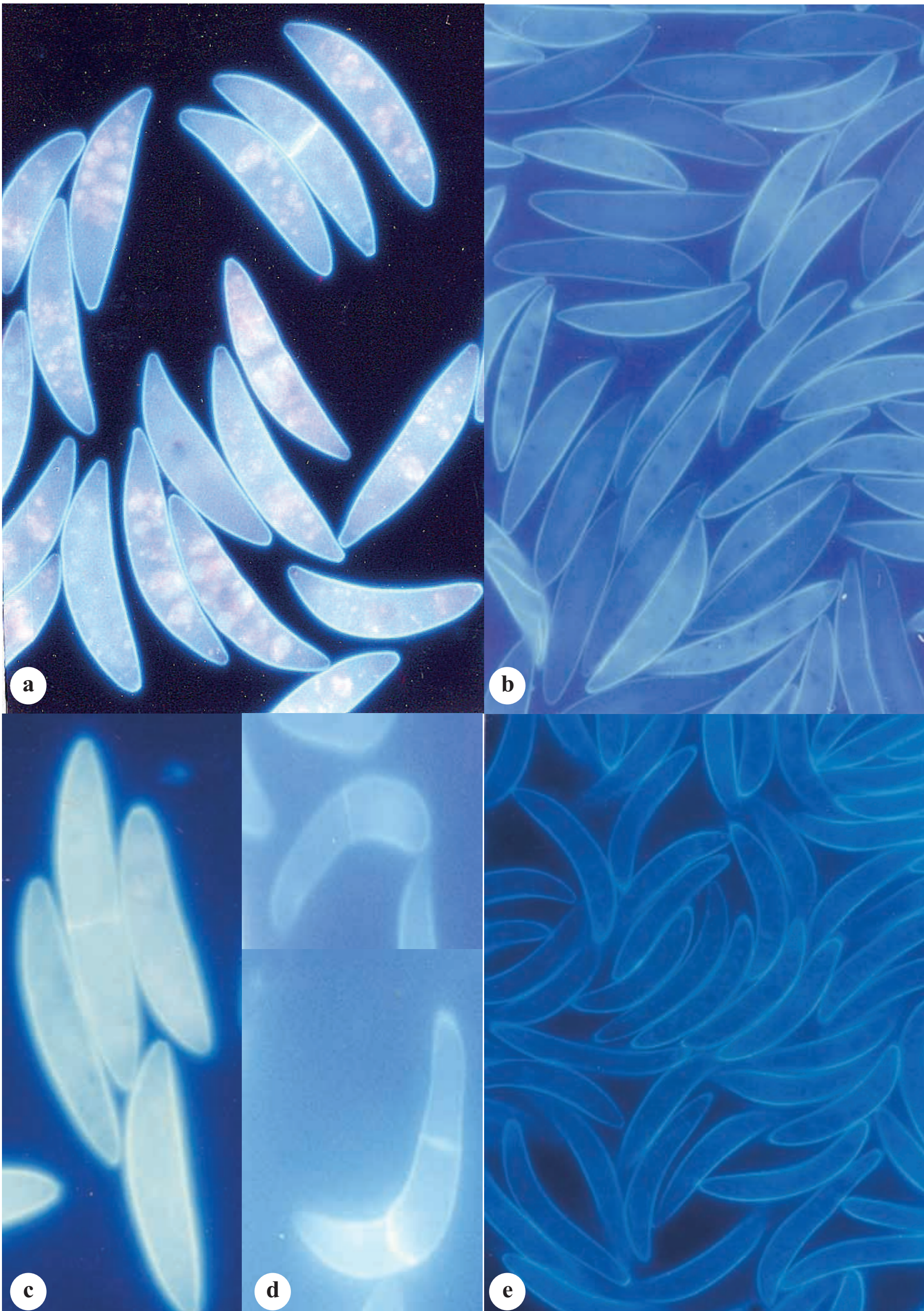

Plate 19 : Conidial sbape and size

(a) Typical conidia (b) & (e) Variation in conidial shape (a), (c) (d) Note the formation of septa in conidia even in non-germinating conidia.

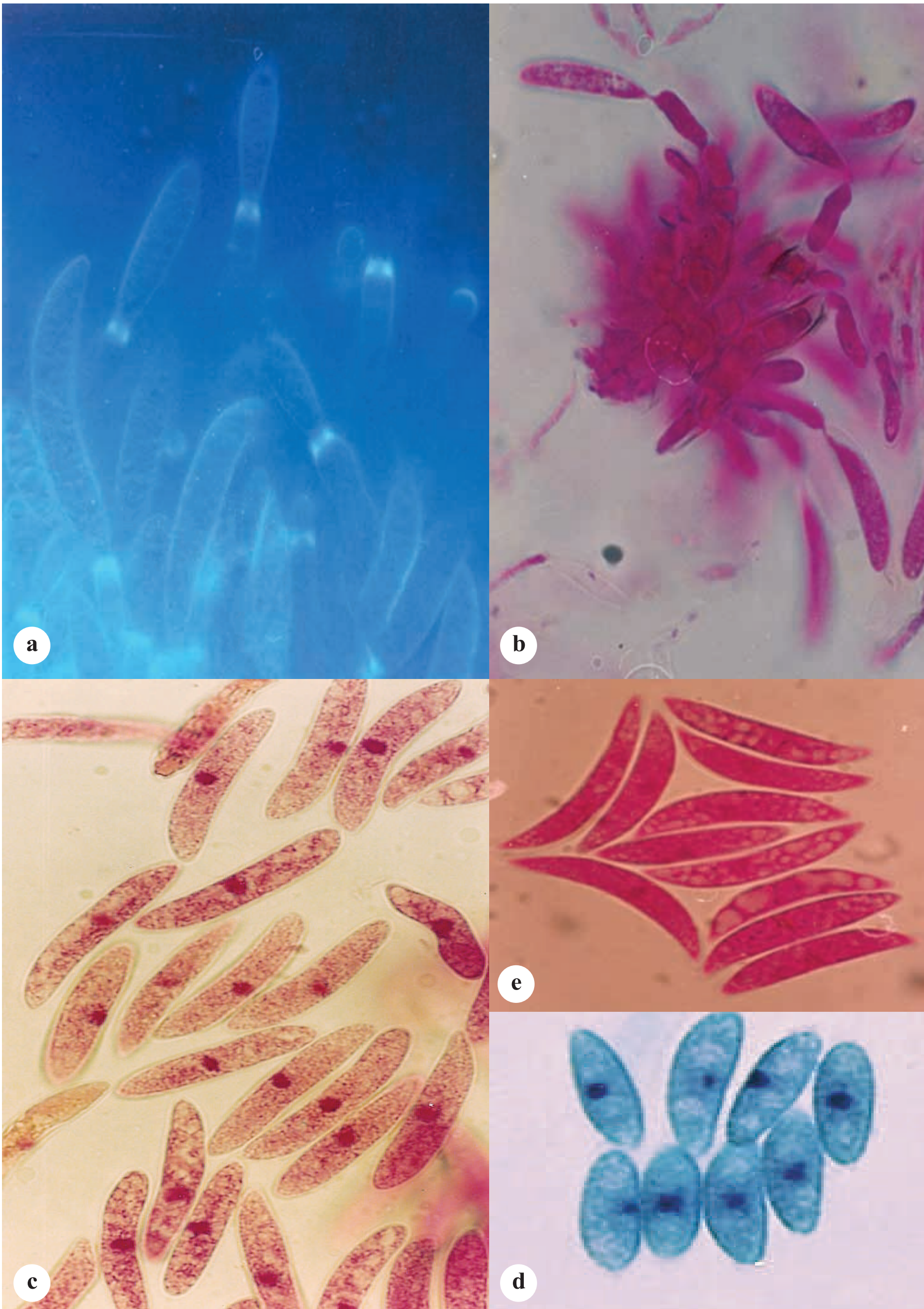

Plate 20 : Conidial development and nuclear condition

(a) & (b) Development of conidia on conidiophore (c) Single central nucleus in conidia (d) Single central nucleus in ascospore (c) & (e) Note the granular content of conidia.

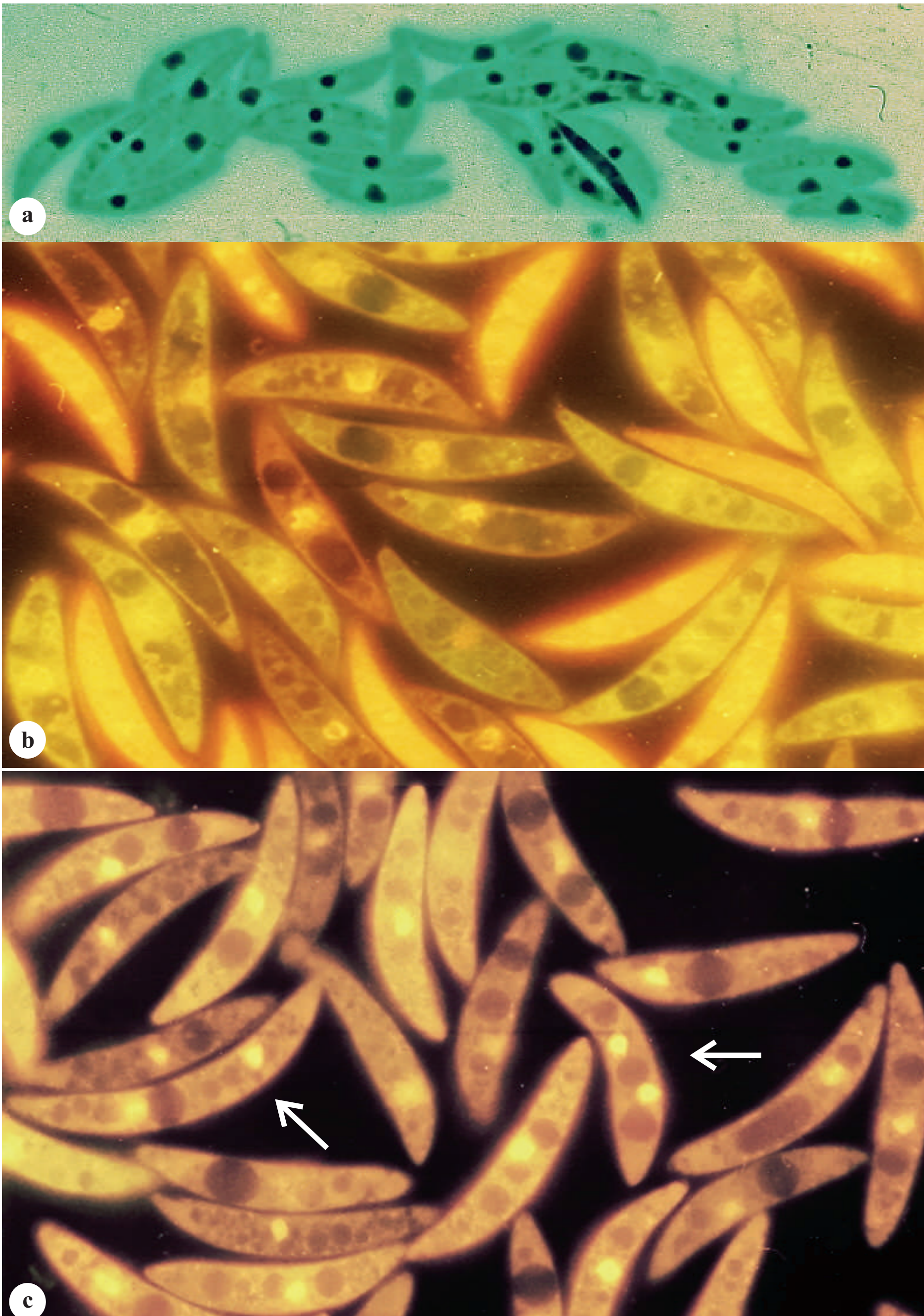

Plate 21 : Nuclear condition in conidia (variation)

(a) & (b) Single central nucleus (c) Conidia possessing two nuclei, note the shape of conidia (acridine orange fluorescence).

Plate 22 : **Germination of conidia**

(a), (b), (c) & (d) Subpolar germination, formation of septum in conidium, immediate formation of appressoria (a), (e), (f) & (g) Bipolar germination (h) Multipolar germination.

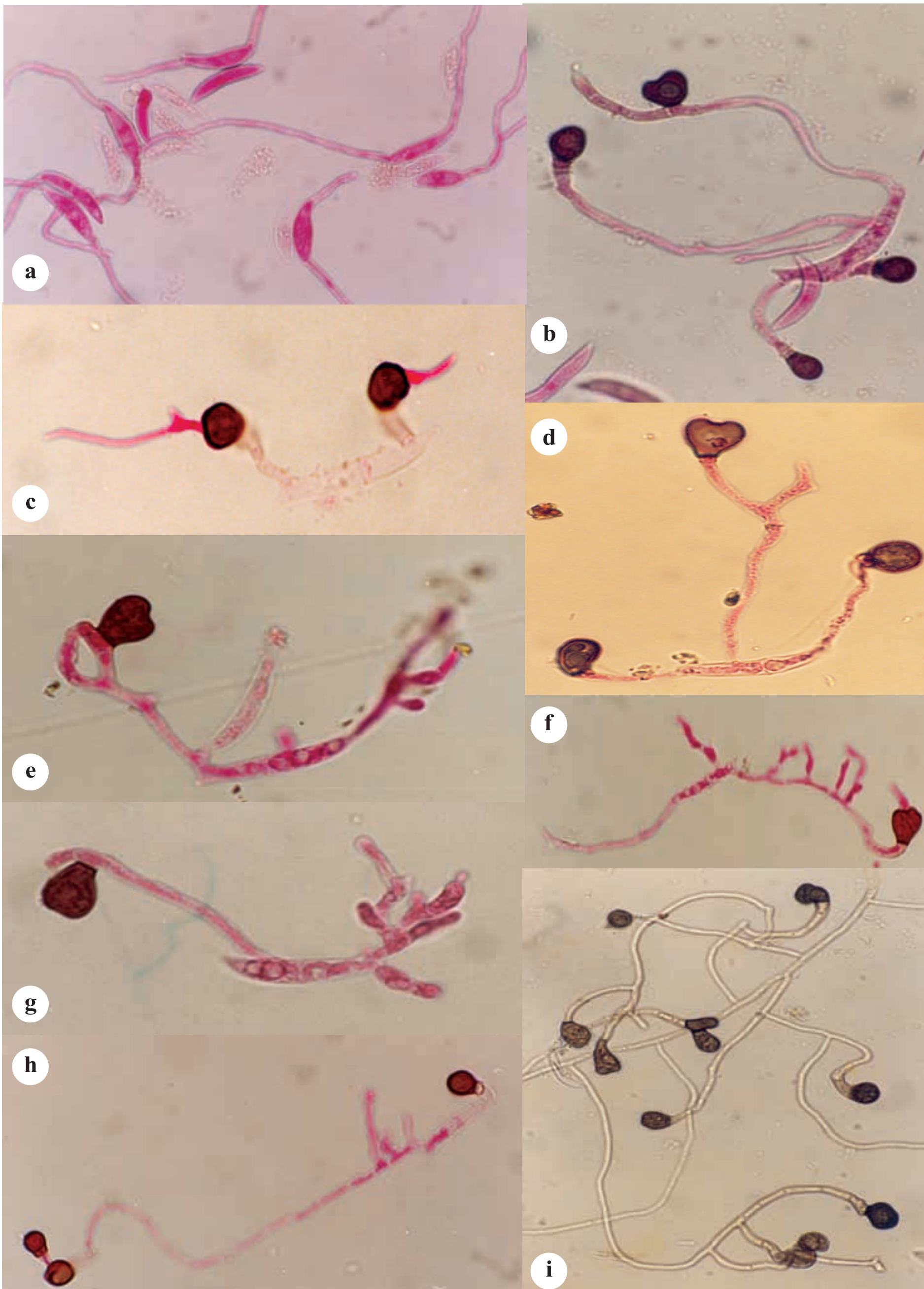

Plate 23 : Germination of conidia (contd.)

(a) Bipolar germination without immediate formation of appressoria (b), (d) Multipolar germination (c), (f) Appressoria have no resting period, immediate development of infection peg (b), (c), (d), (e) & (g) Appressoria may be terminal or intercalary (h) Appressorium forming appressorium instead of infection peg (i) Multiple appressoria formation.

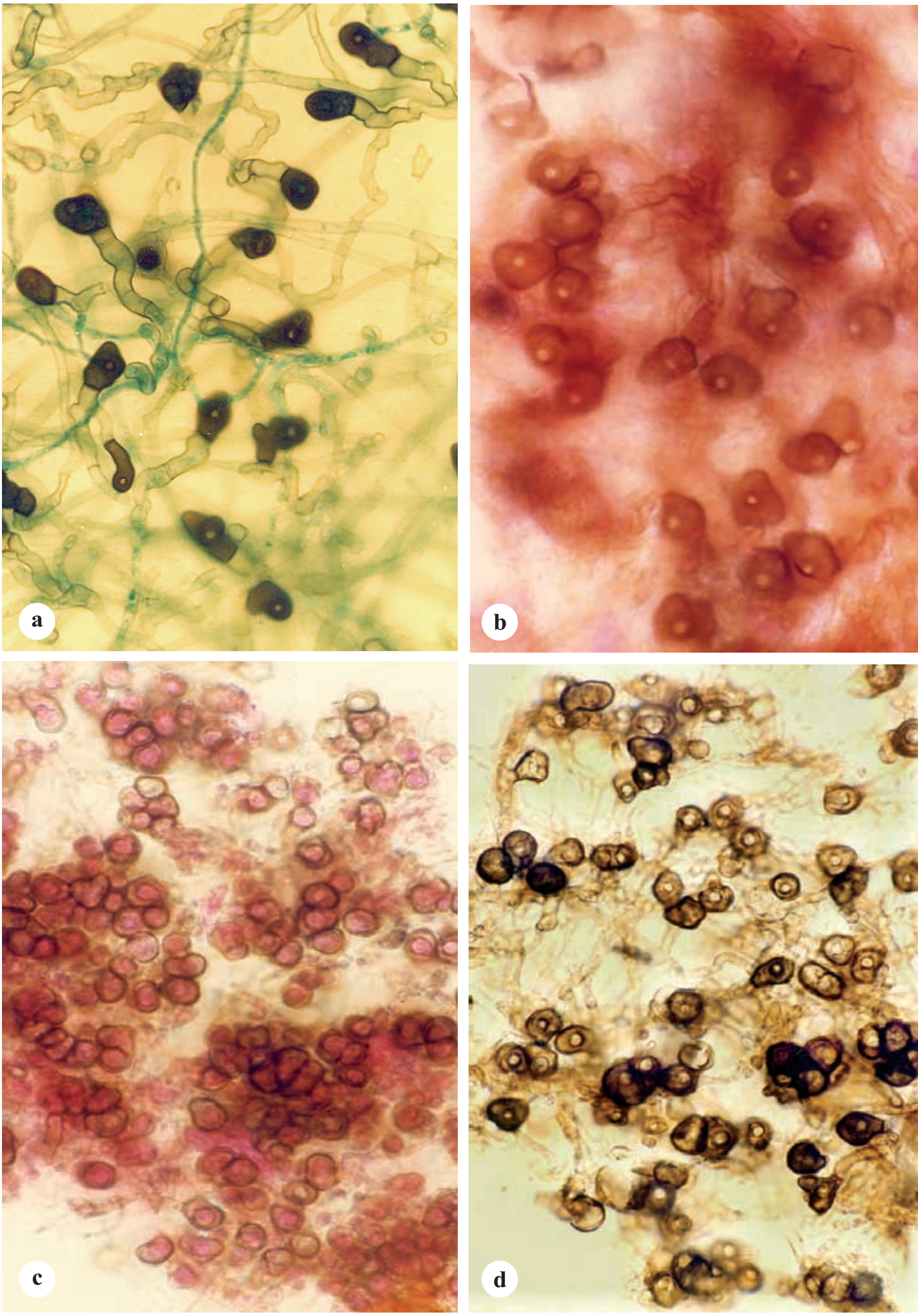

Plate 24 : Appressoria/chlamydospore of *C. falcatum*
(a) Appressoria formation at the bottom of media where it touches the glass (petridish)
(b), (c) & (d) Appressoria/chlamydospore formation in the leaf fold (spindle infection).

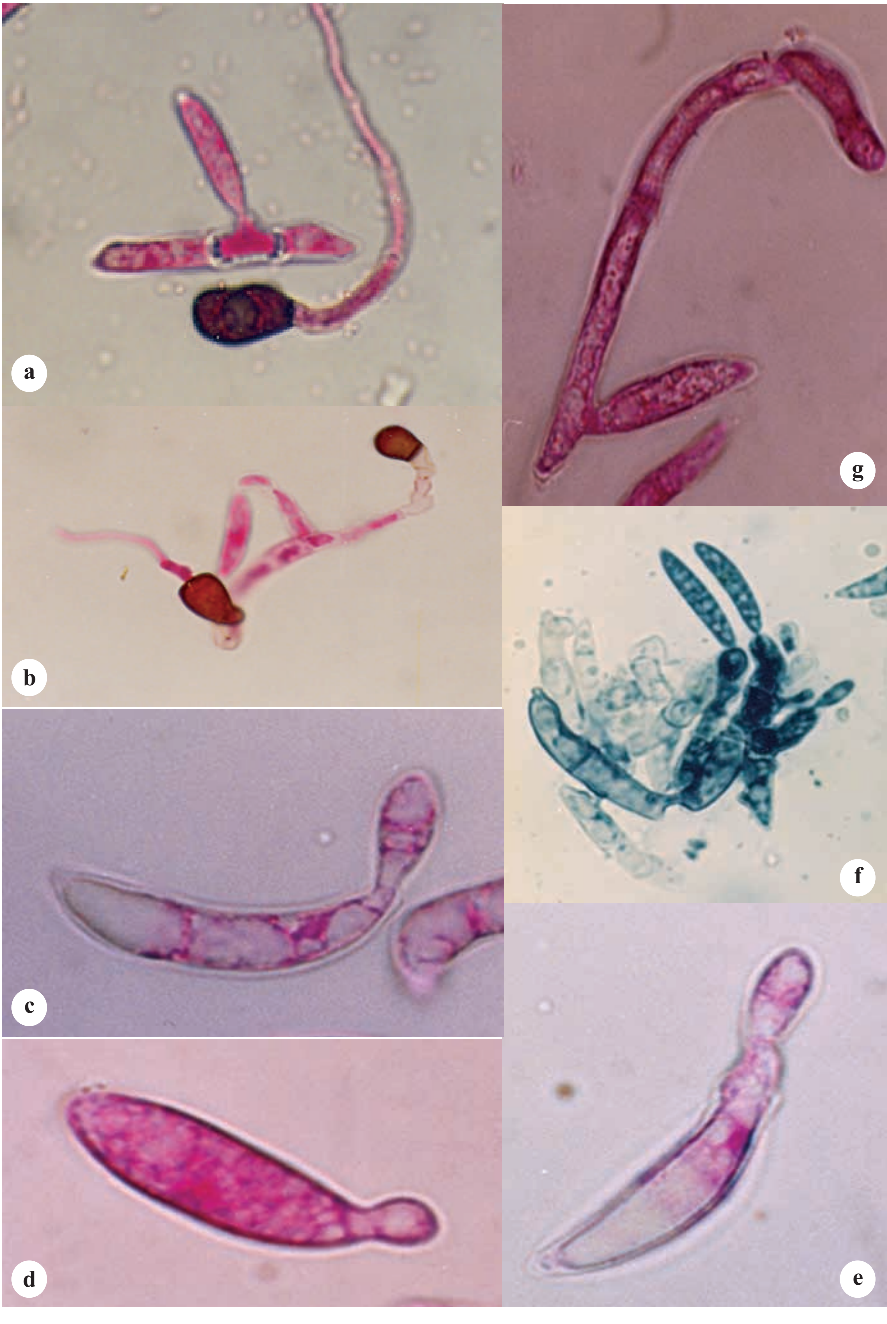

Plate 25 : Transformation of conidia into secondary conidia bearing structures

(a) Central portion of conidium giving rise to daughter conidium (b), (c), (d) & (e) Formation of daughter conidia (f) conidia formation after conidial fusion (g) Large sized conidium (aberrant).

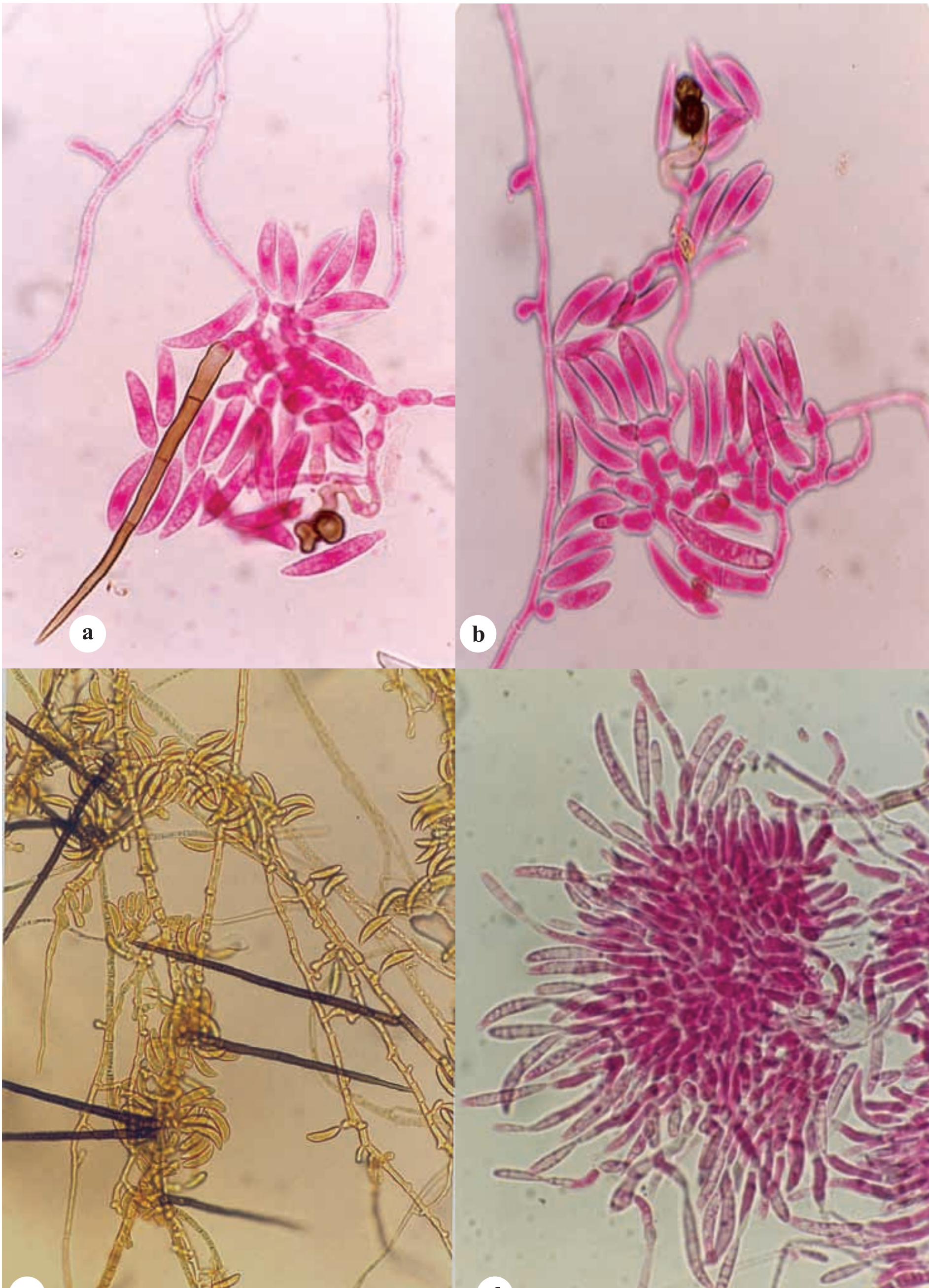

Plate 26 : Development of acervulus in culture

(a) & (b) Initial phase of acervulus development after germination of conidium (note appressoria & setae) (c) Acervulus development on media (note the transformation of hyphae into conidiophores and setae) (d) An acervulus.

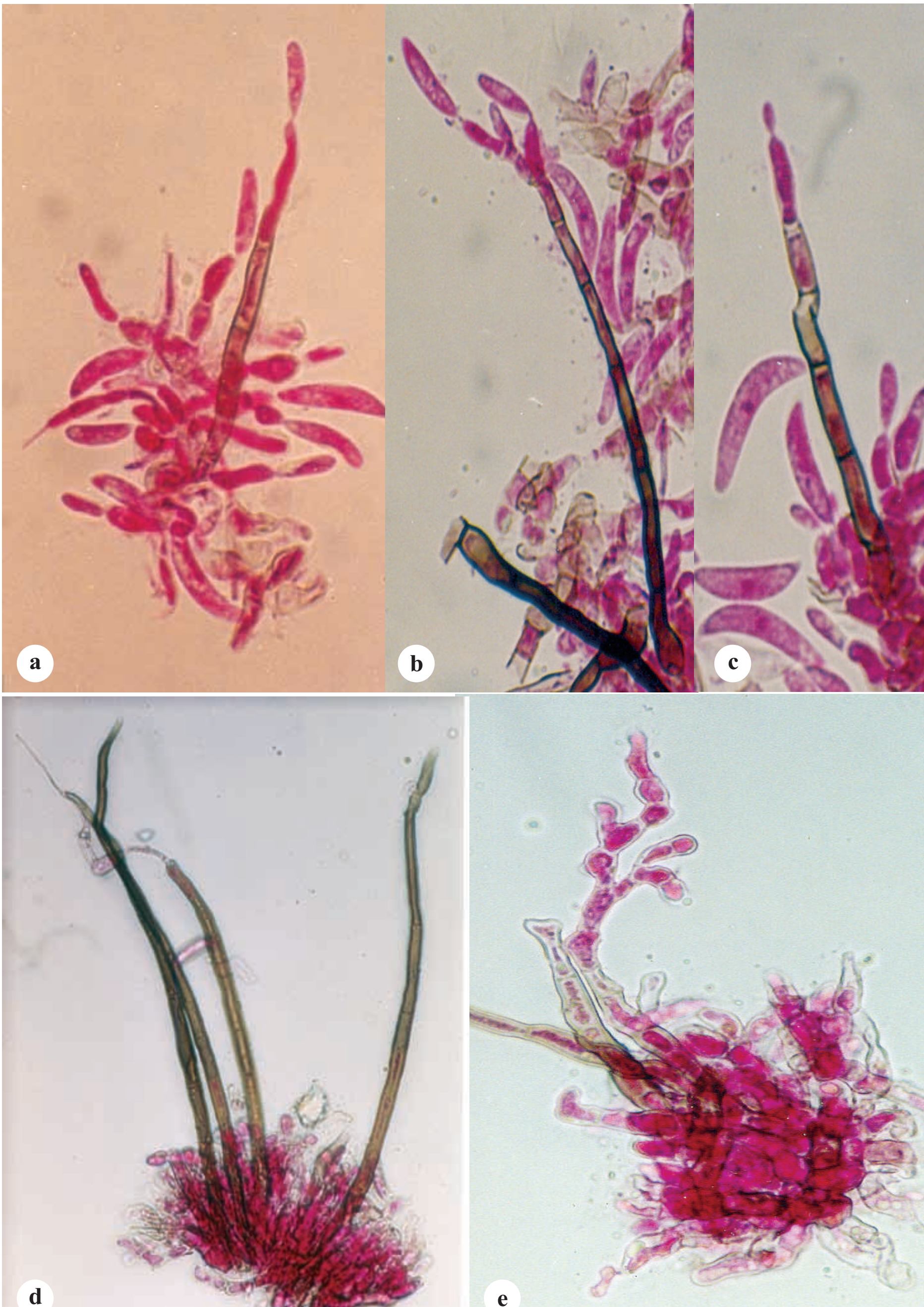

Plate 27 : Transformation of setae to perform different functions
(a), (b) & (c) Formation of conidia at the hyaline tip (single & branched)
(d), (e) Changed to hyphal structures.

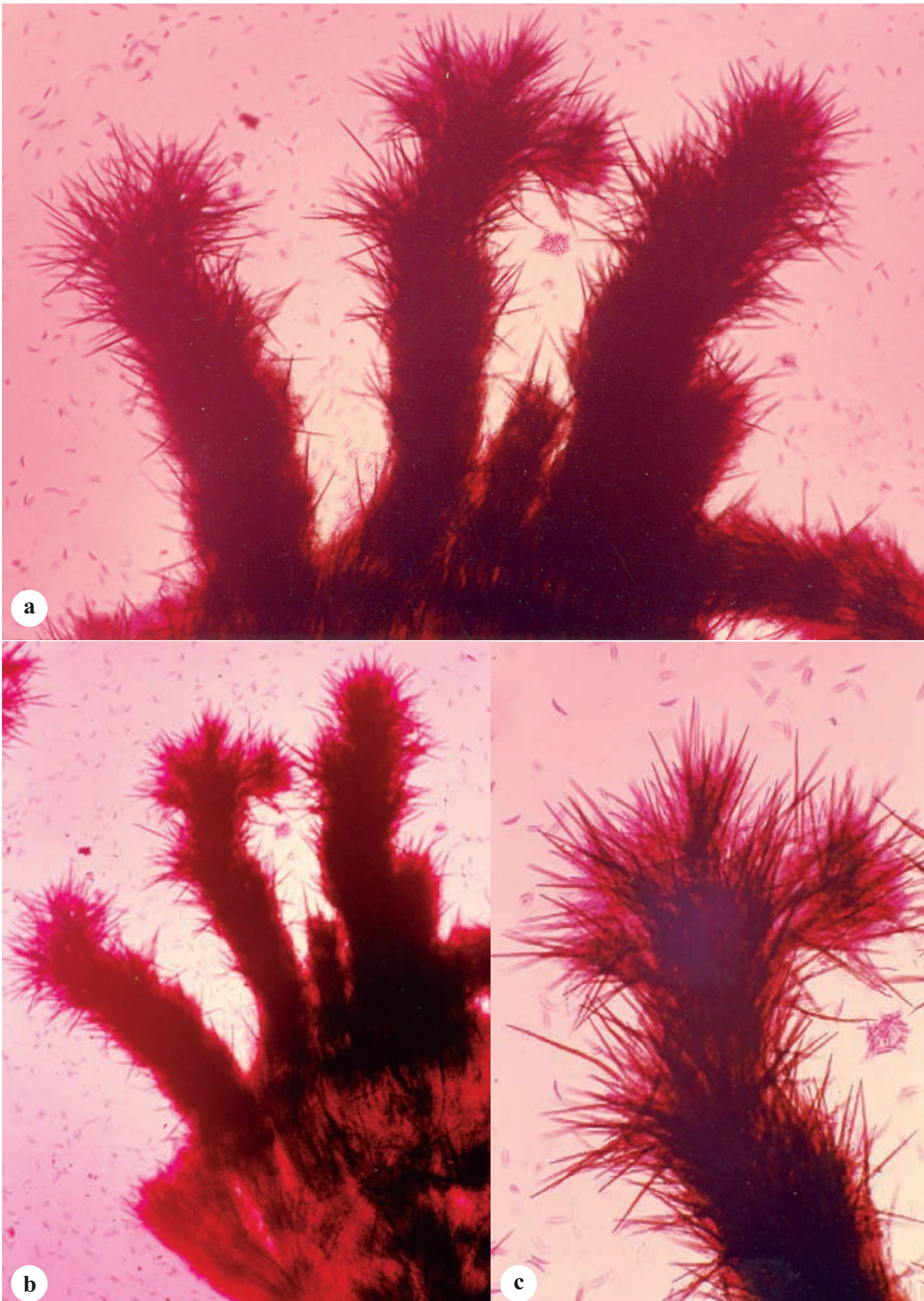

Plate 28 : **Synnamatous development of setae**

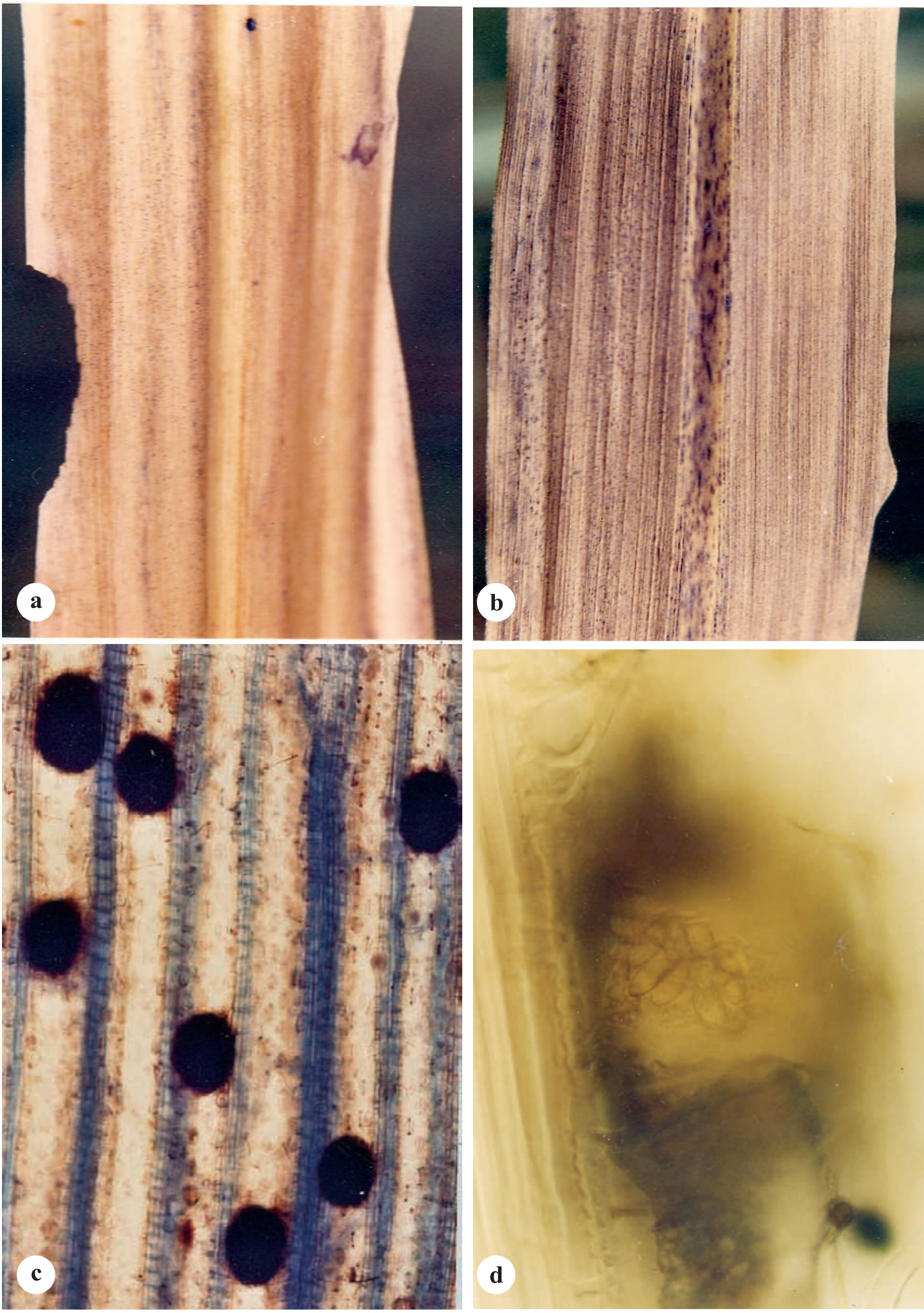

Plate 29 : Sexual stage of *C. falcatum*

(a) & (b) Perithecia on dry/dead cane leaves (black dots) (c) Perithecia (round to oval appearance) develop in between vascular bundles (magnified abaxial view) (d) Ostiole showing exit of ascospores from perithecium.

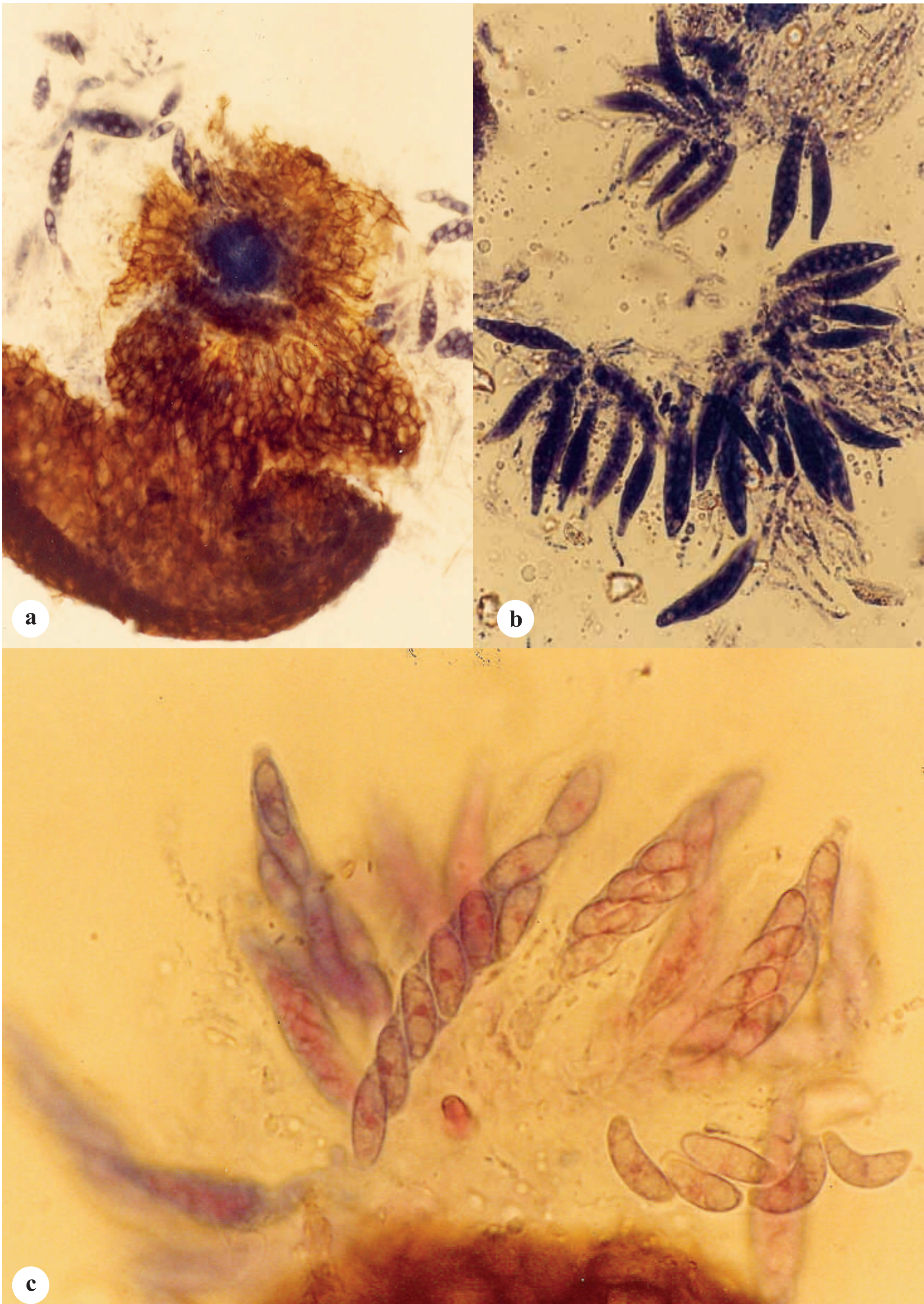

Plate 30 : **Perithecium, asci and ascospores *(Glomerella tucumanensis)***
(a) Perithecium with ostiole (b) Content of perithecium-asci and paraphyses (c) 8-spored ascus, ascospores arranged in a biseriate pattern.

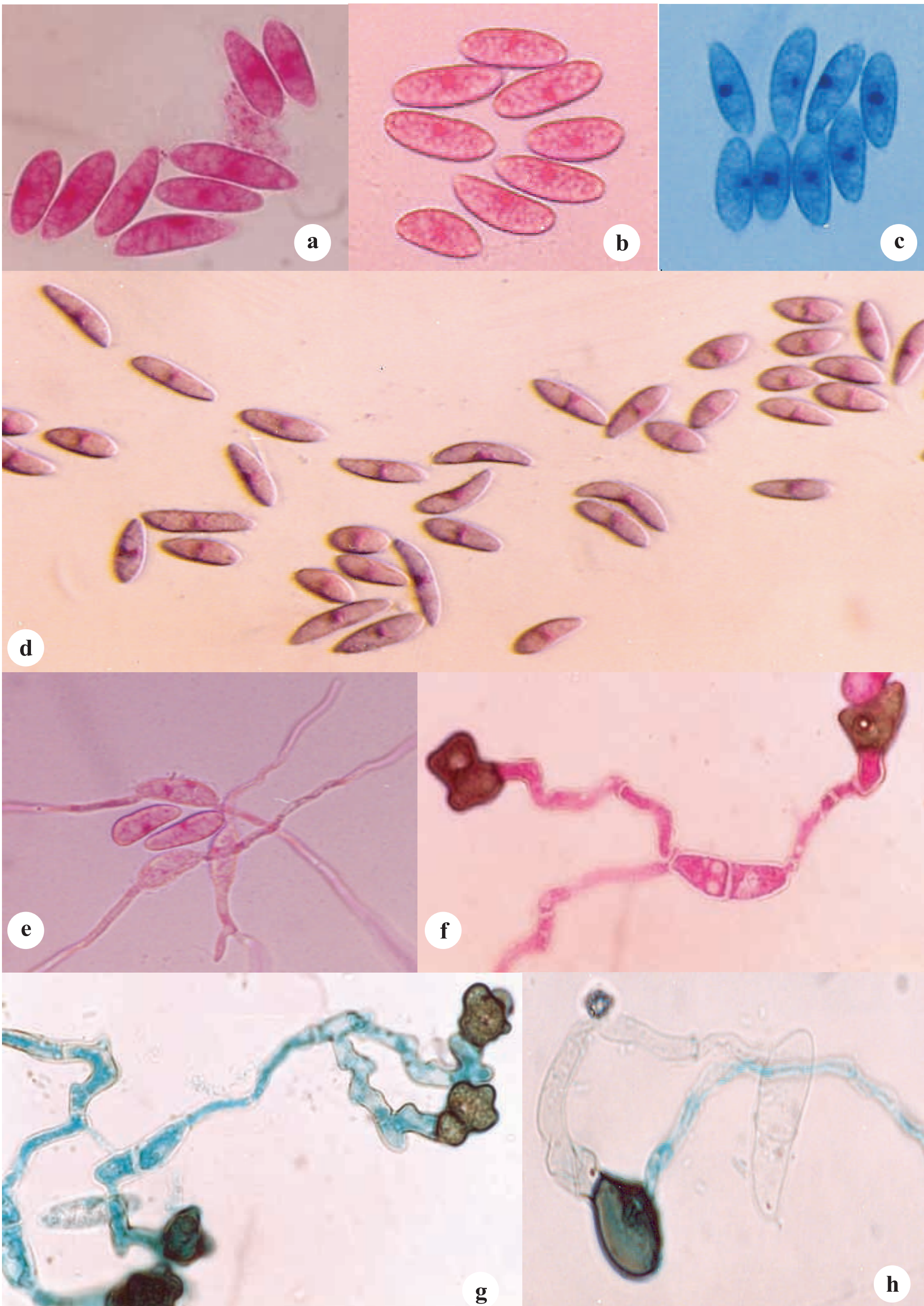

Plate 31 : **Ascospore-shape, size, nuclear condition and germination**
(a), (b), (c) and (d) Single celled ascospores, single central nucleus (e), (f) & (g) Germination subpolar - with or without septa formation, development of appressoria (h) Appressorium has no resting period; immediately infection peg develops.

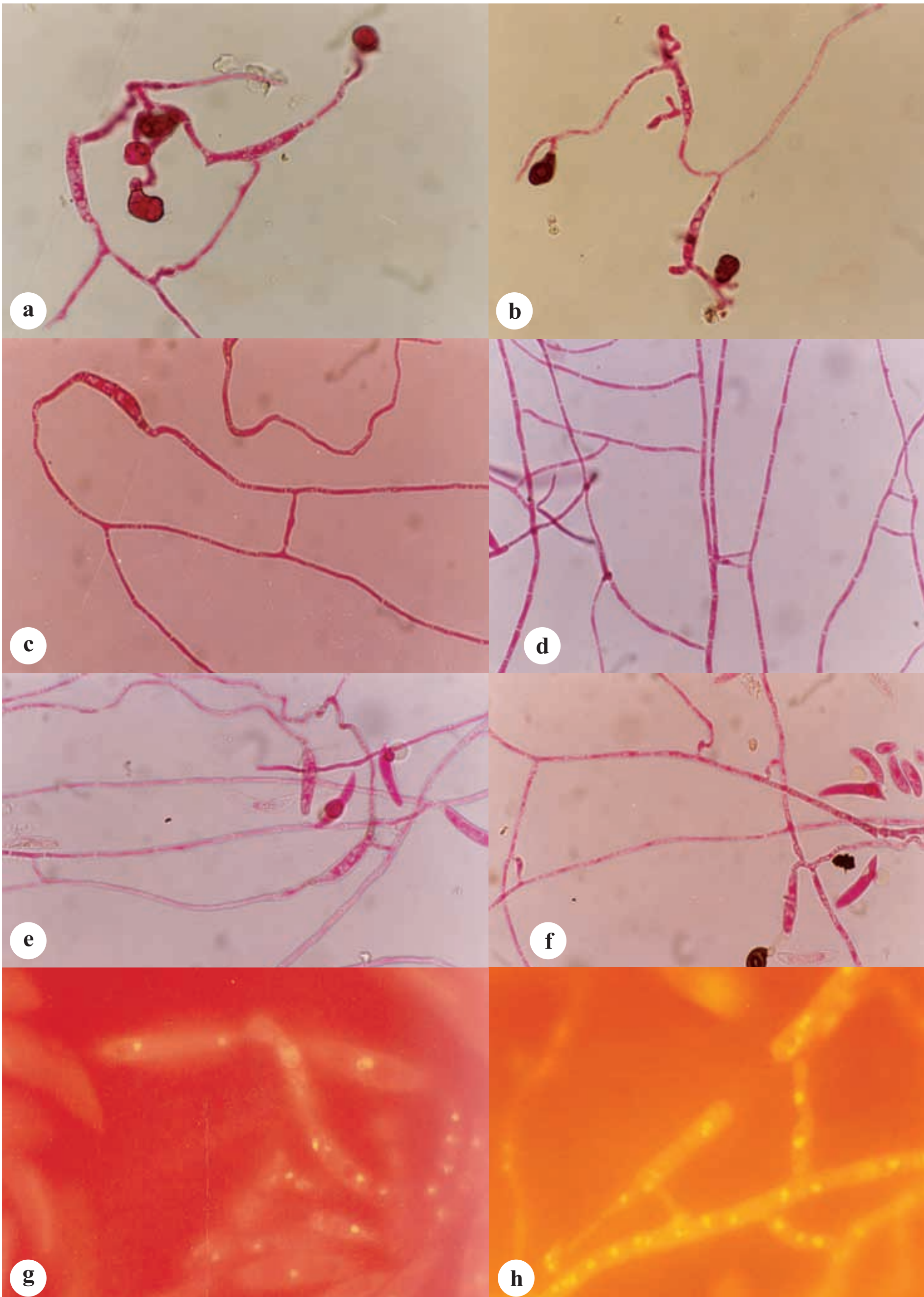

Plate 32 : Hyphal fusion and nuclear migration

(a) & (b) Fusion of hyphae of two conidia at germination (c) Fusion of hyphae of same conidia (d), (e) & (f) Multiple hyphal fusion (g) Nuclear migration from hyphae to conidium (h) Nuclear migration between hyphae.

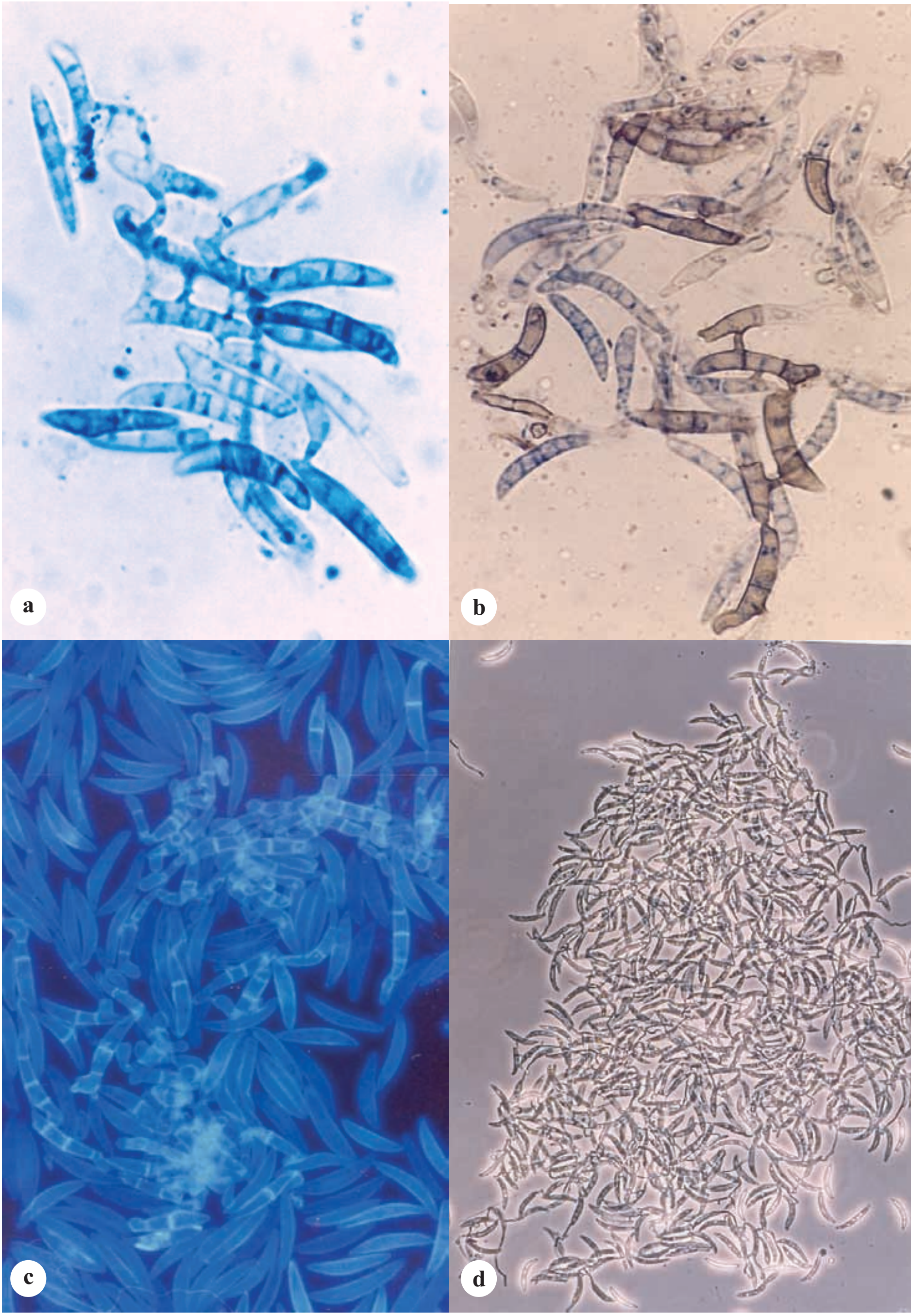

Plate 33 : **Conidial fusion in *C. falcatum***
(a) Multiple fusion among conidia (b) Fusion of conidia and transformation into dark coloured thick walled type (c) Multiple fusion (d) A fusion aggregate of conidia.

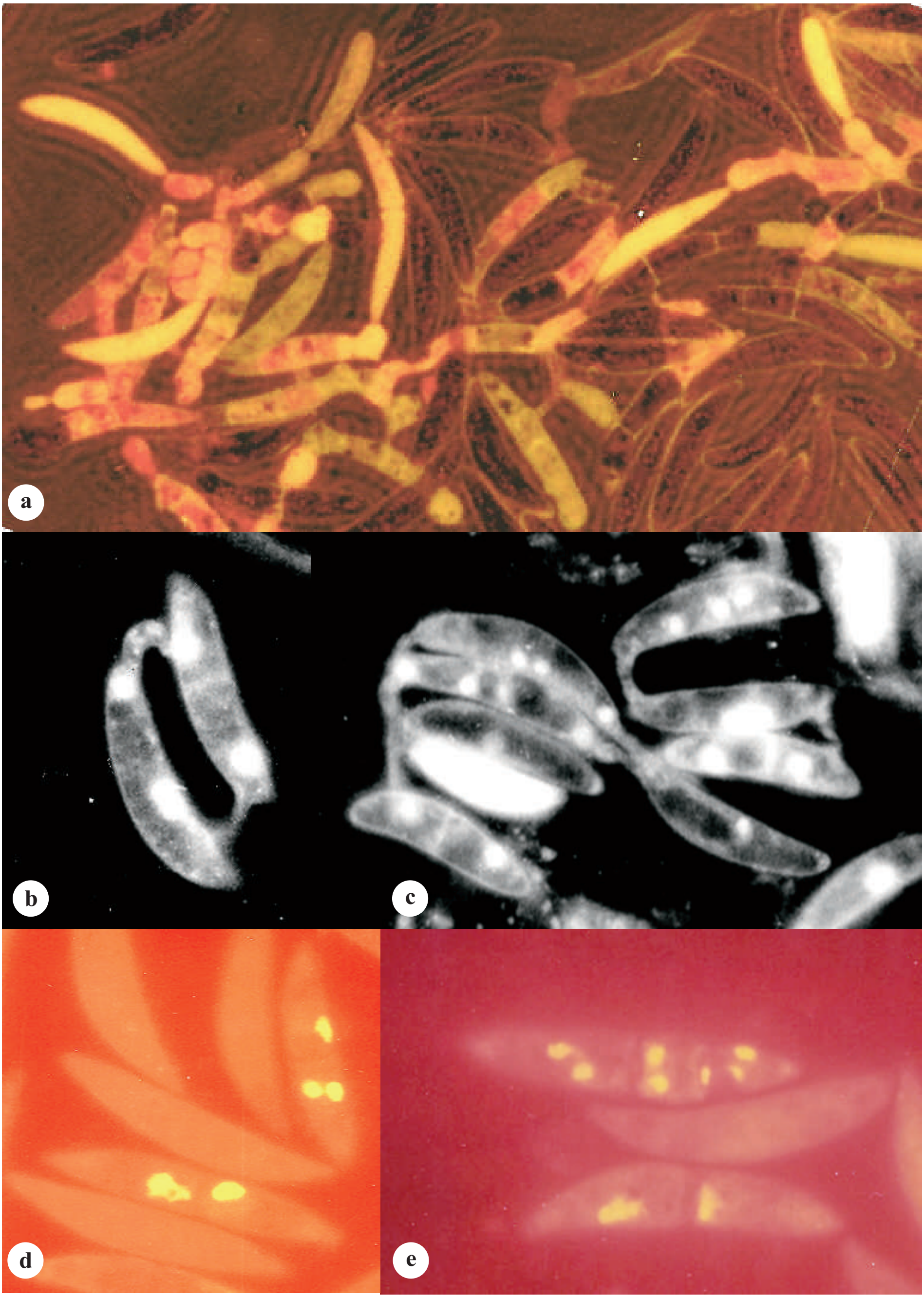

Plate 34 : Nuclear condition and conidial development
(a) Formation of daughter conidia from the fusion aggregate (b) & (c) Nuclear migration in conidial fusion (d) Division of nucleus in conidium (e) Multiple nuclei formation in conidium.

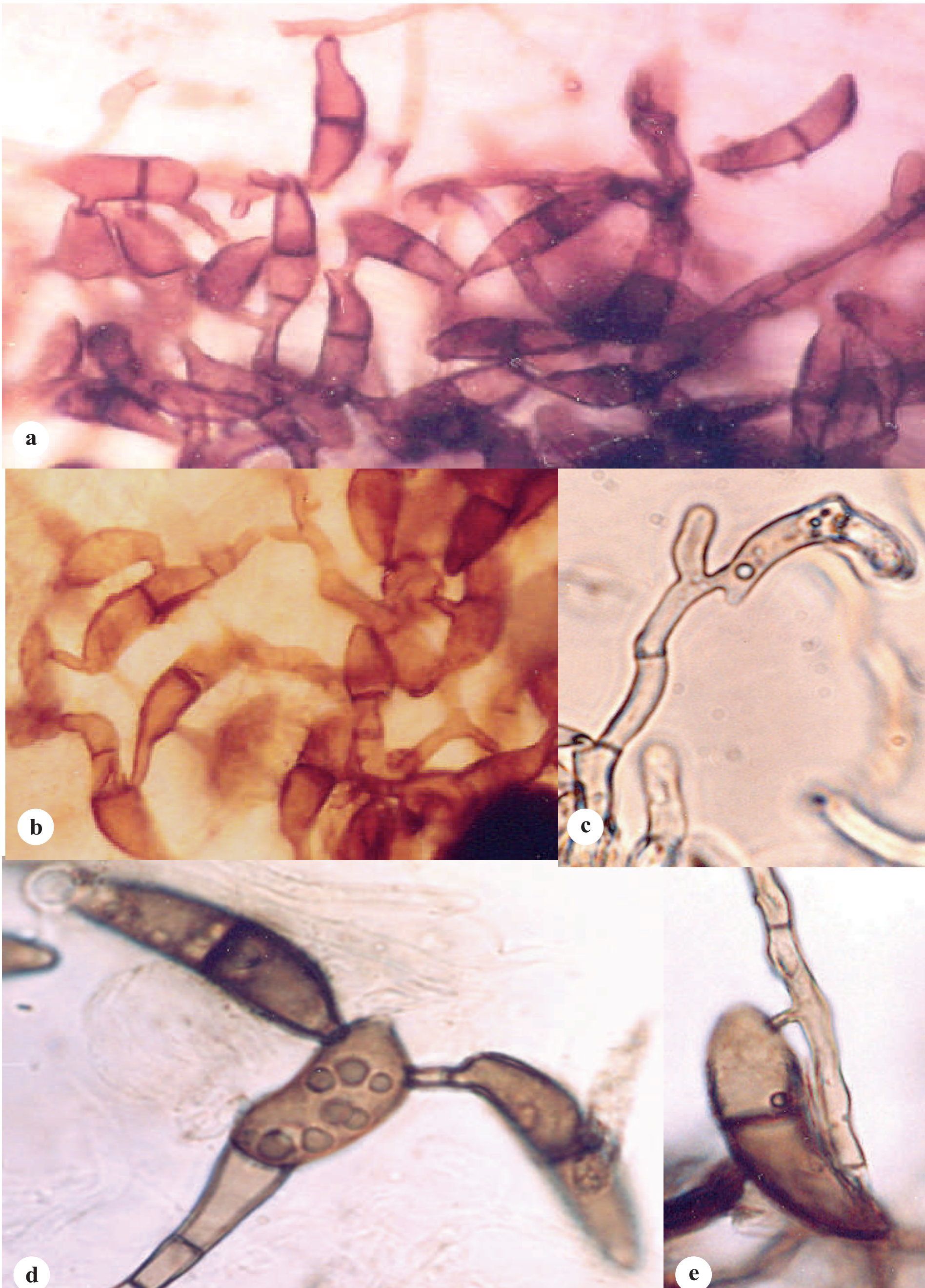

Plate 35 : Variation in conidial fusion

(a) & (b) Between conidia (c) & (e) Between conidium and hypha (d) Between conidium and appressorium (Note the change in colour of conidia)

Plate 36 : **Nodal morphology of cane**
(a) Typical node - wax ring, leaf scar, root band, bud, growth ring (c) Sunken root primordia (root eye) (d) & (e) Protruding root primordia (f) Bud groove. Note different bud morphologies - shape, size of bud, and bud scales.

Plate 37 : Development of roots in sugarcane
(a) & (b) Aerial roots, on mature standing cane (c) Sett roots, at germination (d) Damage of sett roots due to pathogen attack (germination failure).

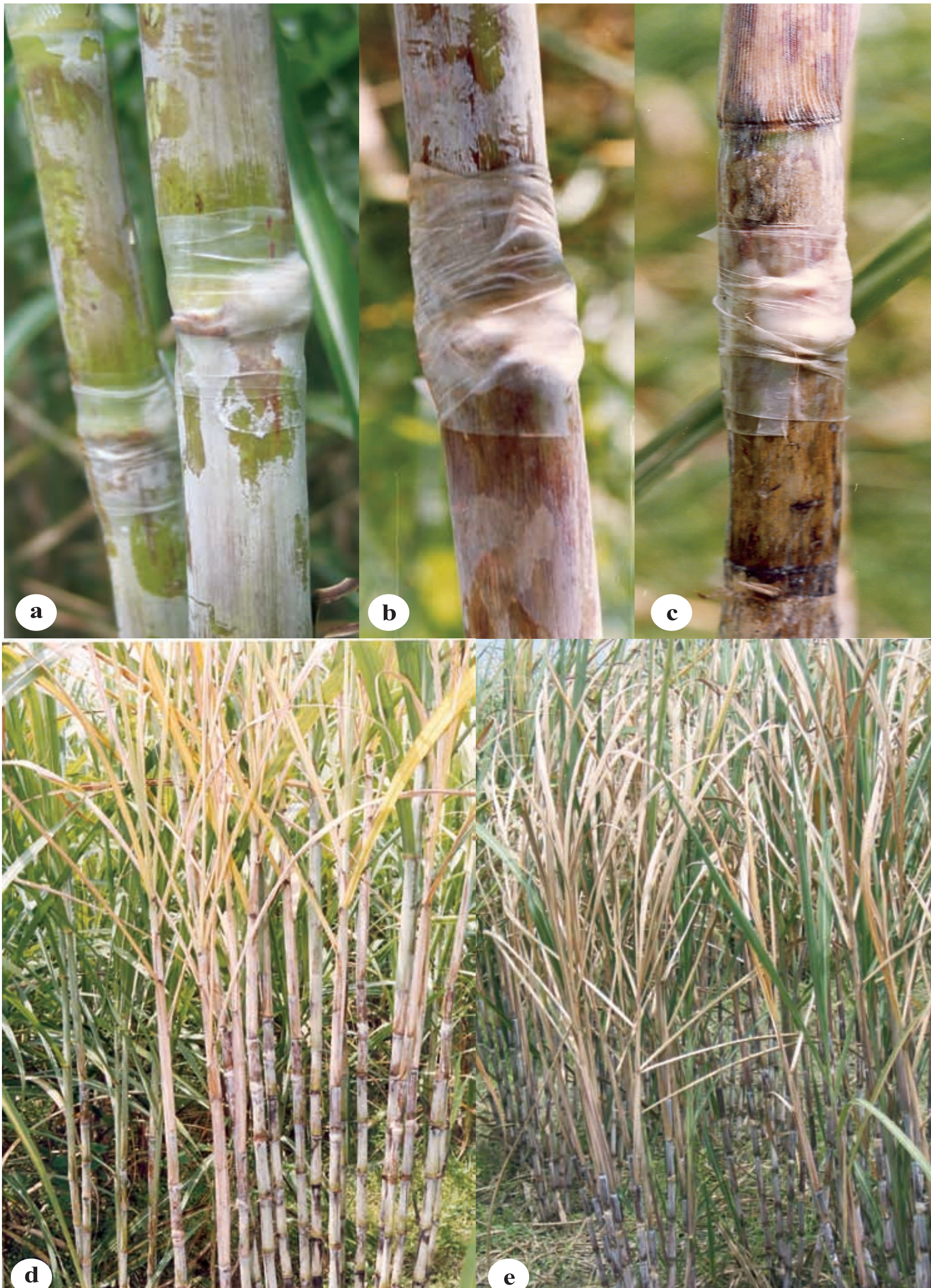

Plate 38 : Parafilm technique of inoculation

(a) Wrapping of parafilm with inoculum in cotton swab (b) & (c) Development of disease (note the discolouration of rind) (d) Yellowing of crown (f) Dying/death of plants.

Plate 39 : Inoculators and plug method

(a) Piercer (b) IISR inoculator (c) Portable drill (d) Hole in the internode (e) Hole with plug (f) Hole and plug covered with modelling clay.